chapter-1

Rajib
che-50

2020, 2017, 2015, 2016

(A) Sketch a diagram of a grating spectrometer? State the purpose of the use of the slits in grating spectrometer?

⇒

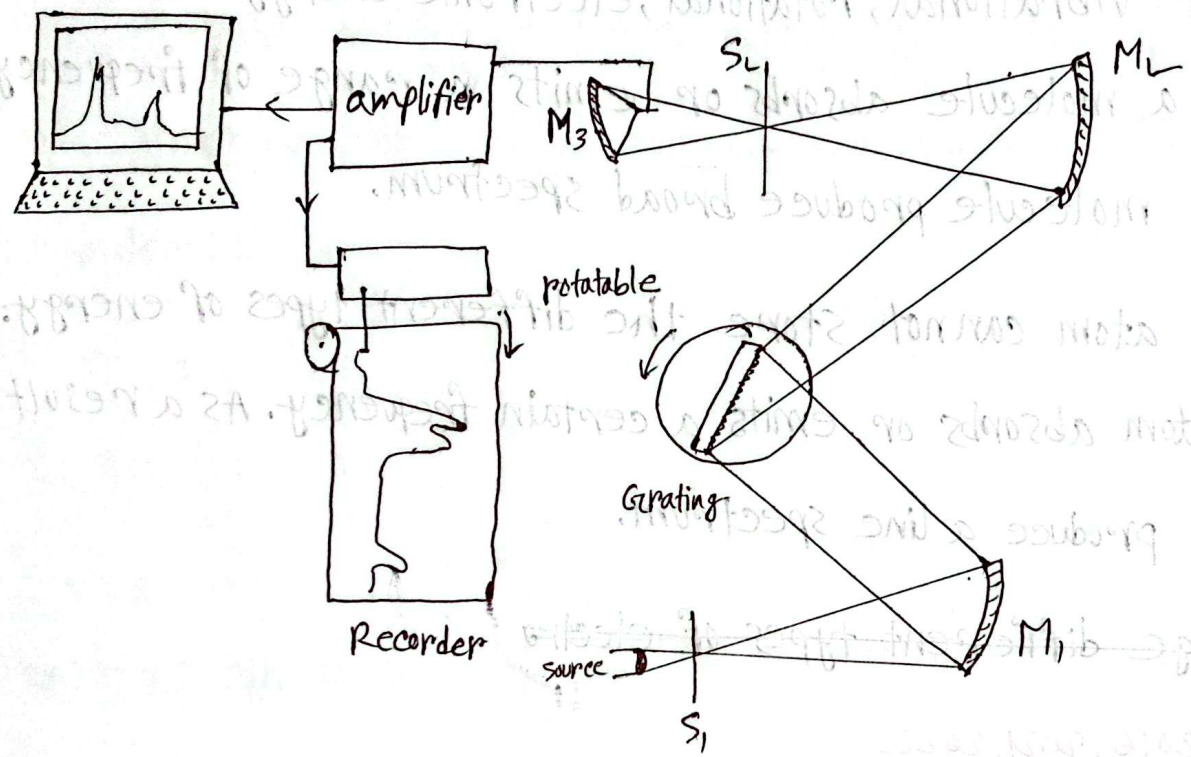


Fig: schematic diagram of a grating spectrometer.

⇒Purpose: (i) Firstly they provide a sharply defined image which can be focused on the detector. (ii) secondly The narrower the slits, the smaller is the frequency range which passes through & consequently the resolving power of the instrument is increased. (iii) Finally they help to prevent stary radiation perhaps scattered from various component of spectrometer, or arriving from outside, getting to the detector & thus being falsely recorded as part of the spectrum.

CamScanner

(★) 2018, 2022
Why does a molecule produce a band spectrum but an atom produce line spectrum?

⇒ A molecule in space can have ~~not~~ many types of energy, such as- vibrational, rotational, electronic energy etc. Because of this, a molecule absorbs or emits a range of frequency. So the molecule produce broad spectrum.

But an atom cannot store the different types of energy. So an atom absorbs or emits a certain frequency. As a result an atom produce a line spectrum.

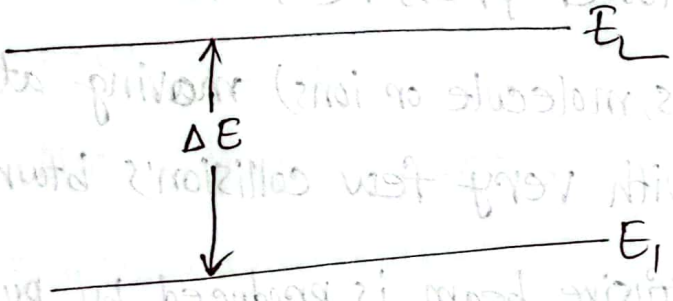
(★) 2017, 2016, 2021, 2022
What is meant by quantization of energy? Illustrate how all forms of energies are quantized?

⇒ quantization of energy is the concept that energy can only be gained or lost in discrete packets, rather than in a continuous manner. This means that energy exist in specific fixed amounts. These energy packets are called quanta.

Ex. The energy levels of an e^- in an atom are quantized, meaning e^- can only occupy certain discrete energy levels. Another

example is photoelectric effect.

⇒ a particular molecule can exist in a variety of rotational, vibrational etc, energy levels & can move from one level to another only by a sudden jump involving a finite amount of energy. Consider two possible energy states of a system - two rotational energy levels of a molecule, E_1 & E_2.



Transitions can take place btwn the levels E_1 & E_2 provided the appropriate amount of energy, $\Delta E = E_2 - E_1$, can be either absorbed or emitted by the system. which is suggested by planck

$$\Delta E = h\nu$$

Here the energy value for each energy level (E_1 & E_2) is fixed & their energy difference (ΔE) also fixed valve. so all forms of energies are quantized. Molecules vibrate at specific frequencies, & their vibrational energy is quantized. Rotating molecules also exhibits quantized energy Levels.

CamScanner

2017

Explain how molecular effusive beam can be produced? Describe effusive molecular beam method to remove pressure broadening?

⇒ A molecular effusive beam is produced by allowing a gas at higher pressure to expand through a small orifice into chamber at lower pressure to form a beam of particles (atoms, free radicals, molecule or ions) moving at approximately equal velocities with very few collision's btwn the particles.

OR, An molecular effusive beam is produced by pumping atom or molecules through a narrow slight, typically 20 μm wide & 1cm long with a pressure of a few torr on the source side of the slit. The beam may be further collimated by suitable apertures along it.

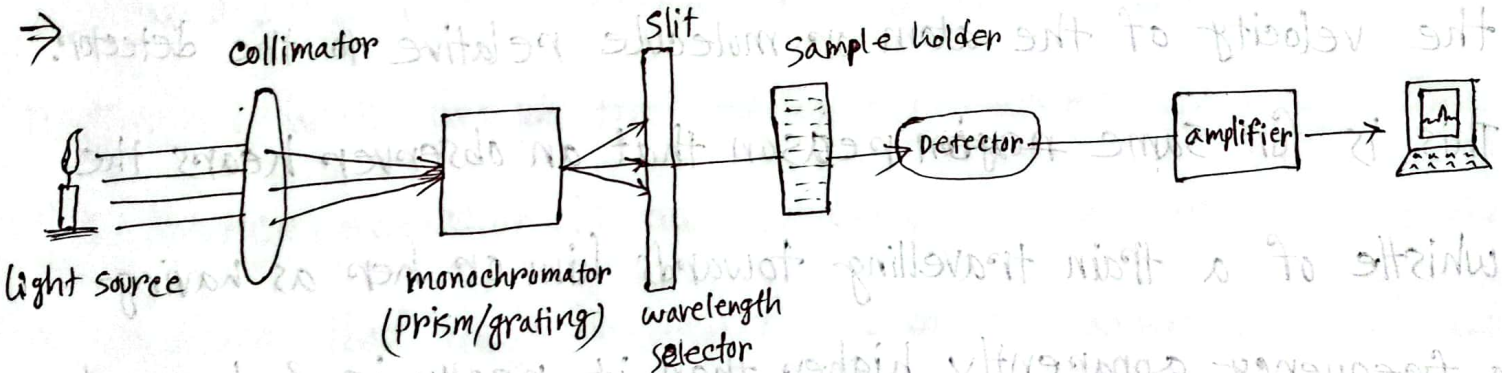
pressure broadening of spectral lines is removed in an effusive beam & if observations are made perpendicular to the direction of the beam, Doppler broadening is considerably reduced because the velocity component in the direction of observation is very small.

(X) 2015 Explain why sample is placed near the exit slit rather than enhance slit in the grating spectrometer?

⇒ when the sample is placed at exit slit, bathed in only a narrow range of frequencies at any time rather than the entrance slit. As a result the decomposition of sample is less when it is placed at exit slit rather than entrance slit. So sample placed near exit slit.

(X) 2024 Draw & label block diagram of single beam spectrometer & describe each component mentioning its function?

⇒



Function: ① Light source: Emits a broad spectrum of electromagnetic radiation (e.g. tungsten lamp for visible light, deuterium lamp for UV light).

(III) Monochromator: Isolates a narrow band of λ from light source.

(II) Entrance slit: controls the amount of light entering the monochromator.

(IV) Exit slit: Determines the spectral bandwith of the light reaching the sample.

CamScanner

5. Sample holder: contains the sample cuvette, which holds the Sln to be analyzed.

6. Detector: converts the transmitted light into an electrical signal.

7. Readout Device: Displays the measured absorbance or transmittance values.

⊗ 2020, 2016 what is Doppler broadening? How could you remove Doppler broadening (Lamb dip method) ?

⇒ Doppler broadening: whether radiation is being absorbed or emitted the frequency at which it takes place depends on the velocity of the atom or molecule relative to the detector. This is for same ~~reason~~ reason that an observer hears the whistle of a train travelling towards him or her as having a frequency apparently higher than it really is & lower when it is travelling away from him or her. This effect is known as the doppler effect. If an atom or molecule is travelling towards the detector with a velocity (V_z), then the frequency (ν_0) at which a transition is observed to occur is related to the actual transition frequency ν, in a stationary atom or molecu

$$\nu_a = \nu\left(1 - \frac{v_a}{c}\right)^{-1}$$

where c is speed of light. Because of unusual maxwell velocity distribution there is a ~~speed of v_a~~ spread of values v_a & a characteristic Doppler broadening is given by-

$$\Delta\nu = \frac{\nu}{c}\left(\frac{2kT\ln 2}{m}\right)^{1/2}$$

where m = mass of atom or molecule, $\Delta\nu$ is normally far greater than the natural line width.

⇒ lamb dip method : Fig-1 shows a Doppler-broadening line & Fig-2 illustrates an absorption experiment in which the source radiation is reflected back through the cell by a reflector, R. we assume that the source radiation is narrow compared with the line width, a condition that may be satisfied with, for example, microwave or laser radiation.

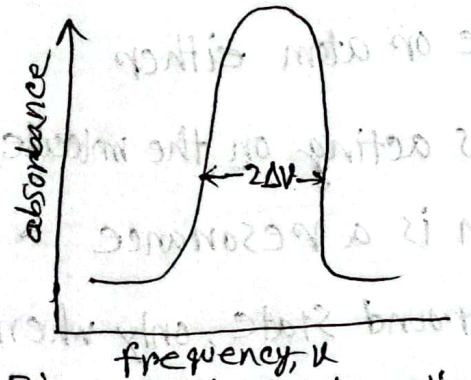


Fig-1: TYPical absorption line showing $\Delta\nu$ & a lamp dip

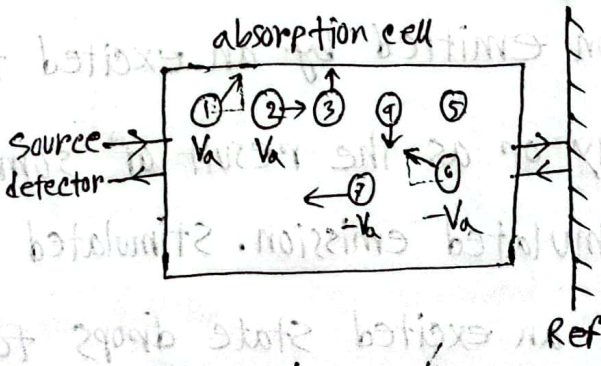


Fig-2: lamp dip experiment

Tuning the source to a frequency V_a, higher than the resonance frequency, at the line center, results in only molecules such as molecules 1 & 2 in Fig-2, which have a velocity component V_a away from the source, absorbing radiation. Molecules 6 & 7. Which have a velocity $-V_a$ away from the source. Molecules 3, 4 & 5 have zero velocity component away from the source, behave uniquely in that they absorb radiation of the same frequency V_{res} wheather the radiation is travel towards or away from the R, this results in saturation. If saturation occurs then no further absorption takes place. The result is that a dip in the absorbance curve is observed at V_{res}. This is known as lamb dip spectroscopy.

(*) 2020, 2015
what is meant by stimulated emission? with diagram & example

⇒ Radiation emitted by an excited molecule or atom either spontaneously or as the result of some stimulus acting on the molecule called stimulated emission. Stimulated emission is a resonance phenomenon - an excited state drops to the ground state, only when a photon of the same frequency ν interacts with the system.

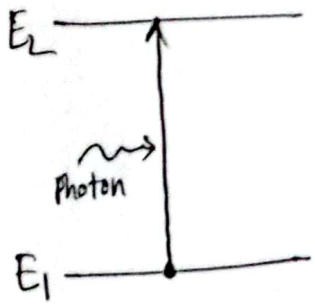
Fig: stimulated emission from an excited energy state.

Figure of stimulated emission show a photon of frequency $\nu_{stim.}$ interacts with the excited state & causes radiation of frequency $\nu_{emit.}$ to be released.

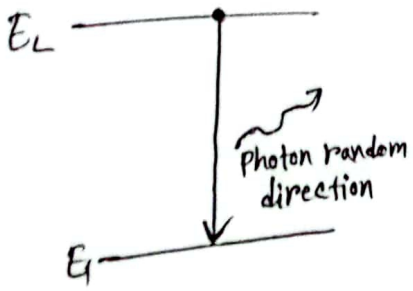
Ex. Laser (light amplification by stimulated emission of radiation) is the most common example of stimulated emission. In laser an optical cavity is used to contain & amplify light via stimulated emission. [Stimulated means induced]

Extra
⊗ Draw diagram of Stimulated (induced) absorption, spontaneous emission & stimulated emission?

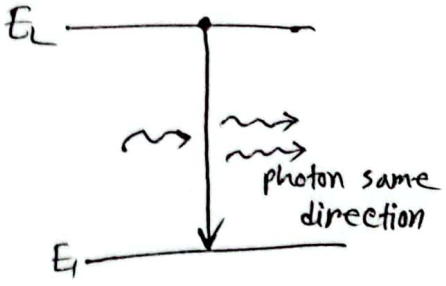
⇒



Stimulated absorption



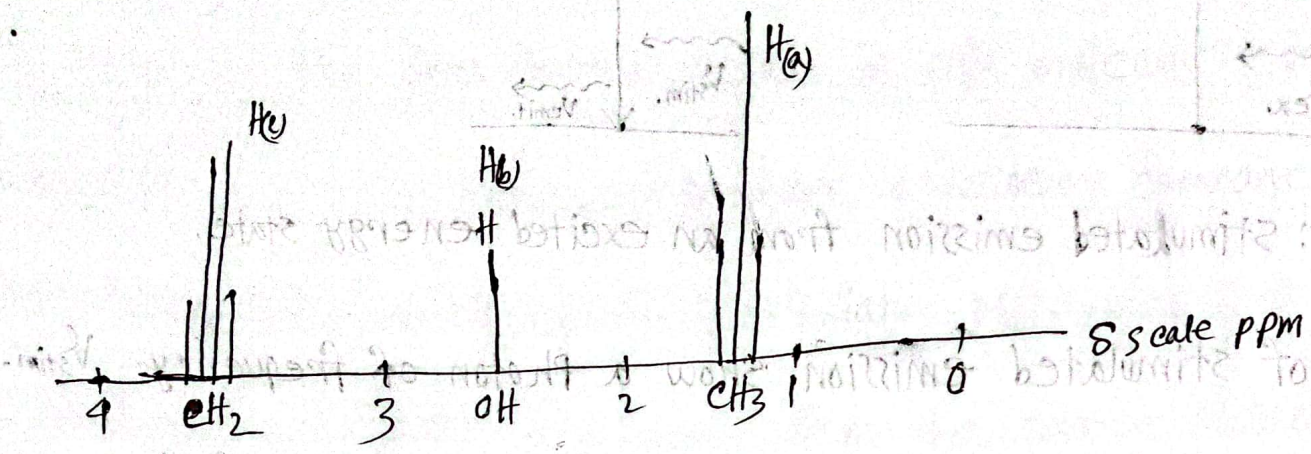
Spontaneous emission

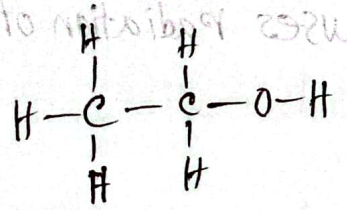
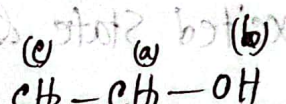


Stimulated emission

2016
Sketch the ^1H-NMR spectrum of ethanol?

⇒



ethanol →  → 

2015
What is meant by pressure broadening?

⇒ pressure/collision broadening: when collisions occur btwn gas phase atoms or molecules, there is an exchange of energy which leads effectively to a broadening of energy levels. This is known as pressure broadening. If τ is mean time btwn collision & each collision for transition btwn two states, then pressure broadening, $\Delta\nu = (2\pi\tau)^{-1}$.

2015
Describe effusive molecular beam method to remove line broadening?

⇒ The effusive m. beam method is a technique used to minimize line broadening effects, particularly pressure broadening & Doppler broadening. The effusive molecular beam method

creates a collimated stream of gas molecules in a vacuum. By isolating & directing the molecules, it significantly reduces collisions & minimizes thermal motion effects, thereby mitigating line broadening mechanisms.

Method: A gas sample is introduced into a chamber containing a small orifice. The orifice seperates the high pressure gas source from a low pressure vacuum chamber. Molecules escape through the orifice due to thermal motion, forming molecular beam. the effusion process ensures minimal collisions btwn molecules, as the mean free path of the particle is larger than the dimensions of the orifice. The beam is collimated using slits to create a directed stream of molecules with minimal angular divergence. The collimated molecular beam is exposed to the spectroscopic probe. Measurements are taken under conditions where pressure broadening & doppler broadening are reduced.

2nd chapter

(★) 2024, 2018, 2017, 2016

Write down the energy expression in cm^{-1} for a diatomic molecule based on rigid rotator model. Show that the absorption spectrum of a rigid rotator is expected to consist of a series of equidistant lines?

⇒ Eenergy Expression:

$$\varepsilon_J = BJ(J+1)\ cm^{-1} \ ; J=0,1,2,3,4\ldots\ldots \text{——①}$$

where, $B = \frac{h}{8\pi^2 Ic}$ = rotational constant, J = rotational quantum number

putting $J=0,1,2,3\ldots\ldots$ in eqn ① we get,

when $J=0$, $\varepsilon_J = 0\ cm^{-1}$

when $J=1$, $\varepsilon_J = 2B\ cm^{-1}$

when $J=2$, $\varepsilon_J = 6B\ cm^{-1}$

when $J=3$, $\varepsilon_J = 12B\ cm^{-1}$

A. to selection rule, for rigid diatomic molecule $\Delta J = \pm 1$ means that, a transition can occur from an particular level to its immediate neibouring state.

for $J=0$ to $J=1$ transition we get -

$$\varepsilon_{J=1} - \varepsilon_{J=0} = 2B - 0 = 2B\ cm^{-1}$$

And therefore - $\bar{\nu}_{J=0 \to J=1} = 2B\ cm^{-1}$

For $J=1$ to $J=2$ transition -

$$\bar{\nu}_{J=1\to J=2} = \varepsilon_{J=2} - \varepsilon_{J=1} = 6B - 2B = 4B \text{ cm}^{-1}$$

In general, to rise the molecule from the state, J to state (J+1), we would have,

$$\bar{\nu}_{J\to J+1} = \varepsilon_{J+1} - \varepsilon_J = B(J+1)(J+2) - BJ(J+1) = 2B(J+1) \text{ cm}^{-1}$$

Thus a stepwis raising of the rotational energy results in an absorption spectrums consisting of equidistant lines at 2B, 4B, 6B... cm^{-1} which is shown in Fig-1.

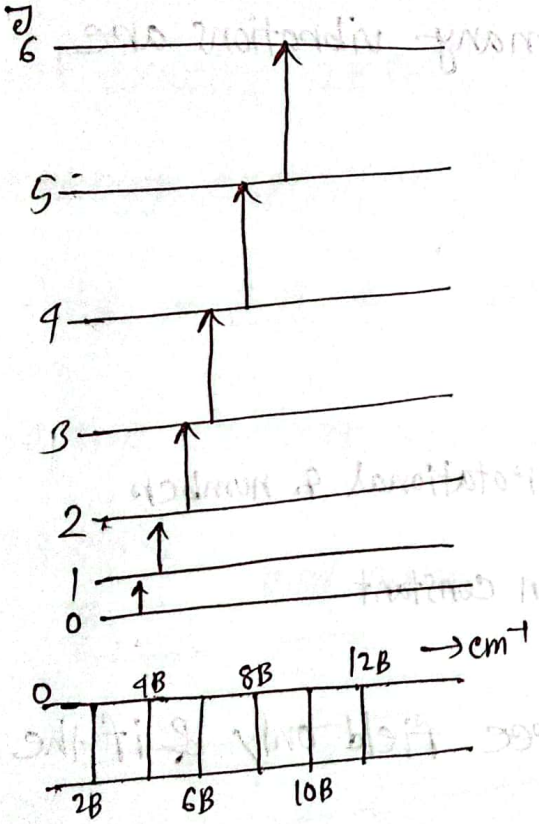


Fig: allowed transitions btwn the energy levels of a rigid diatomic molecule & the spectrum which arises from them.

2021, 2018, 2016
⊗ What are the limitations of the rigid rotator model? Write E for non rigid rotator in cm^{-1}? Compare the energy spacing btwn these two models? Discuss the isotope effect on rotational spectra?

⇒ Limitations: ① The bond btwn two atoms considered as rigid bond

in rigid rotator model which is not correct. since all bonds are elastic to some extent.

(ii) It is considered that the spectrum btwn succesive two lines are $2B\ cm^{-1}$ which is not correct. The seperation btwn succesive lines decrease steadily with increasing I.

For HF, $J=0$; $B = 20.56$

$J=1$; $B = 20.98$

$J=2$; $B = 20.93$

(iii) Here vibration is not considered, many vibrations are occured during a rotation.

⇒ energy for non rigid molecule is,

$$\mathcal{E}_J = BJ(J+1) - DJ^2(J+1)^2\ cm^{-1} \quad \text{---(i)}$$

where, $B = \frac{h}{8\pi^2 Ic}$ = rotational constant, J = rotational q. number

$D = \frac{h^3}{32\pi^4 I^2 r^2 Kc}$ = centrifugal distortion constant

eqn (i) applies for a simple harmonic force field only & if the field is anharmonic, the expression becomes-

$$\mathcal{E}_J = BJ(J+1) - DJ^2(J+1)^2 + HJ^3(J+1)^3 + KJ^4(J+1)^4 \text{ ----} cm^{-1} \quad \text{---(ii)}$$

where, H, K --- etc are small constant depend upon the geometry of the molecule & they are negligible in comparism to D. then eqn (i) reduces to eqn (i),

⇒ ~~comparison o~~ comparison of energy spacing btwn rigid & non-rigid:

Energy for non rigid -

$$\varepsilon_J = BJ(J+1) - DJ^2(J+1)^2 \ cm^{-1}$$

Now transition from J to J+1

$$\bar{V} = \varepsilon_{J+1} - \varepsilon_J = B(J+1)(J+2) - D(J+1)^2(J+2)^2 - BJ(J+1) + DJ^2(J+1)^2 \ cm^{-1}$$

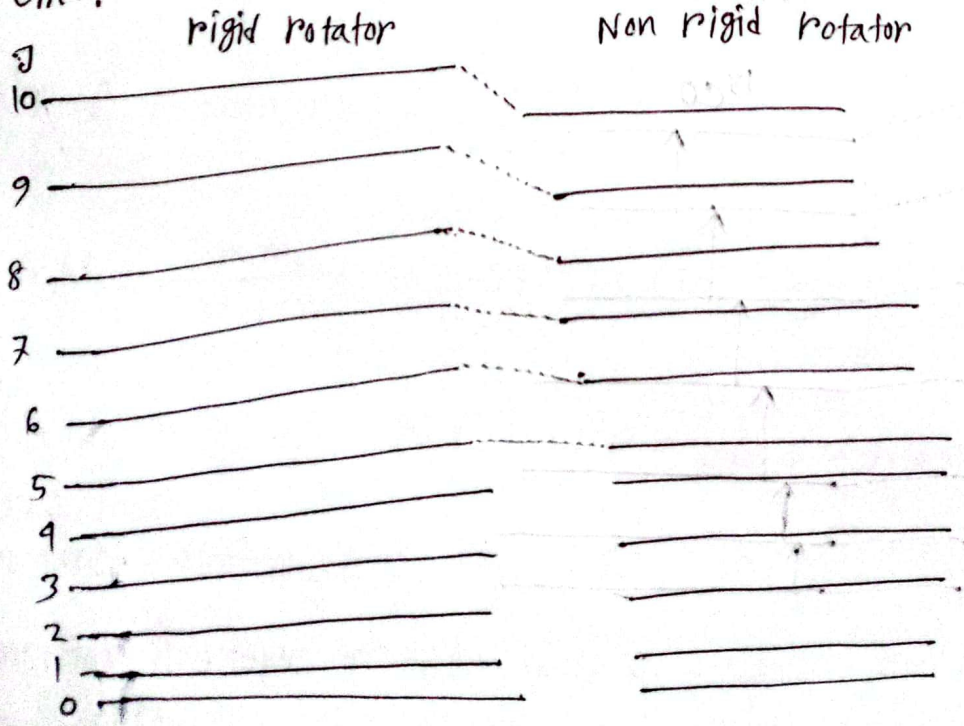
$$= 2B(J+1) - 4D(J+1)^3$$

So the energy spacing btwn two succesive energy levels is not exactly 2B cm^{-1}, it is smaller than 2B cm^{-1}.

energy expression for rigid rotator is - $\varepsilon_J = BJ(J+1) \ cm^{-1}$

$$\therefore \bar{V}_J = \varepsilon_{J+1} - \varepsilon_J = B(J+1)(J+2) - BJ(J+1) = 2B(J+1)$$

So the energy spacing btwn two successive energy level is 2B cm^{-1}.



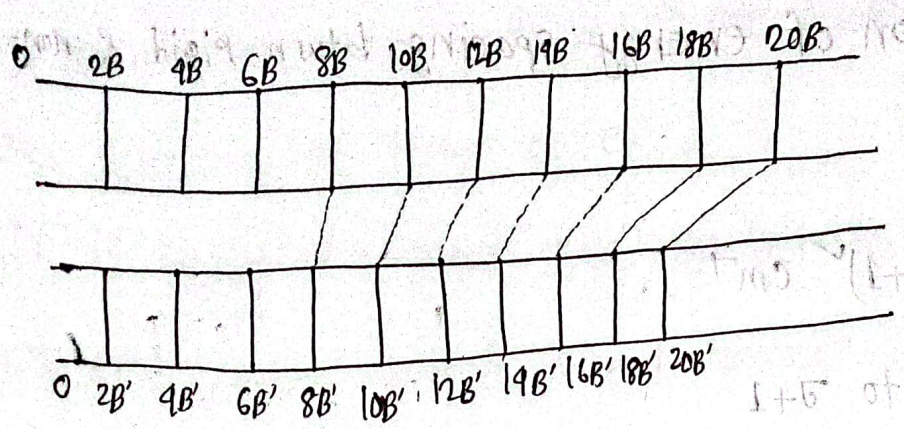
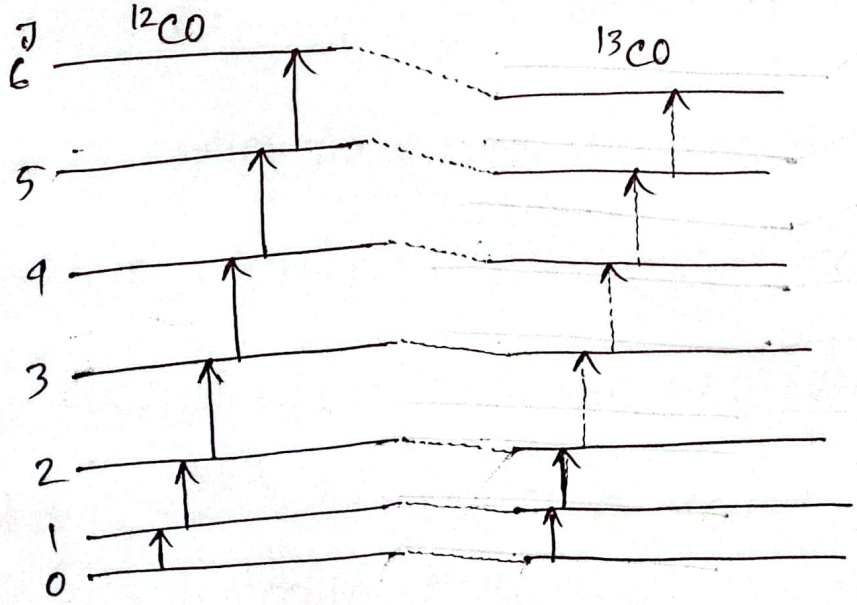


Fig: The change in rotational energy levels & spectrum when passing from a rigid to a non-rigid diatomic molecule.

⇒ Isotope effect on rotational spectra: When a particular atom in a molecule is replaced by its isotope, there occurs a change in total mass & moment of inertia & B value for the molecule. But there is no appreciable change in intermolecular distance on isotopic substitution. For CO, we see that on going from $^{12}C^{16}O$ to $^{13}C^{16}O$, there is a mass increase & hence a decrease in the B value. $B = \frac{h}{8\pi^2 Ic}$; $I = mr^2$



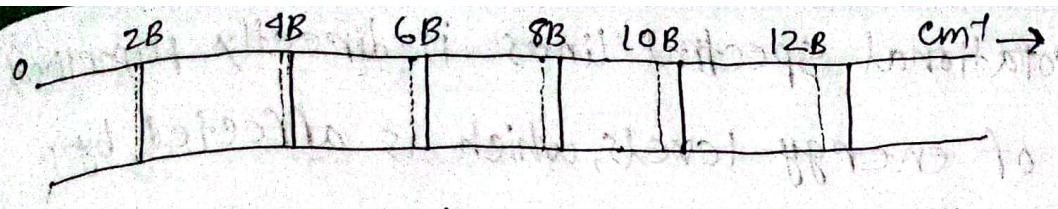


Fig: The effect of isotopic substitution on the energy levels & rotational spectrum of a diatomic molecule such as CO.

(★) 2016 The rotational spectrum of $^{79}Br^{19}F$ shows a series of equidistance lines $0.71933\ cm^{-1}$ apart. Calculate rotational constant B & hence moment of Inertia & bond length of the molecule?

⇒ Given, $2B = 0.71933\ cm^{-1} \Rightarrow B = 0.357165\ cm^{-1}$ Ans.

we know, $B = \frac{h}{8\pi^2 Ic} \Rightarrow I = \frac{h}{8\pi^2 Bc} = \frac{6.626\times10^{-34}}{8\times(3.1416)^2\times 0.357165\times3\times10^{10}}$

[As B value is in cm^{-1} so c value is in $cm\ (3\times10^{10}\ cm)$] $= 7.8399\times10^{-46}\ kgm^2$ Ans.

Now,

mass of Br, $m_1 = \frac{79\times10^{-3}}{6.023\times10^{23}} = 1.311638\times10^{-25}\ kg$

mass of F atom, $m_2 = \frac{19\times10^{-3}}{6.023\times10^{23}} = 3.154579\times10^{-26}\ kg$

$\therefore \mu = \frac{m_1 m_2}{m_1+m_2} = 2.592973\times10^{-26}\ kg$

We know, $I = \mu r^2 \Rightarrow r = \sqrt{\frac{I}{\mu}} = 1.75589\times10^{-10}\ m$ Ans.

(★) 2022, 2020 In the rotational spectrum of CO at 25°C, the $J=3 \rightarrow J=4$ transition is more intense than $J=0 \rightarrow J=1$ – explain?

⇒ Intensity of rotational spectral lines is directly proportional to the population of energy levels, which is affected by:

① Bolzmann distribution, ② Number of degenerate energy levels.

A.to Bolzmann distribution law,

$$\frac{N_J}{N_0} = e^{-BJ(J+1)hc/kT}$$

where N_J = population at any energy level

N_0 = " " first "

Population decreases as the energy level J value increase but the number of degeneracy is, $2J+1$. So

For, $J=0$, Number of degeneracy is, $2J+1=1$

$J=3$, " " , $2J+1=7$

which means each energy level is $2J+1$ fold degenerate.

Thus although the molecular population decreases exponentially, the number of degenerate level available increases rapidly with J.

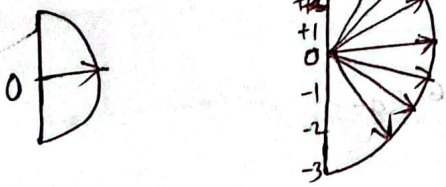


Fig: The one and 7 degenerate rotational orientations for a molecule with $J=0$ & $J=3$ respectively

The total relative population at an energy E_J will plainly be,

population $\propto (2J+1)\, e^{-BhcJ(J+1)/KT}$ ——— (11)

when this plotted again J, it shows that the population rises to a maximum the diminishes.

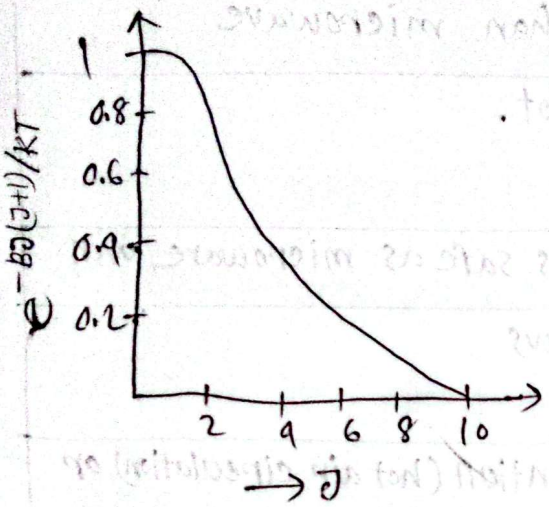


Fig: The total relative population without considering degeneracy.

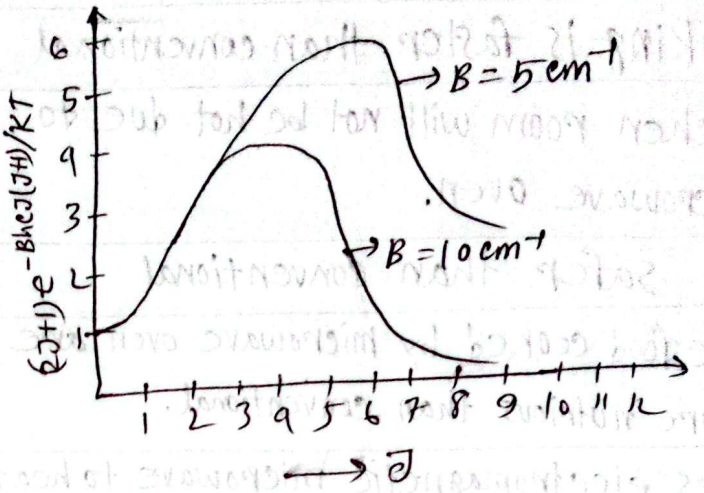


Fig: The total relative population including degeneracy of the rotational energy level of a diatomic molecule.

Differentiation of eqn (11) shows that the population is maximum at the nearest integral J value to, maximum population, $J = \sqrt{\frac{KT}{2hcB}}$ —½

Here, $T = 25°C = 298K$, assuming $B = 5$

$$J = \sqrt{\frac{1.38\times10^{-16}\times298}{2\times6.626\times10^{-27}\times3\times10^{10}\times5}} - ½ = 4$$

In this condition for $J = 4$, the population is maximum. The intensities is directly proportional to population, Hence it is plain that transitions btwn leve with very low or very high J value will have small intensities. For $J = 0 \to 1$ tran the value of J is low $J = 3 \to 4$ transition the value of J is moderate. So th rotational spectrum of CO at 25°C, the $J = 3 \to 4$ transition is more inten than $J = 0 \to 1$.

2017
What are the difference btwn microwave oven & conventional wave?

⇒

Microwave oven	conventional wave
Heating is internal for cooking food	heating from outside
cooking is faster than conventional	lower than microwave
Kitchen room will not be hot due to microwave oven.	will be hot.
Safer than conventional	It is not as safe as microware oven
The food cooked by microwave oven are more nutrious than conventional.	less nutrious
Uses electromagnetic microwave to heat & cook food	uses convention (hot air circulation) or radiant heat
Best for reheating & simple cooking	suitable for roasting & complex recipes
consumes less energy due to quicker cooking times	consumes more energy due to longer cooking times & preheating
easier to clean	requires more effort to clean
more energy efficient	less energy efficient
Generally more affordable	Generally more expensive

2017
Discuss the mechanism of action of microwave oven?

⇒ Its mode of operation depends entirely upon the absorption by the food of the microwave radiation in which it is bathed. only water molecules absorbs the radiation & so become raised into high rotational energy states, while the biological molecules in food are too large to be can not able to rotate. As with

many other excited states, the excess rotational energy of the water molecules is re-emitted as heat & the food becomes cooked.

(*) 2022, 2023(T)
How do you classify molecules A. to rotation along 3 directions ?

⇒ classifying molecules based on their rotational properties along 3 directions involves considering their rotational symmetry & moment of inertia along the principal axes.

1. Linear Molecules: All atoms are arranged in a straigh line. Two moments of inertia are equal ($I_B = I_C$) & the third is ~~negle~~ negligible ($I_A = 0$). Ex. HCl, H_2, CO_2, OCS

2. spherical tops: Two moment of ~~of~~ inertia inertia are equal & the third is different ($I_B = I_C \neq I_A$). ~~S~~ymmetric top can be further divided into two types: If $I_B = I_C > I_A$, then the molecule is called prolate symmetric top molecule. Ex. CH_3Cl.

If $I_B = I_C < I_A$, then the molecule is called oblate symmetric top molecule. Ex. C_6H_6

3. Spherical Top molecules: All three moment of inertia are equal ($I_A = I_B = I_C$). Ex. CH_4, SF_6. They have high symmetry.

CamScanner

4. Asymmetric top molecules: All three ~~molecules~~ moments of inertia are different ($I_A \neq I_B \neq I_C$). They have low symmetry. Ex. H_2O, CH_2Cl_2.

2015
What is rotational constant (B)? Describe different factors on the values of B?

⇒ B is a physical quantity used in molecular spectroscopy to describe the rotational energy levels of a diatomic or polyatomic molecule.

$$B = \frac{h}{8\pi^2 I c}$$

where, μ = reduced mass of the molecule.

$$\Rightarrow B = \frac{h}{8\pi^2 I c} = \frac{h}{8\pi^2 \mu r^2 c}$$

values of B depends on:

① Bond length(r): B is inversely propotional to r^2. Longer bond length(r) increas I & reducing B. & vice varsa.

2. Reduced mass(H): B is inversely proportional with M. heavier atoms has larger μ which lead to smaller B. & vice-versa.

3. Molecular isotopes: Different isotopes of atoms changes M & thus alter B.

4. Molecular geometry: In linear or symmetric top molecules,

CamScanner

the symmetry & distribution of mass affect I & hence B. For non-linear molecules, rotational constants along different axes (A,B,C) vary. Different Molecular shapes can lead to different B value

2017
⁂ Describe briefly the effect of different factors on the rotational energy & spectrum ?

⇒ Factors are: ① Moment of Inertia: rotational energy levels depend on I, which is determined by $I = \mu r^2$. Larger I lowers the energy spacing btwn levels, shifting the spectrum toward lower frequencies. $E = \frac{h^2}{8\pi^2 I} J(J+1)$ Joules ; where $J = 0, 1, 2 \ldots\ldots$

② Bond length & Atomic mass: Heavier atoms or longer bond length increase I, resuting in smaller energy level spacing & lower rotational frequencies in the spectrum. $I = \mu r^2$

③ Temperature: Higher T. populate higher rotational energy levels, leading to more transitions being observed in the spectrum.

④ Molecular geometry: Linear molecules have a simple rotational spectrum with equally spaced lines. symmetric top molecules have more complex spectra due to rotational additional degrees of freedom. Asymmetric top molecules have even more complex

Spectra with irregularly spaced lines.

(V) Vibrational state: can slightly increase the average bond length, affecting I & consequently the rotational energy levels.

(VI) Isotopic substitution: substituting atoms with different isotopes can alter the molecular mass & bond length, affecting I & the rotational energy levels. This can lead to slight shift in the rotational spectrum.

340-Solve (1&2)

RAJIB
Che-50

2021, 2022, 2018

(*) What is electromagnetic radiation? Mention its important characteristics? → 2016, 2017, 2020

⇒ EMR is a form of energy that propagates as both electrical & magnetic waves travelling in packets of energy called photons. It includes various types of radiation such as radio waves, microwaves, Infrared, visible light, UV, x-ray, ~~γ-ra~~ & γ-rays.

⇒ (I) EMR travel at the speed of light.

(II) All waves have amplitude, λ, velocity & frequency.

(III) EMR can travel through empty space.

(IV) EMR has both electric & magnetic components.

(V) The energy of EMR is directly proportional to its frequency. $E = h\nu$.

(VI) frequency of EMR is inversely proportional to its wavelength. $\nu = \frac{c}{\lambda}$.

(VII) Different types of EMR have different wavelength

(VIII) EMR exibits both wavelike & particle like behaviour.

CamScanner

(IX) EMR doesnot require a medium to travel. It can propagate through the vacuum of space.

(★) 2022, 2020, 2017, 2015
Mention limitations of microwave spectroscopy?

⇒ (I) MS is only for molecule having permanent dipole moment. (II) The spectra of large molecules can become very complex & difficult to interpret due to the numerous rational transitions. (III) the sample must be in gaseous state. (IV) MS generally provides lower resolution compared to other spectroscopic techniques like IR. (V) some transitions may have low intensities, making them difficult to detect. (VI) MS requires specialized equipment, including microwave sources, waveg & detectors which are expensive. (VII) MS requires high concentration sample molecules (VIII) constructing & maintaining microwave spectroscopy instrumentation can be challenging & expensive.

(★) 2022, 2019(t), 2021(t)
write EMR that has highest & lowest λ?

⇒ highest → Radio wave, lowest → γ rays.

2022, 2020, 2017, 2015

(*) State criteria for a molecule to be microwave active. Give 4 example which show rational microwave absorption Spectrum?

⇒ criteria: (I) molecule must have a permanent electric dipole moment. (II) There needs to be a uneven distribution of ~~energy~~ e⁻ density with the molecule (heteronuclear diatomic molecule). (III) Dipole moment should be changed during the rotation of the molecule (IV) the molecule must be able to undergo rotational transitions.

⇒ Example: H_2O, NH_3, CO, HCl, CH_3OH, CH_3Cl, CH_2Cl_2

2021, 2018, 2016,

(*) which molecule show microwave rotational spectrum: H_2, HCl, CH_4, CH_3Cl, CH_2Cl_2, H_2O, SF_6 ?

⇒ ~~activ~~ microwave active → HCl, CH_3Cl, H_2O, CH_2Cl_2

microwave inactive → H_2, CH_4, SF_6

2021, 2018

(*) classify on the basis of m. of inertia: HCl, CO_2, SiH_4, SF_6, NH_3, CH_3Cl, H_2O, CH_3OH ?

⇒ linear → HCl, CO_2 symmetrical tops: NH_3, CH_3Cl

spherical tops: SiH_4, SF_6 Asymmetrical tops: H_2O, CH_3OH.

CamScanner

2016
Classify on basis of m. of inertia: HCl, CH_4, CH_3F, H_2O, OCS, BCl_3 ?

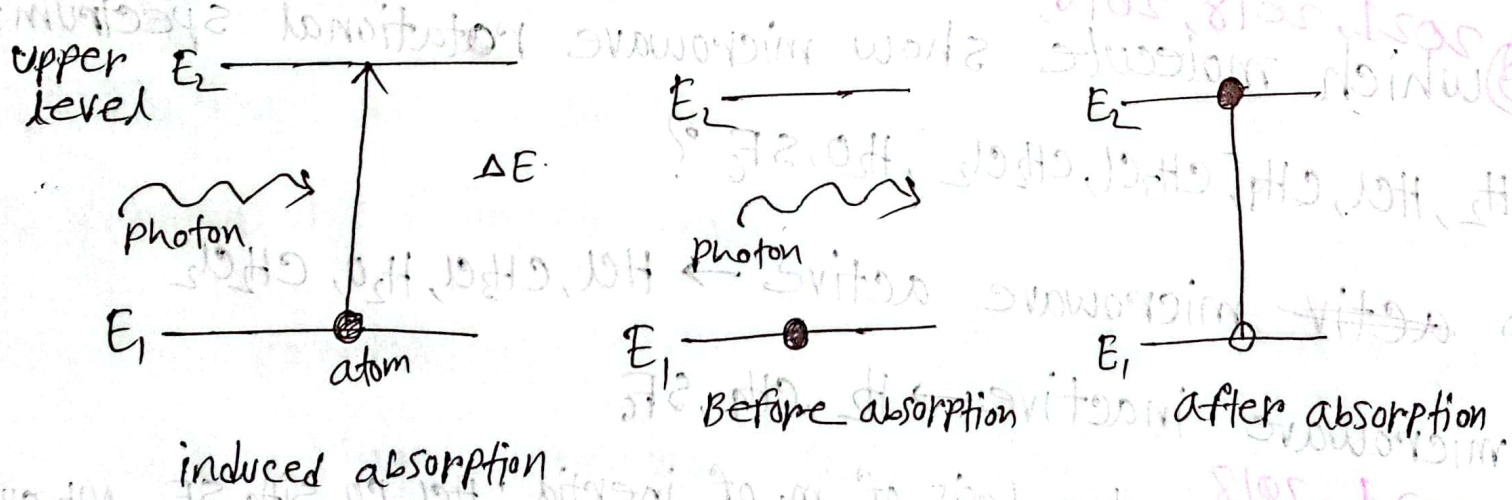
⇒ linear → HCl, OCS spherical tops: CH_4

symmetric tops → CH_3F, BCl_3 asymmetric tops: H_2O

(prolate) (oblate)

2020, 2016
What is induced absorption? (2016)

⇒ Induced absorption is the process in which an atom sized system in lower energy state is raised in to higher energy state by electromagnetic radiati which is quanta of energy to the difference of energy of the two states.



Examples of induced absorption include: Stark effect, zeeman effect, Two-photon absorption etc.

⊛ 2016 Difference btwn spontaneous & stimulated emmission?

spontaneous	Stimulated
The process where an excited e- randomly falls to a lower energy level due to natural tendency of the atom to attain minimum energy.	The process where excited e- falls to lower energy level due to the presence of external photon.
Light obtained is not coherent.	coherent light can be obtained
Results in ordinary light	can be utilized to get laser beam.
Single photon is emitted	Two photons are emitted.
polychromatic radiation	Monochromatic radiation.
Low intense & less directional	High intense & more directional
More angular spread	Less angular spread.
used in natural light sources, fluorescence, phosphorescence	used in lasers, amplifiers & quantum electronics.
It is uncontrollable process	It is controllable process

⊛ 2015, 2017 State the factors which cause the broadening of spectral lines?

⇒ Doppler Broadening, natural B., Pressure B., Stark B., Zeeman B., Instrumental Broadening.

2015, 2017

Difference btwn induced absorption & stimulated emission Process ?

induced absorption	stimulated emission
The process where an atom or molecule absorbs a photon & transitions to a higher energy state.	The process where an excited atom or molecule is stimulated by an incoming photon to emit a second photon & transition to a lower energy state.
A photon is absorbed	a photon is emitted
transition from lower to higher energy state.	t. from higher to lower energy state.
The absorbed photon ceases to exist in its original form.	The emitted photon has the same phase, frequency, polarization and direction as the stimulating photon.
More particles in the lower energy state.	More ~~population~~ particles in the excited state.
Does not directly contribute to laser action	Fundamental process for laser operation
example: absorption of light by a dye in a solution	Emission of light in a laser
can occur spontaneously	Requires an external photon
Energy of the photon matches the energy gap btwn States	Energy of the emitted photon matches the energy of the incident photon.

CamScanner

2021(t), 2019(t)

⊛ What is electromagnetic spectrum?

⇒ The electromagnetic spectrum is the range of all types of e. radiation. The spectrum is divided into various regions based on λ & ν such as radiowave, microwave, Infrared, visible light, UV, X-ray, γ-ray.

2021(t), 2019, (t), 2017, 2018, 2020, 2021

⊛ What are various types of e- radiation? write ν of micro & radiowave? Arrang them A. to increasing energy/frequency?

1. Radiowave: $\lambda \longrightarrow$ 1om to 1cm

$\nu \longrightarrow 3 \times 10^6$ to 3×10^{10} Hz

uses→ communications (tv, radio, cell phone), radar, astronomy.

2. Micro wave: $\lambda \longrightarrow$ 1cm to 100 μm.

$\nu \longrightarrow 3 \times 10^{10}$ to 3×10^{12} Hz.

uses: Microwave oven, wi-fi, radar, satellite.

3. IR: $\lambda \longrightarrow$ 100 μm to 1 μm.

$\nu \longrightarrow 3 \times 10^{12}$ to 3×10^{14} Hz

Uses: Remote controls, night vision, heating, spectroscopy.

4. visible & UV: $\lambda \longrightarrow$ 1 μm to 10 nm

$\nu \longrightarrow 3 \times 10^{14}$ to 3×10^{16} Hz

CamScanner

Uses: Visible is used in human vision, photography & UV is used in fluorescent lamps, UV curing, detecting fake currency

5. X-ray: $\lambda \longrightarrow$ 10 nm to 100 pm.

$\nu \longrightarrow 3 \times 10^{16}$ to 3×10^{18} Hz

Uses: Medical imaging, security scanning, material analysis.

6. γ-ray: $\lambda \longrightarrow$ 100 pm to 1 pm.

$\nu \rightarrow 3 \times 10^{18}$ to 3×10^{20} Hz

Uses: cancer treatment, astrophysical observation.

Arrange: γ-ray > x-ray > UV > visible > IR > Microwave > Radiowave.

(2021(t)) What are the different colors of visible light?

⇒ colors are: Violet, indigo, Blue, Green, Yellow, orange, Red.

Violet → 380-424 nm.	yellow → 575-590 nm
Indigo → 424-450 nm	orange → 590-647 nm
Blue → 450-500 nm	Red → 647-780 nm.
Green → 500-575 nm	

(2021(t)) Explain why CH_4 does not show rotational spectrum but show IR spectrum?

⇒ For a molecule, to exibit rotational spectrum, it must have a permanent dipole moment. But CH_4 is highly symmetrical tetrahedral molecule. The four

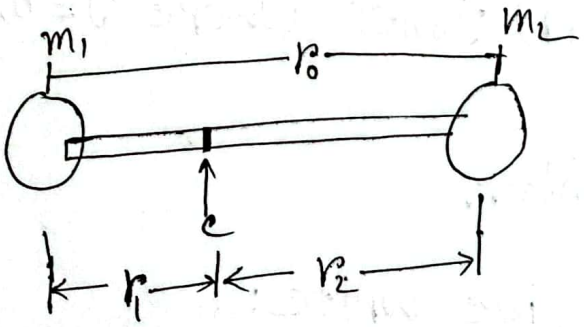
CamScanner

C-H bonds are arranged symmetrically, this symmetry leads to a cancellation of dipole moment, resulting zero net dipole moment. So it does not show rotational spectrum.

Even CH_4 has no permanent dipole moment, vibrational modes can induce a temporary dipole moment, allowing interaction with IR Radiation and show IR spectrum.

⊛ 2023(T) Derive the eqn to estimate the bond length of a rigid ~~no~~ diatomic molecule?

⇒ A rigid diatomic molecule treated as two masses m_1 and m_2 joined by a rigid bar of length, $r_0 = r_1 + r_2$



The molecule rotates end-over-end about a point 'c', the centre of gravity. Now,

$$m_1 r_1 = m_2 r_2$$

The moment of inertia about c is defined by: $I = m_1 r_1^2 + m_2 r_2^2$

$\Rightarrow I = m_2 r_2 r_1 + m_1 r_1 r_2 = r_1 r_2 (m_1 + m_2)$

Now, $m_1 r_1 = m_2 r_2 = m_2 (r_0 - r_1)$ $[\because r_0 = r_1 + r_2]$

So, $r_1 = \frac{m_2 r_0}{m_1 + m_2}$ and $r_2 = \frac{m_1 r_0}{m_1 + m_2}$

$$I = \frac{m_1 m_2}{m_1 + m_2} r_0^2 = \mu r_0^2$$

where, $\mu = m_1 m_2 / (m_1 + m_2)$ and μ is called reduced mass of the system.

By the use of schrodinger eqn it may be shown that the rotational energy levels allowed to the rigid diatomic molecule are given by the expression:

$$E_J = \frac{h^2}{8\pi^2 I} J(J+1) \text{ joules where } J = 0, 1, 2, 3 \ldots$$

J = rotational quantum number.

In term of wavenumber, we write:

$$\epsilon_J = \frac{E_J}{hc} = \frac{h}{8\pi^2 I c} J(J+1) \text{ cm}^{-1} \quad (J = 0, 1, 2 \ldots)$$

where 'c' is velocity of light, we can write,

$\Rightarrow \epsilon_J = B J(J+1)$ cm^{-1}

where B is rotational constant, $B = \frac{h}{8\pi^2 I_B c}$ cm^{-1}.

2022, 2020, 2019(T), 2018(T), 2021(T), 2017

(★) The first rotational line of CO is observed at 3.89235 cm^{-1}. calculate 'Moment of inertia and bond length of CO. The relative atomic weight C = 12 & O = 15.9994, the absolute mass of H = 1.67343×10^{-27} kg ? also calculate rotational constant (B) ?

$\Rightarrow \bar{\nu}_{0\to1} = 3.89235 = 2B\ cm^{-1}$

$\Rightarrow B = 1.92118\ cm^{-1}$ Ans

Now, $I = \frac{h}{8\pi^2 BC} \Rightarrow I_{CO} = \frac{6.626 \times 10^{-34}}{8 \times (3.1416)^2 \times 1.92118 \times 3 \times 10^8 \times 10^2}$

$= 1.46 \times 10^{-46}\ kgm^2$ Ans.

Given, $m_1 = 12$, $m_2 = 15.9994$, $\mu = \frac{m_1 m_2}{m_1 + m_2} = \frac{12 \times 15.9994}{15.9994 + 12} \times 1.6605 \times 10^{-27}$

$= 1.1396 \times 10^{-26}\ kg$

Now, $r = \sqrt{\frac{I}{\mu}} = 1.131 \times 10^{-10}\ m$ Ans.

2021, 2018, 2018(T) 2021

(★) Calculate 'I' of water. HOH = 109.5° & Bond length (OH) = 95.7pm (H = 1.0079, amu = 1.6606×10^{-27}) ?

$\Rightarrow I = \sum m_i r_i^2 = m_H r_H^2 + 0 + m_H r_H^2 = 2 m_H r_H^2$

$\Rightarrow I = 2 m_H R \sin\theta = 2 \times 1.6606 \times 10^{-27} \times (95.7 \times 10^{-12}) \times (\sin 52.25°)^2$

$= 1.901 \times 10^{-47}\ kgm^2$ Ans.

$\sin\theta = \frac{r_H}{R} \Rightarrow r_H = R \sin\theta$

2018(t), 2019(t), 2018

Calculate energy associated with 2 mol of photons of green light (555 nm) ($c = 3\times10^8$ m, $h = 6.626\times10^{-34}$ Js)

⇒ Energy per 1 photon $E = \frac{hc}{\lambda}$

$$= \frac{6.626\times10^{-34}\times3\times10^8}{555\times10^{-9}}$$

$$= 3.58\times10^{-19}\ J$$

∴ energy in 1 mole $= 3.58\times10^{-19}\times6.023\times10^{23}\ J\,mol^{-1}$

$$= 2.16\times10^5\ J\,mol^{-1}$$

∴ energy in 2 mole $= 2.16\times10^5\times2 = 4.32\times10^5$ J Ans

2019(t)

Classify the rigid rotors ?

⇒ rigid rotors classified based on their moment of inertia.

1. linear rotors: Molecules with a linear geometry, meaning all atoms lie along a straight line. Eamples: CO, HCl, CO_2, OCS.

2. spherical rotors: Molecules with perfectly symmetrical structures, where all atoms are equidistant from a

central point. Examples: CH_4, SF_6, SiH_4 etc.

3. Symmetric top rotors: Molecules with two equal moments of inertia. ~~Two axes of rotation are identical~~, while the third is different. Examples: CH_3Cl, NH_3, CH_2Cl_2, CH_3F, BCl_3

4. Assymmetric top rotors: Molecules with three different moment of inertia. Examples: H_2O, CH_3OH

2021(t), 2020, 2021(t)

Explain the reason why rotational constants of $^{12}C^{16}O$ and $^{13}C^{16}O$ are 1.9211 cm^{-1} & 1.88669 cm^{-1} respectively?

⇒ Rotational constant (B) is a key parameter in rotational spectroscopy. B is inversely proportional to the moment of inertia (I).

$$B = \frac{h}{8\pi^2 I c}$$

and the moment of inertia depends on the masses of the atoms.

$$I = \mu r_0^2$$

where $\mu = \frac{m_1 m_2}{m_1 + m_2}$ = reduced mass.

reduced mass of $^{12}C^{16}O$ is $\mu(^{12}C^{16}O) = \frac{12 \cdot 16}{12+16}$

$= 6.857\ u$

and $\mu(^{13}C^{16}O) = \frac{13 \cdot 16}{13+16} = 7.172\ U$.

∴ As reduced mass increased in $^{13}C^{16}O$, so moment of inertia of $^{13}C^{16}O$ is increased and also for this reason rotational constant B is decreased in $^{13}C^{16}O$.

2020, 2024(1), 2018

⁂ Explain the factors that are responsible for the intensity of the spectral line?

⇒ The intensity of a spectral line is influenced by several factors:

1. population of Energy levels: Higher populations of atoms ~~molecules~~ in the excited state lead to more intense spectral lines.
2. Transition probability: Transitions with higher probabilit produce more intense spectral lines.
3. concentration of species: A higher concentration results in more intense spectral lines.
4. path length: Intensity of absorption or emission is proportional to the path length.

5. Pressure and collisional Broadening: higher pressure sometimes intensify spectral lines.

6. Instrumental factors: It can affect the observed intensity.

7. Self absorption and saturation effect: emitted photon can be reabsorbed and can reduce the observed intensity.

8. Doppler Broadening: It sometimes affect the peak intensity of spectral lines.

(*) 2021 The λ of photon is 500 nm. calculate $\bar{\nu}$, ν, E ?

$$\Rightarrow \bar{\nu} = \frac{1}{\lambda} = 2 \times 10^{6}\ m^{-1}.$$

$$\nu = \frac{c}{\lambda} = \frac{3 \times 10^{8}}{5 \times 10^{-7}} = 6 \times 10^{14}\ Hz$$

$$E = \frac{hc}{\lambda} = \frac{6.626 \times 10^{-34} \times 3 \times 10^{8}}{500 \times 10^{-9}} = 3.98 \times 10^{-19}\ J$$ Ans.

(*) 2022, 2015 What is signal to noise ratio ? How do you resolve it ?

⇒ Since almost all modern spectrometer use some form of electronic amplification to magnify the signal produced by

the detector, every recorded spectrum has a background of random fluctuation caused by spurious electronic signal produced by the detector or generated in the amplifying device. These fluctuation are called noise. In order that a real spectral peak must have a 3 or 4 times greater intensity than that of the noise fluctuations. This requirement places a lower limit on the intensity of observeable signals. Normally signal to noise ratio is 3:1 or 4:1.

⇒ In optical spectroscopy, the minimum $\bar{\nu}$, λ or ν difference btwn two lines in a spectrum that can be distinguished is called resolution. Resolving power R is a measure of the ability of a spectrometer to seperate two close λ, ν or $\bar{\nu}$ and is defined by— $R = \frac{\lambda}{\Delta\lambda}$ (In term of λ)

where λ = average of the two wavelength & $\Delta\lambda$ = difference btwn two wavelength. greate the R valve better the resolving power broadened since collision interfere with this motions too. There is more collision in liq. phase than gas phase. Here gas phase spectra usually exhibit shaper lines. Again the motion of solid particles, are more limitted. so the solid phase spectra are often sharp but show evidence of interactions by the splitting at lines in to two more components.

chapter-3

RAJIB
che-50

2021, 2018, 2019(H)

(H) Sketch schematic diagram to represent the energy levels of a diatomic molecule regarding as a harmonic & an anharmonic oscillators. show allowed transitions & the expected spectrum?

⇒ schematic diagram for harmonic oscillators:

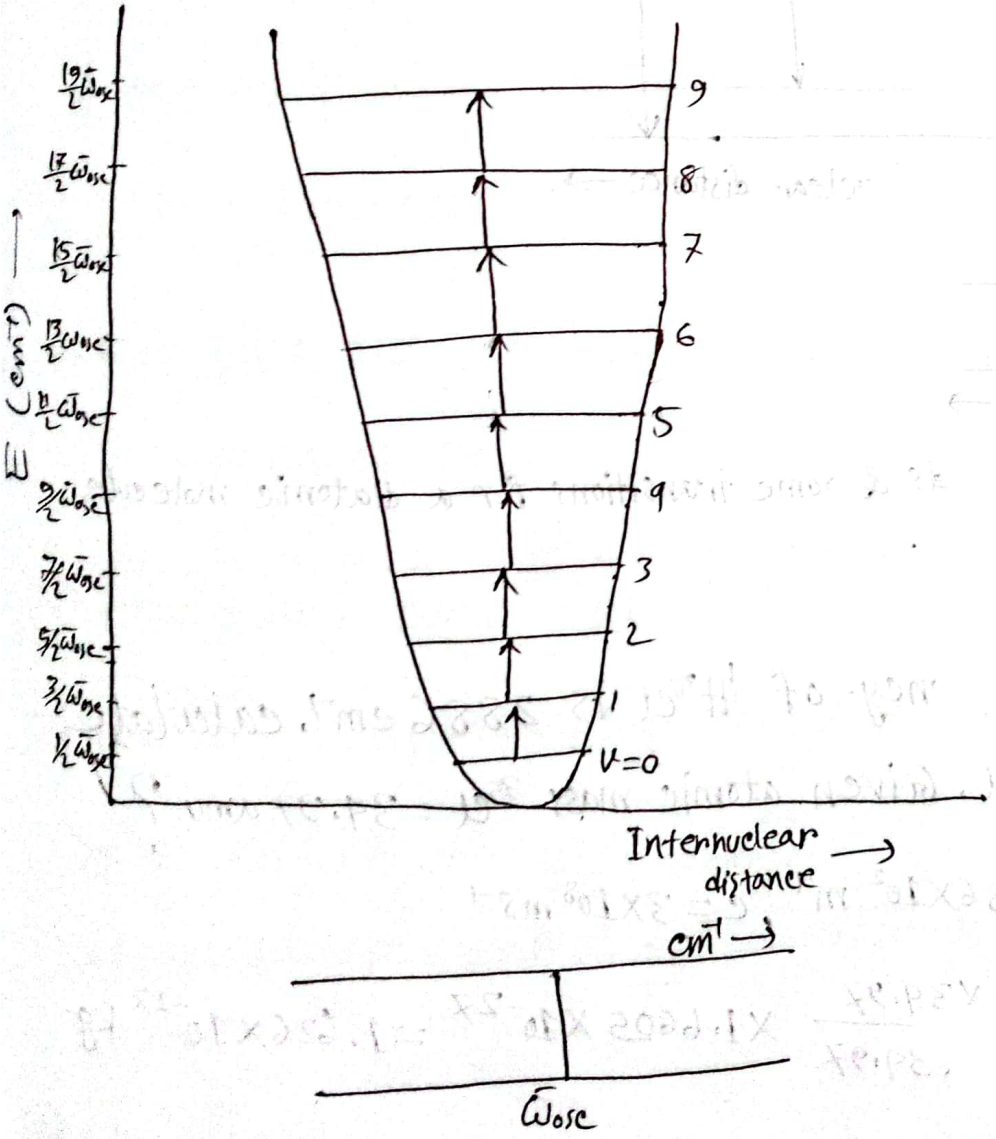


Fig: The vibrational energy levels & allowed transitions btwn them for a diatomic molecule undergoing simple harmonic motion.

CamScanner

Schematic diagram for anharmonic oscillator:

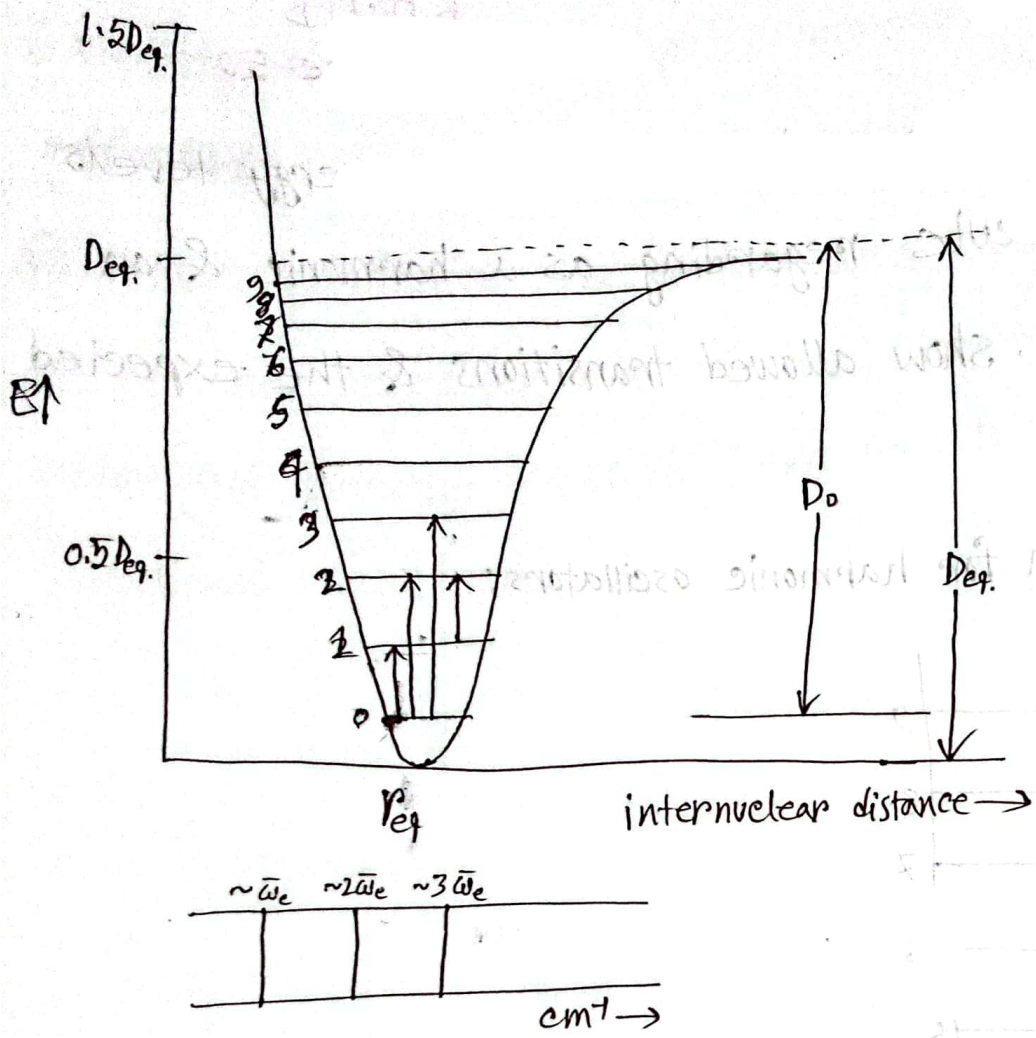


Fig: The vibrational energy levels & some transitions for a diatomic molecule undergoing anharmonic oscillation.

2016, 2018(t)

⊛ The fundamental frequency of $^1H^{35}Cl$ is 2886 cm^{-1}. calculate force constant of $^1H^{35}Cl$. Given atomic mass ^{35}Cl = 34.97 amu ?

$\Rightarrow \bar{\omega}_e = 2886\ cm^{-1} = 2886 \times 10^2\ m^{-1}$, $c = 3 \times 10^8\ ms^{-1}$

$$\mu = \frac{m_1 m_2}{m_1 + m_2} = \frac{1.0078 \times 34.97}{1.0078 * 34.97} \times 1.6605 \times 10^{-27} = 1.626 \times 10^{-27}\ kg$$

$\therefore$ Force constant, $K = 4\pi^2 \bar{\omega}_e^2 c^2 \mu = 481.219\ Nm^{-1}$ Ans.

2017, 2020, 2021(t)

(*) The equilibrium oscillation frequency for NO is 1904 cm^{-1}. calculate force constant of the molecule?

$\Rightarrow \bar{\omega}_e = 1904 \times 10^2 m^{-1}$, $c = 3 \times 10^8 ms^{-1}$,

$$\mu = \left(\frac{14 \times 16}{14+16}\right) \times 1.6605 \times 10^{-27} = 1.24 \times 10^{-26} kg$$

$\therefore K = 4\pi^2 \bar{\omega}_e^2 c^2 \mu = 1597.19 \ Nm^{-1}$ Ans.

2018, 2021, 2019(t)

(*) the fundamental band of CO in inert CCl_4 is 2143 cm^{-1}. calculate K for CO?

$\Rightarrow \bar{\omega}_e = 2143 \times 10^2 m^{-1}$, $\mu = \left(\frac{12 \times 16}{12+16}\right) \times 1.6605 \times 10^{-27} = 1.14 \times 10^{-27} kg$

$\therefore K = 4 \times (3.1416)^2 \times (214300)^2 \times (3 \times 10^8)^2 \times 1.14 \times 10^{-27}$

$= 1860.16 \ Nm^{-1}$

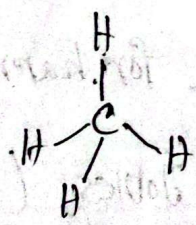
2017, 2020, 2022, 2021(t), 2018(t)

(*) How many normal modes of vibration are possible of : $CH \equiv CH$ CH_4, HBr, O_2, OCS (linear), SO_2 (bent), $CH_2 = CH_2$, NH_3, C_6H_6, H_2O, BCl_3, CH_3I ?

$\Rightarrow$ (i) for $CH \equiv CH$, $\rightarrow$ $H - C \equiv C - H$

It is linear molecule. So normal mode of vibration $= 3N - 5$
$= 3 \times 4 - 5$
$= 7$

(ii) For CH_4,



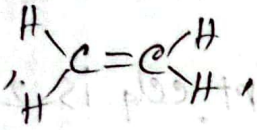
It is non linear. So mode of vibration $= 3N - 6 = 3 \times 5 - 6 = 9$

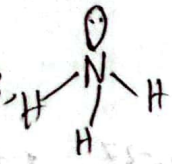
(III) For HBr, H–Br, it is linear. So, mode = 3N−5
= 3×2−5
= 1

(IV) For O_2, O=O, it is linear, so mode = 3N−5 = 3×2−5 = 1.

(V) For OCS, O=C=S, it is linear. So mode = 3×3−5 = 4

(VI) For SO_2, , it is non linear. So mode = 3×3−6 = 3

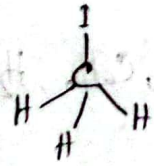
(VII) For $CH_2=CH_2$,  , it is non linear, so mode = 3×6−6
= 12

(VIII) For NH_3,  , it is non linear. So mode = 3×4−6 = 6

(IX) For C_6H_6, , it is non linear. So mode = 3×12−6 = 30

(X) For H_2O, , it is non linear. So mode = 3×3−6 = 3

(XI) For BCl_3, , it is non linear. So mode = 3×4−6 = 6

(XII) For CH_3I,  , it is non linear, so mode = 3×5−6 = 9

(X) 2020, 2017, 2015, 2021(t) write eqn ~~of~~ vibrational energy of a simple harmonic oscillator? From this eqn show diatomic molecule can never have zero vibrational energy? selection rule 2019(t) for harmonic oscillator?

⇒ vibrational energy, $E_v = \left(v + \frac{1}{2}\right) h\omega_{osc}$ Joules $(v = 0, 1, 2, \ldots)$

where, V = vibrational quantum number. converting to the spectroscopic units, cm^{-1}, we have:

$$\varepsilon_v = \frac{E_v}{hc} = \left(V + \frac{1}{2}\right)\frac{\omega_{osc.}}{c} = \left(V + \frac{1}{2}\right)\bar{\omega}_{osc.}\ cm^{-1}$$

⇒ we should notice that the lowest vibrational energy, obtained by putting V = 0,

$$\therefore E_0 = \frac{1}{2} h\omega_{osc.} \text{ Joules } (\omega_{osc.} \text{ in Hz})$$

or, $$\varepsilon_0 = \frac{1}{2}\bar{\omega}_{osc.}\ cm^{-1} \quad (\bar{\omega}_{osc.} \text{ in } cm^{-1})$$

So we can say that, the diatomic molecule can never have zero vibrational energy.

⇒ selection rule for harmonic oscillator, $\Delta V = \pm 1$

2016
~~⊗~~ Sketch the Morse curve for anharmonic oscillator? Give selection rule for anharmonic oscillator? Explain fundamental, hot band & overtone transitions with energy expressions?

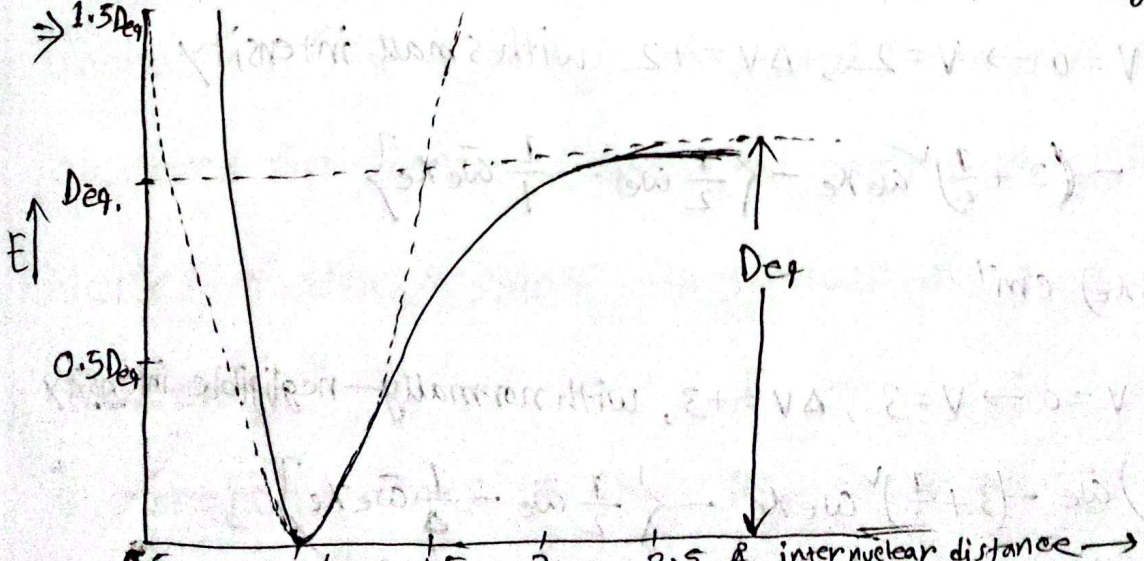


Fig: The Morse curve of diatomic molecule undergoing anharmonic extensions & compressions.

⇒ selection rule for anharmonic oscillator, $\Delta V = \pm 1, \pm 2, \pm 3, \ldots\ldots$

⇒ vibrational energy for anharmonic oscillator,

$$\varepsilon_v = \left(V+\frac{1}{2}\right)\bar{\omega}_e - \left(V+\frac{1}{2}\right)^2 \bar{\omega}_e x_e \ \text{cm}^{-1} \quad (V = 0, 1, 2, \ldots\ldots)$$

where, $\bar{\omega}_e$ = oscillation frequency, x_e = anharmonicity constant

Fundamental absorption: Fundamental vibrational frequencies of a molecule correspond to transition from $V=0$ to $V=1$ with considerable intensity.

$$\Delta\varepsilon = \varepsilon_{V=1} - \varepsilon_{V=0} = \left(1+\frac{1}{2}\right)\bar{\omega}_e - \left(1+\frac{1}{2}\right)^2 \bar{\omega}_e x_e - \left\{\frac{1}{2}\bar{\omega}_e - \frac{1}{4}\bar{\omega}_e x_e\right\}$$

$$= \bar{\omega}_e (1 - 2x_e) \ \text{cm}^{-1}$$

overtone band: An overtone band is the spectral band that occur in vibrational spectrum of a molecule when the molecule makes a transition from the ground state $V=0$ to the second or 3rd excited state ($V=2$ or $V=3$).

For 1st overtone: $V=0 \rightarrow V=2$, $\Delta V = +2$ with small intensity

$$\Delta\varepsilon = \left(2+\frac{1}{2}\right)\bar{\omega}_e - \left(2+\frac{1}{2}\right)^2 \bar{\omega}_e x_e - \left\{\frac{1}{2}\bar{\omega}_e - \frac{1}{4}\bar{\omega}_e x_e\right\}$$

$$= 2\bar{\omega}_e (1 - 3x_e) \ \text{cm}^{-1}$$

For 2nd overtone: $V=0 \rightarrow V=3$, $\Delta V = +3$, with normally negligible intensity

$$\Delta\varepsilon = \left(3+\frac{1}{2}\right)\bar{\omega}_e - \left(3+\frac{1}{2}\right)^2 \bar{\omega}_e x_e - \left\{\frac{1}{2}\bar{\omega}_e - \frac{1}{4}\bar{\omega}_e x_e\right\}$$

$$= 3\bar{\omega}_e (1 - 4x_e) \ \text{cm}^{-1}$$

CamScanner

Hot bands: The weak absorption arise at slightly lower wave number than the fundamental, such weak absorption is known as hot bands. A high temperature is the preliminary conditions for their appearance. $V=1 \longrightarrow V=2$, $\Delta V=+1$, normally very weak

$$\Delta E = \left(2+\frac{1}{2}\right)\bar{\omega}_e - \left(2+\frac{1}{2}\right)\bar{\omega}_e x_e - \left(1+\frac{1}{2}\right)\bar{\omega}_e + \left(1+\frac{1}{2}\right)\bar{\omega}_e x_e$$

$$= \bar{\omega}_e (1-4x_e) \quad cm^{-1}.$$

2018, 2021, 2019(t)

(*) What are P, Q, R branches in the vibration rotation spectra? Explain their origin?

⇒ P branch: Lines to the lower frequency side of band origin ($\bar{\omega}_0$), corresponding to negative m (that is $\Delta J = -1$) are referred to P branch. The P branch lines appear at lower frequencies than the pure vibrational transition. Appear at lower frequencies than pure vibrational transition.

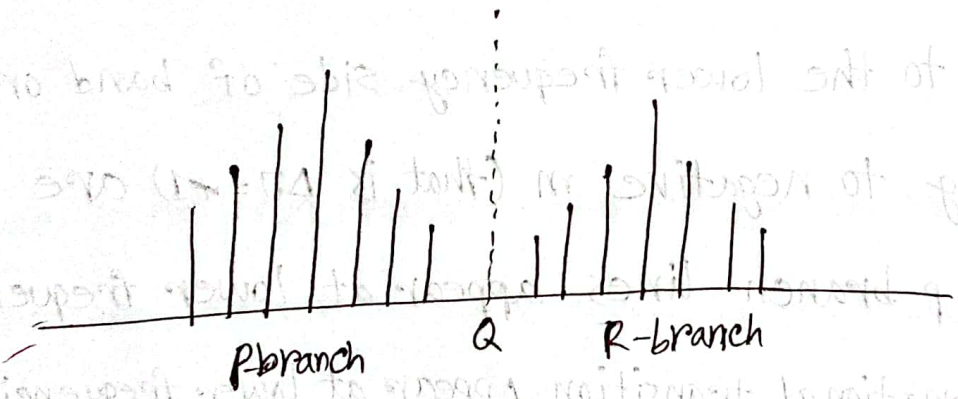
Q branch: In Q branch, rotational quantum number remains unchanged ($\Delta J=0$) during vibrational transition. Q branch is only observed for linear & spherical top molecules. This results in a series of closely spaced lines near the pure vibrational transition frequency.

R branch: Lines to the higher frequency side of band

origin ($\bar{\omega}_0$), corresponding to positive m (that is $\Delta J = +1$) are referred to as R branch. In R branch, J increases by one unit ($\Delta J = +1$) during vibrational transition. this means molecules gains both vibrational & rotational energy. the R branch appear at higher frequencies than the pure Vibrational transition.

Lines arising from $\Delta J = -2, -1, 0, +1, +2$

called O P Q R S branch



2017, 2020

Ⓧ Explain why C=O stretching vibration of Aldehyde gives a strong absorption in IR, although absorption due to C=C vibration in alkene is normally very weak?

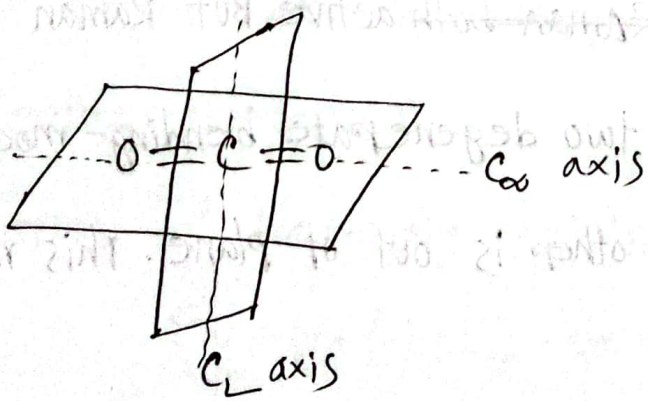
⇒ C=O bond has significant polarity due to electronegativity difference btwn C & O. This results large dipole moment. when the C=O bond stretches, the dipole moment will be changes. The large change in dipole moment during vibration leads to a strong

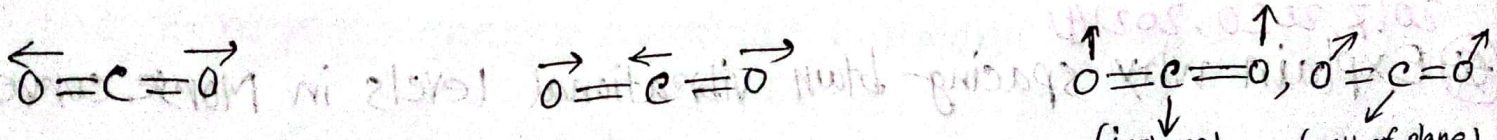
absorption of IR radiation.

the C=C bond in an alkene is not as polar as C=O bond bond since the two C atoms cary almost identical changes; exactly identical if the substituents are same on both C. So ~~when~~ C=C has smaller dipole moment or in some cases almos no dipole moment so when the C=C bond stretches, the changes in dipole moment is very small. This is because the bond length changes but the e^- distribution remains relatively symmetric. Smaller changes in dipole moment during vibration result in a weaker absorption of IR radiation.

~~2017, 2020~~ Sketch symmetry & fundamental vibrations of CO_2 in a figure? explain them clearly?

⇛





(a) symmetric stretch (b) antisymmetric stretch (c) Bending

Fig: The symmetry & fundamental vibrations of CO_2.

total fundamental vibration are = $3N-5$ [as linear]

$= 3\times3-5$

$= 4$

No. of stretching vibration = $N-1 = 3-1 = 2$

No. of Bending mode vibration = $2N-4 = 2$ [as linear]

For CO_2 there are two different sets of symmetry axes. There is an infinite number of two fold axes (C_2) passing through the C atom at right angles to the bond direction, & there is an ∞-fold axis (C_∞) passing through the bond axis itself.

In symmetric stretch Both 'O' atom move toward & away from the C atom simultaneously. This mode is IR inactive but Raman active.

In assymmetric stretch one O moves toward C & other O moves away from C. This mode is IR ~~RAMAN~~ active. But Raman inactive.

Bending mode : There are two degenerate bending modes. one in the plane of the paper & other is out of plane. This mode is both IR & RAman active.

2017, 2020, 2021(T)

(X) Explain why spacing btwn vibrational levels in Morse curve decreases gradually? What are zero point energies in relation to harmonic & anharmonic oscillators?

⇒ The morse curve represents how molecules vibrate. At lower energies, the vibrational levels are evenly apart, as the molecule behaves as a perfect spring. But as the molecule vibrates more, & more, the bonds btwn the atoms stretch & does not behave like a perfect spring anymore. The further it stretches, the less energy it is needed to make it vibrate. Hence the energy difference btwn the vibrational levels get smaller as we go up to higher levels, meaning that the levels are packed more closely together. so the vibrations get stronger, the bond length increases due to stretching & so energy levels get closer. Due to potential energy of a real diatomic molecule is anharmonic. this means that as the vibrational energy increases, the vibrational energy spacing decreases. + Graph.

⇒ energy of harmonic oscillator is -

$$\varepsilon_v = \left(v + \frac{1}{2}\right) \bar{\omega}_{osc} \ cm^{-1} \quad (v = 0, 1, 2, \ldots)$$

So the zero point energy ($v=0$) for harmonic oscillator,

$$\varepsilon_0 = \left(0 + \frac{1}{2}\right) \bar{\omega}_{osc}$$

$$\therefore \varepsilon_0 = \frac{1}{2} \bar{\omega}_{osc}.$$

CamScanner

energy for anharmonic oscillator

$$\varepsilon_v = \left(v+\frac{1}{2}\right)\bar{\omega}_e - \left(v+\frac{1}{2}\right)^2 \bar{\omega}_e x_e \ \text{cm}^{-1} \quad (v=0,1,2,\cdots)$$

∴ zero point energy ($v=0$) for anharmonic oscillator,

$$\varepsilon_0 = \frac{1}{2}\bar{\omega}_e - \frac{1}{4}\bar{\omega}_e x_e$$

⊛ 2022, 2015 Write & compare energy of harmonic & anharmonic oscillator. Using Morse curve, Mention the allowed vibrational energy levels & some transitions in anharmonic. write the name of band observed by this transition?

⇒ energy for harmonic, $\varepsilon_v = \left(v+\frac{1}{2}\right)\bar{\omega}_{osc.}\ \text{cm}^{-1}$

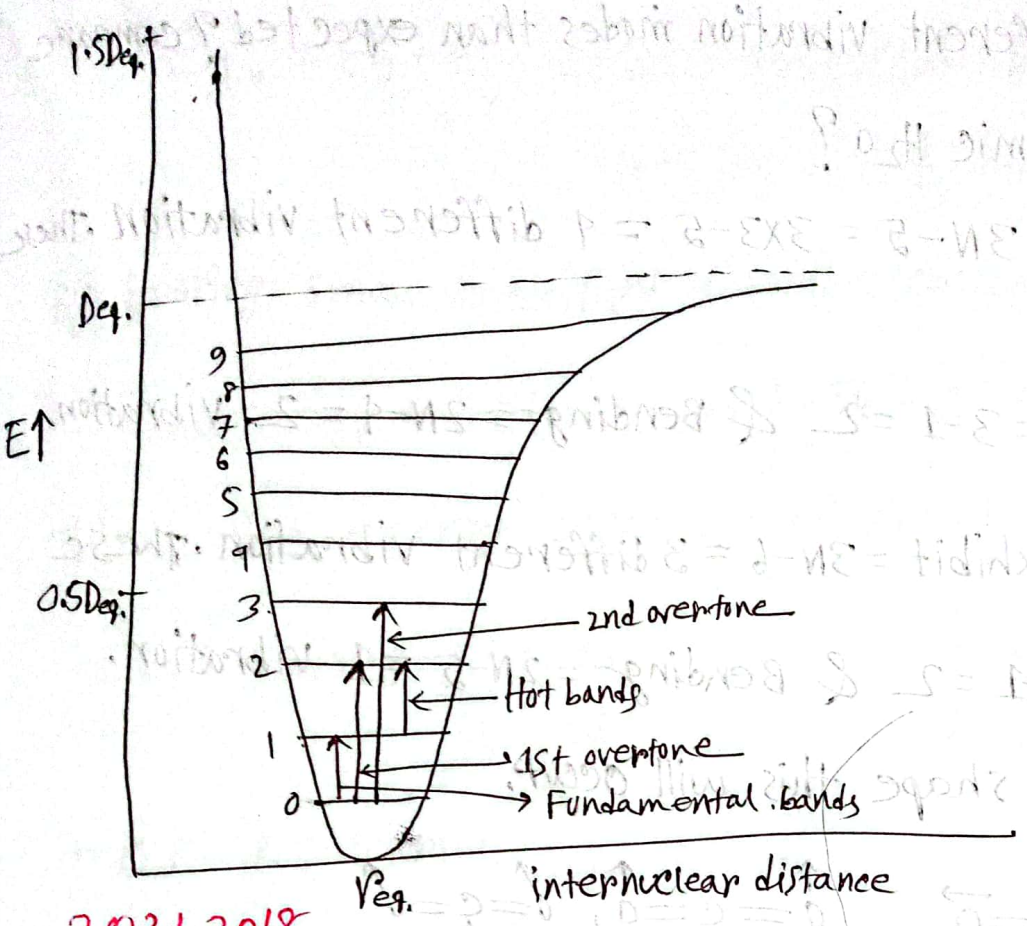
energy for anharmonic, $\varepsilon_v = \left(v+\frac{1}{2}\right)\omega_e - \left(v+\frac{1}{2}\right)^2 \bar{\omega}_e x_e \ \text{cm}^{-1}$

compare the energy of harmonic & anharmonic:

$$\bar{\omega}_{osc.} = \bar{\omega}_e\left\{1 - x_e\left(v+\tfrac{1}{2}\right)\right\}$$

Thus anharmonic oscillator behaves like the harmonic oscillator but with an oscillation frequency which decreases steadily with increasing v. Harmonic oscillator have equally spaced energy levels, while anharmonic oscillators have unequally spaced energy levels.

CamScanner



2021, 2018

(X) How can inter & intramolecular H-bonding indentified by IR?

⇒ Intermolecular H-bonding produces broad bands, while intramolecular H-bonding produces sharp bonding.

Inter & intra-molecular H-bonding can also be identified by dilution. Intra-molecular H-bonding remains unaffected by dilution & so the absorption band also remains unaffected. whereas ~~##~~ intermolecular H-bonds are broken on dilution & so there is a decrease in the bonded O-H absorption.

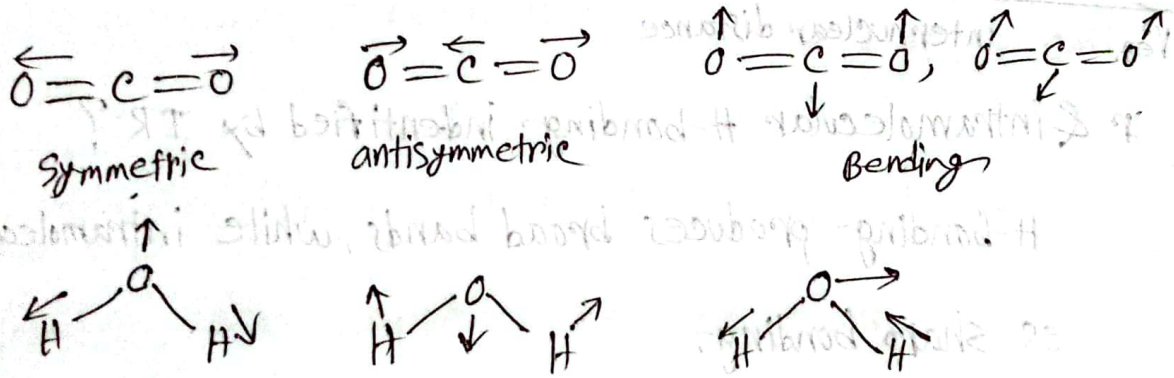
2022
Why CO_2 exhibit different vibration modes than expected? compare with the similar triatomic H_2O?

⇒ Linear CO_2 exhibit $= 3N-5 = 3\times3-5 = 4$ different vibration. These

Stretching $= N-1 = 3-1 = 2$ & Bending $= 2N-4 = 2$ vibration.

And Non linear H_2O exhibit $= 3N-6 = 3$ different vibration. These are, Stretching $= N-1 = 2$ & Bending $= 2N-5 = 1$ vibration.

Due to difference in shape this will occur.



2022
write the energy of a diatomic vibrating rotator? How do you observe the O,P,Q,R,S bands for this molecule?

⇒ A diatomic molecule can execute rotations & vibrations independently. The combined vibrational-rotational energy in the sum of the separate energies, which is:

$$E_{total} = E_{vib.} + E_{rot.} \quad J$$

$$\varepsilon_{total} = \varepsilon_{vib.} + \varepsilon_{rot.} \quad cm^{-1}$$

Taking the separate expressions for $\varepsilon_{rot.}$ & $\varepsilon_{vib.}$, we get,

CamScanner

$$\varepsilon_{J,v} = \varepsilon_J + \varepsilon_v = BJ(J+1) - DJ^2(J+1)^2 + HJ^3(J+1)^3 + \ldots\ldots$$
$$+\left(v+\tfrac{1}{2}\right)\bar{\omega}_e - \left(v+\tfrac{1}{2}\right)^2 \bar{\omega}_e x_e \quad cm^{-1}$$

By ignoring small centrifugal distortion constant D, H, etc. Hence we write,

$$\varepsilon_{total} = \varepsilon_{J,v} = BJ(J+1) + \left(v+\tfrac{1}{2}\right)\bar{\omega}_e - \left(v+\tfrac{1}{2}\right)^2 \bar{\omega}_e x_e$$

this is the energy expression of diatomic vibrating rotator.

Lines arising from $\Delta J = -2, -1, 0, +1, +2$
called : O P Q R S branch

+ O, P, Q, R, S describe.

2016, 2019(t), 2018(t)
⊛ State the basic requirement for substance to be IR active with example?

⇒ For IR active, it must ~~have~~ exhibit a change in the dipole moment during molecular vibrations. Vibration that do not result in a dipole moment change are IR active. Ex: HCl is polar with permanent dipole moment. During vibration bond length changes, which alters the dipole moment. This makes HCl IR-active. CO_2's asymmetric stretch resuts change in dipole moment making this mode is IR-active. But Symmetric stretch of CO_2 does not change dipole moment so this mode of vibration is IR inactive.

CamScanner

Extra

⊗ The fundamental mode of HCl occurs at 2886 cm^{-1}. Predict the frequency of the corresponding mode of DCl?

⇒ Let, K is equal for H-Cl & D-Cl

$$\frac{\nu_{H-Cl}}{\nu_{D-Cl}} = \frac{\sqrt{\frac{K_1}{\mu_1}}}{\sqrt{\frac{K_2}{\mu_2}}} = \sqrt{\frac{\mu_2}{\mu_1}} \quad \text{—— (1)}$$

$$\mu_1 = \frac{1\times35}{1+35} = \frac{35}{36}, \quad \mu_2 = \frac{2\times35}{2+35} = \frac{70}{37}$$

$$\therefore \frac{\nu_{H-Cl}}{\nu_{D-Cl}} = \sqrt{\frac{\frac{70}{37}}{\frac{35}{36}}} = \sqrt{\frac{70\times36}{37\times35}} = 1.39$$

$$\Rightarrow \nu_{D-Cl} = \frac{\nu_{H-Cl}}{1.39} = \frac{2886}{1.39} = 2068\,cm^{-1} \text{ Ans.}$$

2016

⊗ Explain why all transitions in vibrational spectroscopy are restricted to originate from ground state (V=0)?

⇒ because at room temperature, only the lowest energy vibrational stat is populated. Due to population distribution & selection rules. The combination of Boltzmann distribution & selection rule results in the dominance of transitions from ground state (V=0) to the 1st excited state. Higher transitions such as V=0 to V=2 or V=1 to V=2 are less probable due to lower population of higher vibrational levels. while hot bands can be observed at higher temperature. they are significantly weaker base: The intensity of these transitions decreases accordingly. It occur due to the low population of higher vibrational states at standard temperatures & large gap btwn vibrational levels.

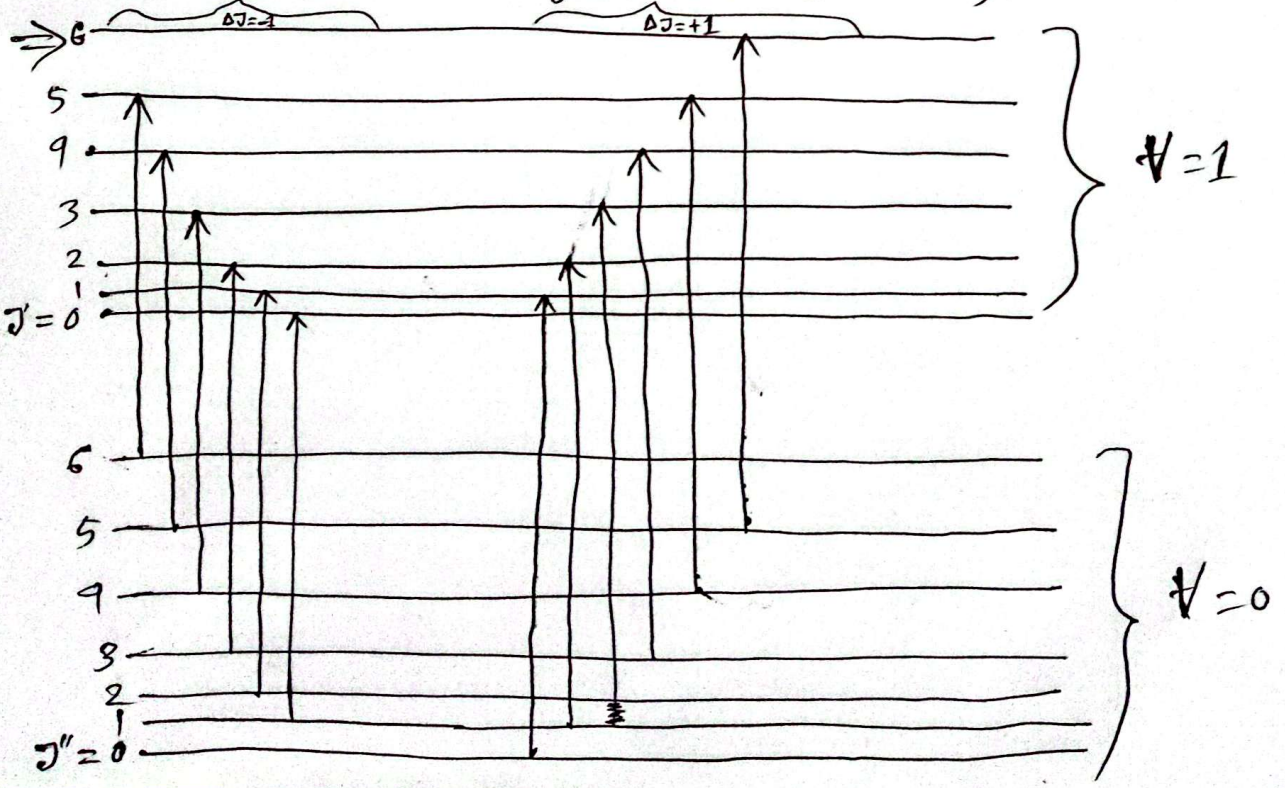
2016

(8) what are P & R branch lines? Sketch vibrational-rotational energy levels & show transitions btwn them to appear P & R branch lines in vibration-rotation spectrum?

⇒ P & R branch lines refer to specific types of spectral lines observed in molecular spectroscopy, particularly in rotational-vibrational spectroscopy of diatomic molecules.

P branch corresponds to transitions where the rotational quantum number decreases by 1 ($\Delta J = -1$). Transition occur from higher rotational state to lower state ($J \rightarrow J-1$).

R branch corresponds to transitions where the rotational quantum number increases by 1 ($\Delta J = +1$). Transition occur from a lower rotational state to a higher one ($J \rightarrow J+1$).



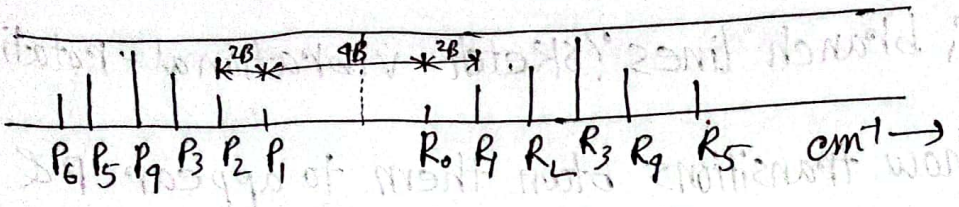


Fig: some transitions btwn the rotational vibrational energy levels of a diatomic molecule, together with the spectrum arising from them.

2018(t)

A soln gives absorbance 0.459 at 359 nm in a 1 cm cell. the molar absorptivity is 1.25×10^4 cm^{-1} $mol^{-1}L$. what is concentration?

$$\Rightarrow A = \varepsilon c l \Rightarrow c = \frac{A}{\varepsilon l} = \frac{0.459}{1.25 \times 10^4 \times 1} = 3.632 \times 10^{-5} M$$

Ans.

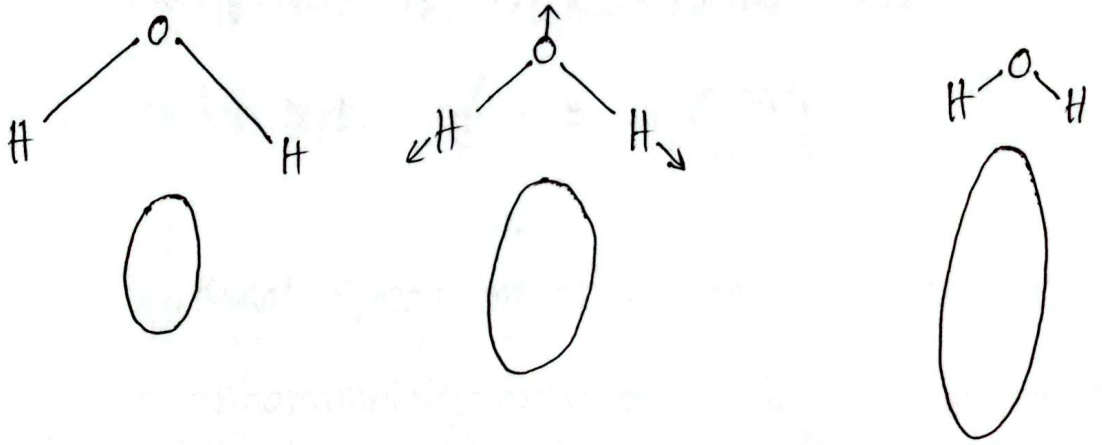
Chapter-4

Rajib
Che-50

(*) 2022, 2016
Explain the Raman activity of the fundamental mode of H_2O?

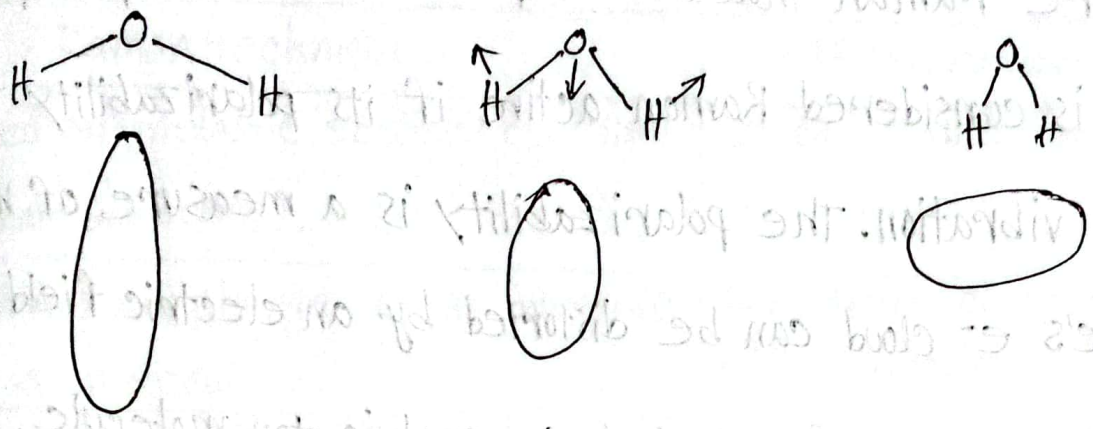
⇒ Fundamental mode of H_2O are $= 3N-6$
$= 3\times3-6 = 3$

3 fundamental modes of H_2O are symmetric, asymmetric stretching & Bending.

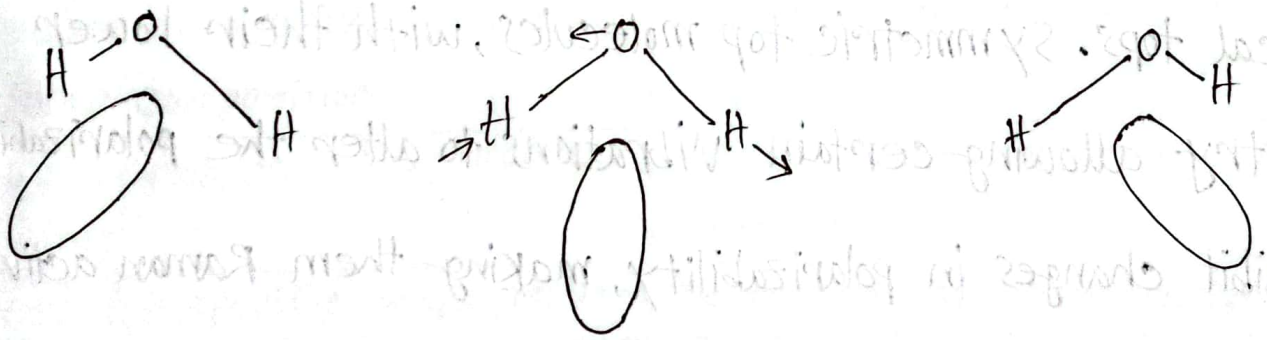


this is symmetric stretching mode of H_2O. During ~~stretching~~ symmetric stretching the molecule increases & decreases in size. when a bond is stretched, the e^- forming it are less firmly held by the nuclei & so the bond becomes more polarizable. Thus the polarizability ellipsoid of H_2O may be expected to decrease in size while the bond stretch, & to increase while they compress but to maintain an approximately constant shape. As there is change in ~~magnitude~~ size of polarizability ellipsoid, so symmetric

stretching mode of H_2O is Raman active



This is bending mode of H_2O. In bending motion the shape of the ellipsoid changes most. For change in shape of polarizability ellipsoid bending mode of H_2O is Raman active.



This is asymmetric mode of H_2O. In this mode both the size & shape remains approximately constant but the direction of the major axis changes markedly. For change in direction this mode is Raman active.

[Change in size/shape/direction = Raman active]

N.B.

(2022) Symmetric top molecules are Raman active but spherical top molecules are raman inactive - explain with example?

⇒ A molecule is considered Raman active if its polarizability changes during vibration. The polarizability is a measure of how easily a molecule's e⁻ cloud can be distorted by an electric field.

The polarizability ellipsoid for typical symmetric top molecule, plainly rotation about the top axis produces no change in the polarizability but end-over-end rotation will produce a change. Symmetric top molecules exhibit lower symmetry than spherical tops. Symmetric top molecules, with their lower symmetry allowing certain vibrations to alter the polarizability & exhibit changes in polarizability, making them Raman active. such as $CHCl_3$.

The polarizability ellipsoid for spherical top molecule is a spherical surface, & it is evident that rotation of this ellipsoid will produce no change in polarizability. So the pure rotations of spherical top molecules are completely inactive in the Raman. Ex. CH_4, SiH_4.

2021
⊛ State difference btwn Raman technique & IR vibrational measurment

⇒

Raman technique	IR vibrational measurement
Based on inelastic scattering of light (Raman scattering).	Based on absorption of IR radiation.
Depends on changes in the polarizability of a molecule	Depends on changes in the dipole moment of a molecule.
Detects symmetric vibrations effectively	Detects asymmetric vibrations effictively
Requires a monochromatic light source, such as a laser	Uses a broadband IR radiation source
Vibrations are Raman-active if they cause a change in polarizability.	Vibrations are IR-active If they cause a change in dipole moment.
less sensitive to water interference	Higly sensitive to water interference.
Used in structural analysis, identifying molecular compositions.	Used in functional group analysis, identifying impurities.
Provides informations on symmetric vibrations	Provides information on asymmetric vibrations.
No need for specific sample preparation	Requires specific sample preparation.
can use water as a solvent	cannot use water as a solvent
Higly expensive method	comperatively inexpensive
Raman intensity depend on polarizability of the sample.	Doesnot depend on the sample's polarizability

2021
⊛ Define molecular polarizability?

⇒ M. polarizability is a measure of how easily the e^- cloud of an atom or molecule can be distorted by an external electric field. when a molecule is put into a static electric field it

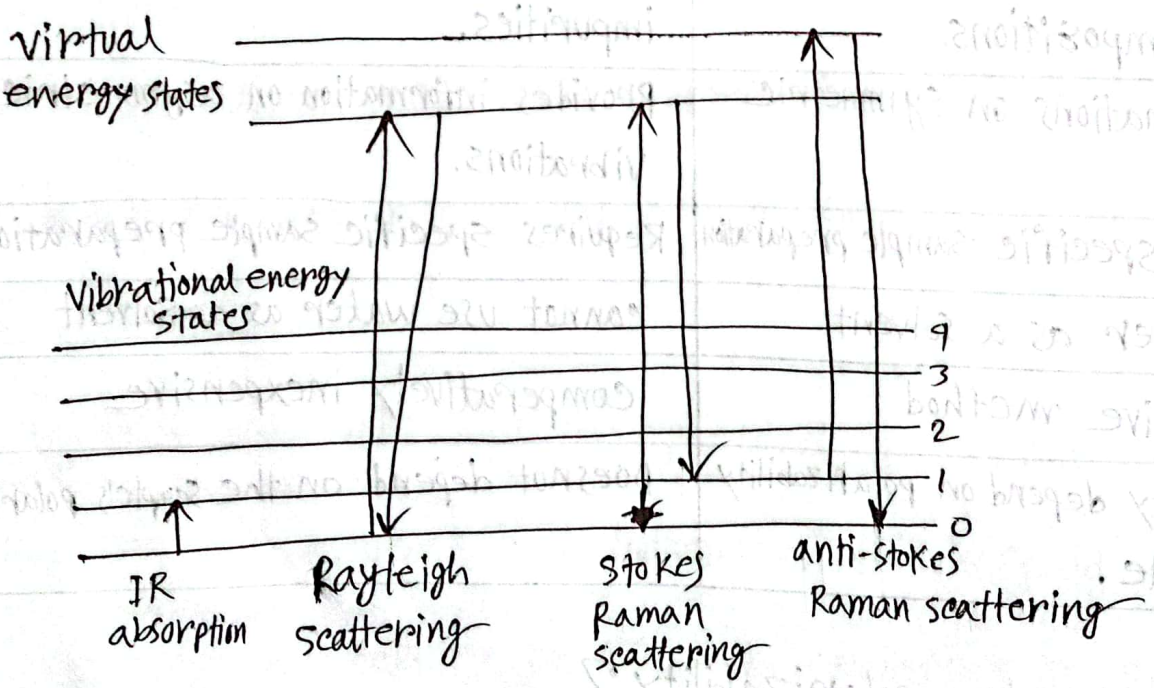
Suffers some distortion, the positively charged nuclei being attracted towards the negative pole of the field, the e^- to the positive pole. This seperation of charge centres causes an induced electric dipole moment to be set up in the molecule & the molecule is said to be polarized. The size of the induced dipole μ, depends both on the magnitude of the applied field, E and on the ease with which the molecule can be distorted. We can write

$$\mu = \alpha E$$

where, α is the polarizability of the molecule.

⊛ 2020 Illustrate quantum theory of Raman spectra?

⇒



⊛ 2020 Write energy expression & selection rule of Raman spectra in case of symmetric top molecules?

energy expression,

$$\varepsilon_{J,K} = BJ(J+1) + (A-B)K^2 \ cm^{-1}, \quad (J=0,1,2,\ldots;\ K=\pm J, \pm(J-1), \ldots)$$

Selection rule for Raman spectra are

$$\Delta K = 0$$

$$\Delta J = 0, \pm 1, \pm 2 \quad (\text{except for } K=0 \text{ states when } \Delta J = \pm 2 \text{ only})$$

where, K = rotational quantum number for axial rotation.

2020

Show that R branch lines appear at $4B, 6B, 8B, 10B \ldots cm^{-1}$ from the exciting line?

⇒ A symmetric top molecules has anisotropic polarizability. This selection rule hold for any K. There are two cases:

① $\Delta J = +1$ (R branch)

$$\Delta E_R = \varepsilon_{J'=J+1} - \varepsilon_{J''=J}$$

$$= 2B(J+1) \ cm^{-1} \quad (J=1,2,3,\ldots \text{ (but } J \neq 0))$$

② $\Delta J = +2$ (S branch)

$$\Delta E_S = \varepsilon_{J'=J+2} - \varepsilon_{J''=J}$$

$$= B(4J+6) \ cm^{-1} \quad (J=0,1,2,\ldots)$$

∴ Allowed R branch lines,

when, $J=1$, $\Delta E_R = 2B(1+1) = 4B \ cm^{-1}$.

When, $\bar{J}=2$, $\Delta\bar{\varepsilon}_R = 2B(2+1) = 6B \ cm^{-1}$

when, $\bar{J}=3$, $\Delta\bar{\varepsilon}_R = 2B(3+1) = 8B \ cm^{-1}$

when, $J=4$, $\Delta\bar{\varepsilon}_R = 2B(4+1) = 10B \ cm^{-1}$

So, R branch lines appear at $4B, 6B, 8B, 10B \ldots\ldots cm^{-1}$ from the exciting line.

~~2016~~ The wavenumber of the incident Radiation in a Raman Spectrometer is 20487 cm^{-1}. What is the wavenumber of the scattered stokes radiation for the $\bar{J}=2 \leftarrow 0$ transition of $^{14}N_2$? [Given $B = 1.9987 \ cm^{-1}$] ?

$\Rightarrow$ transition $\bar{J}=0 \leftarrow 0$ so $\Delta\bar{J} = +2$ that mean it is S branch.

Given, $B = 1.9987 cm^{-1}$, $\bar{\nu}_{ex.} = 20487 \ cm^{-1}$, $\bar{J}=0$

For S branch, $\bar{\nu}_S = \bar{\nu}_{ex.} - \Delta\bar{\varepsilon}_S = \bar{\nu}_{ex.} - B(4\bar{J}+6) \ cm^{-1}$

$= 20487 - (1.9987 \times 6) = 20475 \ cm^{-1}$ Ans.

~~2015, 2017~~ If the bond length of H_2 is 0.07417 nm. What would be the spacing of the lines in the spectrum?

$\Rightarrow$ $\mu = \dfrac{m_1 m_2}{m_1 + m_2} = \dfrac{1.0078 \times 1.0078}{1.0078 + 1.0078} \times 1.6605 \times 10^{-27} = 8.42 \times 10^{-28} \ kg$

$r = 0.07417 \times 10^{-9} m$, $c = 3 \times 10^8 \ ms^{-1}$

$\therefore B = \frac{h}{8\pi^2 I c} = \frac{h}{8\pi^2 \mu r^2 c} = 60.5\ cm^{-1}$

$\therefore$ Line spacing $= 4B = 243.16\ cm^{-1}$ Ans.

2016

(A) Consider the symmetric top molecule $CHCl_3$:

(I) write energy expression?

(II) write selection rule for raman spectra in this case?

(III) write eqn of $\bar{\nu}_R$ & $\bar{\nu}_S$ mentioning their J & ΔJ?

(IV) sketch S branch series only in a figure & show S branch occur at 6B, 10B, 14B, ... cm^{-1} from the exciting line?

$\Rightarrow$ (I) energy, $\varepsilon_{J,K} = BJ(J+1) + (A-B)K^2\ cm^{-1}$ $(J = 0, 1, 2, \ldots;\ K = \pm J, \pm(J-1), \ldots)$

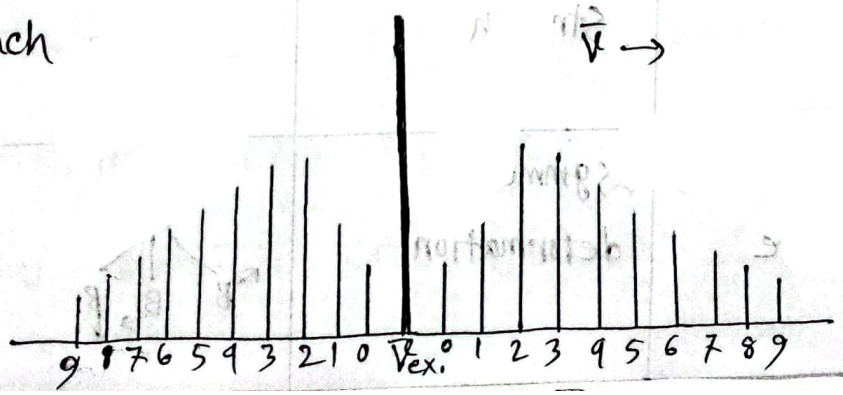
(II) selection rule, $\Delta K = 0$

$\Delta J = 0, \pm 1, \pm 2$ (except for K=0 states when $\Delta J = \pm 2$ only)

(III) $\bar{\nu}_R = \bar{\nu}_{ex.} \pm \Delta\varepsilon_R = \bar{\nu}_{ex.} \pm 2B(J+1)\ cm^{-1}$ $(J = 1, 2, \ldots)$

$\bar{\nu}_S = \bar{\nu}_{ex.} \pm \Delta\varepsilon_S = \bar{\nu}_{ex.} \pm B(4J+6)\ cm^{-1}$ $(J = 0, 1, 2, \ldots)$

(IV) Fig. of S branch



If we consider $\bar{\nu}_S = \bar{\nu}_{ex.} \pm B(4J+6)$ cm^{-1} eqn for S branch then (excepting $\bar{\nu}_{ex.}$), taking $J = 0, 1, 2$ gradually, $\Delta\bar{\nu}_S = B(4B+6)$

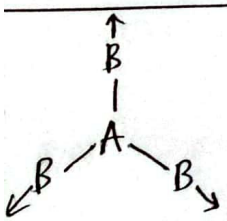
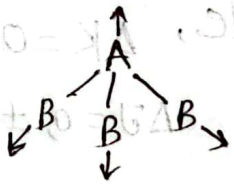
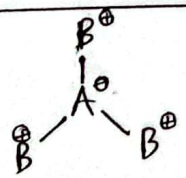
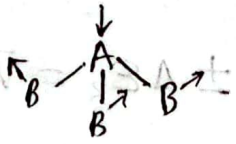
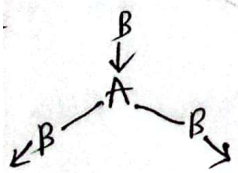
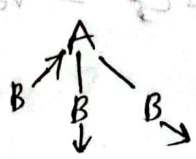
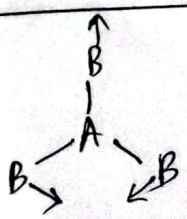
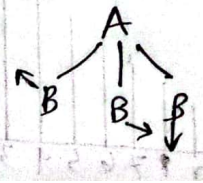
when, $J=0$, $\Delta\bar{\nu}_S = B(4\times0+6) = 6B$ cm^{-1}

when, $J=1$, $\Delta\bar{\nu}_S = B(4\times1+6) = 10B$ cm^{-1}

when, $J=2$, $\Delta\bar{\nu}_S = B(4\times2+6) = 14B$ cm^{-1}

So S branch occur at $6B, 10B, 14B, \ldots$ cm^{-1} from existing line.

~~(X)~~ 2016 which of the ~~following~~ vibrations of AB_3 are IR or Raman active when it is (i) trigonal planar (symmetric planar) (ii) trigonal pyramidal (symmetric pyramidal)?

symmetric planar	activity (R = Raman, I = Infrared)	vibration	symmetric pyramidal	activity (Raman = R, Infrared = I)
	R = active (Pol.) strong I = inactive	Symmetric stretch		R = active (Pol.) strong I = active ∥
	R = inactive I = active ∥	out-of-plane symmetric deformation		R = active (Pol.) medium I = active ∥
	R = active (depol.) weak I = active ⊥	asymmetric stretch		R = active (depol.) weak I = active ⊥
	R = active (depol.) weak I = active	asymmetric deformation		R = active (depol.) weak I = active ⊥

RAJIB
che-50

(*) 2018, 2020
Explain the quantum theory of Raman effect?

⇒ The occurrence of Raman scattering easily understood by quantum theory of radiation. The radiation of frequency ν of photon having energy $h\nu$, where k is plank's constant. photons undergo collisions with molecules & if the collision is perfectly elastic, they will be deflected unchanged. Elastic scattering is like a ball bearing bounches on the table without any loss of energy. since the energy is exchanged btwn photon & molecules during collision, then this collisions are called in-elastic. the molecule can gain or loss amounts of energy ΔE Joules. ΔE represent the change in the vibrational & rotational energy of the molecule. If the molecule gains energy ΔE, the photon will be scattered with energy $h\nu - \Delta E$ & the equivalent radiation will have a frequency $\nu - \frac{\Delta E}{h}$. If molecule loses energy ΔE then the energy of the scattered radiation $h\nu + \Delta E$ & the

frequency $\nu + \frac{\Delta E}{h}$.

2018, 2021, 2020, 2022

⊛ Difference btwn stokes & anti-stokes radiation?

⇒ If the radiation scattered with a lower frequency than that of the incident beam is called stokes radiation. It is accompanied by an increase in molecule energy. But If the radiation scattered with a higher frequency than that of the incident beam is called anti-stokes radiation. It is accompanied by a decrease in molecules energy. stokes radiation Generally more intense than anti-stokes. the intensity of stokes radiation is usually stronger than antistokes radiation because most molecules start from the ground state at typical temperature. stokes can be started from ground state but anti-stokes cannot started from ground state

2018, 2021, 2020, 2017, 2015

⊛ State the criteria for a molecule to be Raman active? Give 3 example which give Raman spectra?

⇒ A molecule is Raman active if it undergoes a change in its polarizability during vibration. Change in polarizability means the e⁻ cloud distribution needs to shift significantly

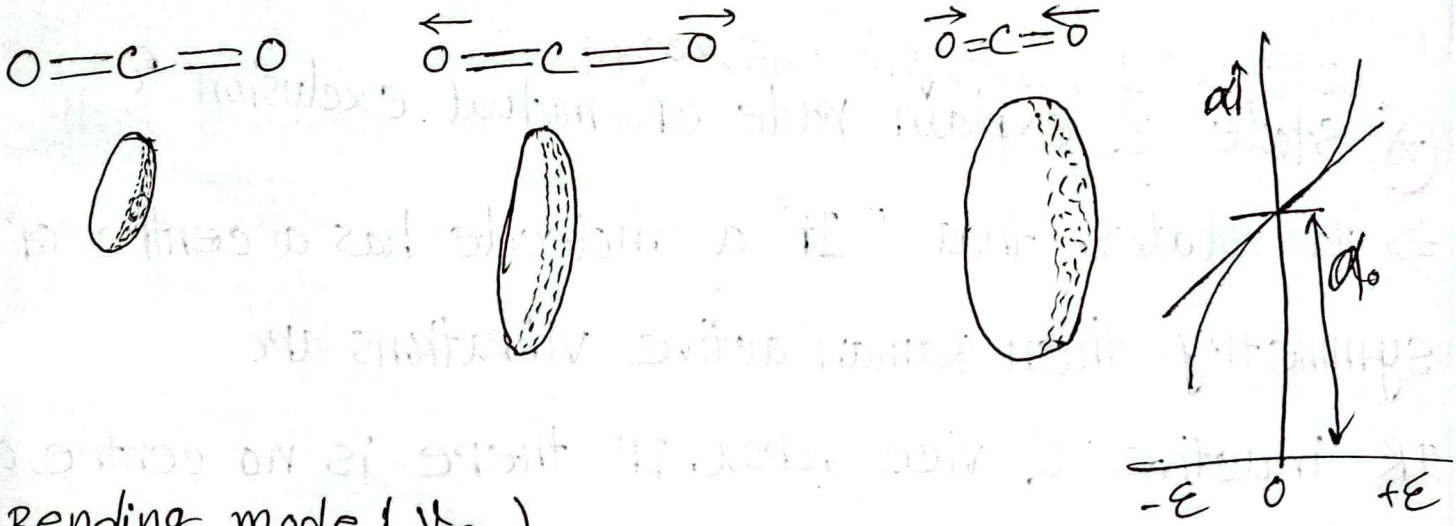
during the vibration. For a molecule to be Raman active it must have a vibrational mode that causes a change in its polarizability.

example : H_2O, C_6H_6, CO_2, N_2, Cl_2, CH_4, CCl_4, $CHCl_3$ etc.

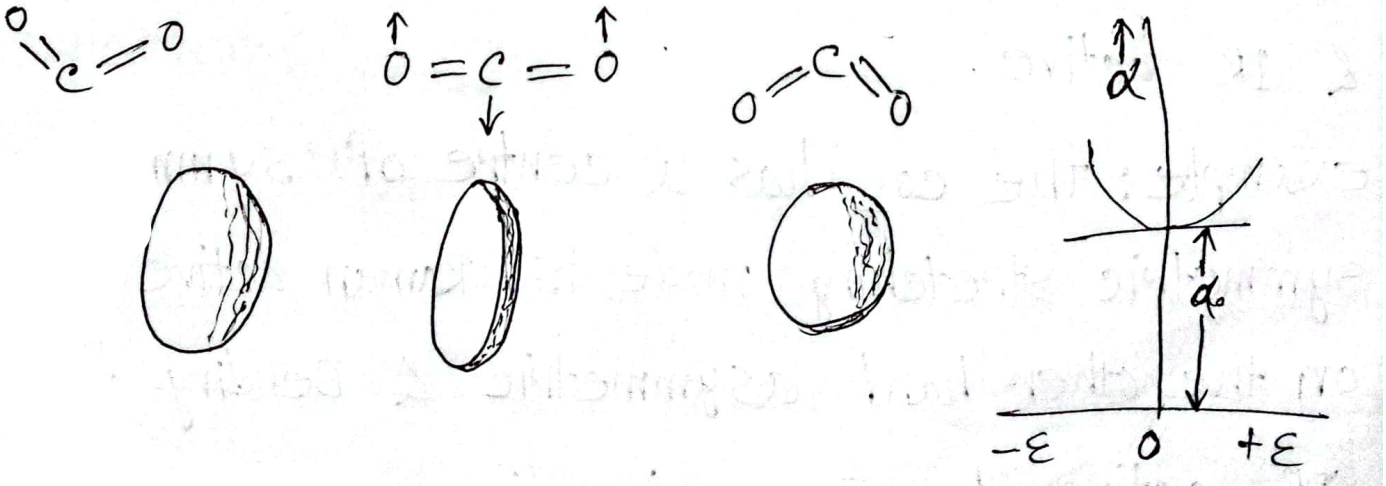
2018, 2015

Ⓐ Symmetric stretching mode of CO_2 is Raman active, but bending & asymmetric modes is inactive. why?

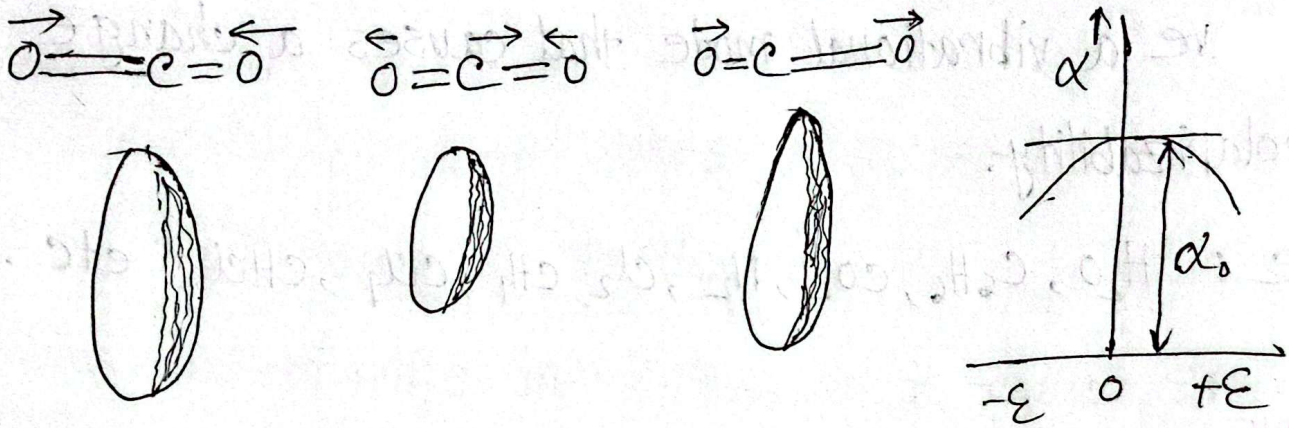
⇒ ⓘ Symmetric stretching mode (ν_1)



ⓘⓘ Bending mode (ν_2)



(iii) asymmetric stretching mode:



But only in symmetric stretching mode change in polarizability so it is Raman active but other are not Raman active.

2018, 2021, 2020, 2017, 2015, 2024(H)

(X) State & explain rule of mutual exclusion?

⇒ It states that "If a molecule has a centre of symmetry then Raman active vibrations are ~~IR inactive~~ IR inactive & vice versa. If there is no centre of symmetry then some vibrations may be both Raman & IR active.

example: The CO_2 has a centre of symmetry. Its symmetric stretching mode is Raman active & IR inactive on the other hand asymmetric & Bending modes are IR active but Raman inactive. N_2O has no centre

of symmetry for this reason N_2O has common bands in IR & Raman spectra.

2018
Explain why the pure rotations of spherical top molecules are completely inactive in Raman spectroscopy?

⇒ Examples of spherical top molecules are those with tetrahedral symmetry such as CH_4 or silane, SiH_4. The polarizability ellipsoid for such molecules is a spherical surface & it is evident that rotation of this ellipsoid will produce no change in polarizability. Therefore the pure rotations of spherical to molecules are completely inactive in the Raman.

2018, 2020
State advantages of Raman over IR in vibrational measurements?

⇒ ① Firstly, because both the incident & scattered radiation are at UV or visible frequencies, conventional optics & sample cells (glass, quartz etc) can be used, so avoiding the problems inherent in NaCl windows,

CamScanner

atmospheric absorption etc.

(ii) secondly, because the beam can be focused extremely finely, very small samples can be studied. This, combined with pulsed techniques which can give very short time resolutions, enables very small quantities of transient species to be studied.

(iii) thirdly, water, which has strong IR absorptions, is rather weak Raman ~~scha~~ scatterer and so aqueous sln can be studied using Raman, because the sample signal is not swamped by that of the solvent.

⊗ 2017, 2022 what is meant by Raman scattering?

⇒ If monochromatic radiation or radiation of a very narrow frequency band is used, the scattered energy will consist almost entirely of radiation of the incident frequency but in addition certain discrete frequencies

above & below that of the incident beam will be scattered; it is this which is Raman scattering.

⊛ 2017 write energy eqn for pure rotational Raman spectra for a linear molecule?

⇒ The rotational energy level for linear molecules,

$$\varepsilon_J = BJ(J+1) - DJ^2(J+1)^2 \ cm^{-1} \quad (J = 0, 1, 2, \cdots)$$

where, B = Rotational constant

D = centrifugal distortion constant.

But, in Raman spectroscopy, ignoring D. then the simple expression for energy level,

$$\varepsilon_J = BJ(J+1) \ cm^{-1} \quad (J = 0, 1, 2, \ldots\ldots).$$

⊛ 2017 Show the rotational energy levels of a diatomic molecule & the rotational Raman spectrum arising from them with showing the relative intensities of lines in the spectrum?

⇒ The rotational energy levels of a linear diatomic

molecule:

$$\mathcal{E}_J = BJ(J+1) \ cm^{-1}$$

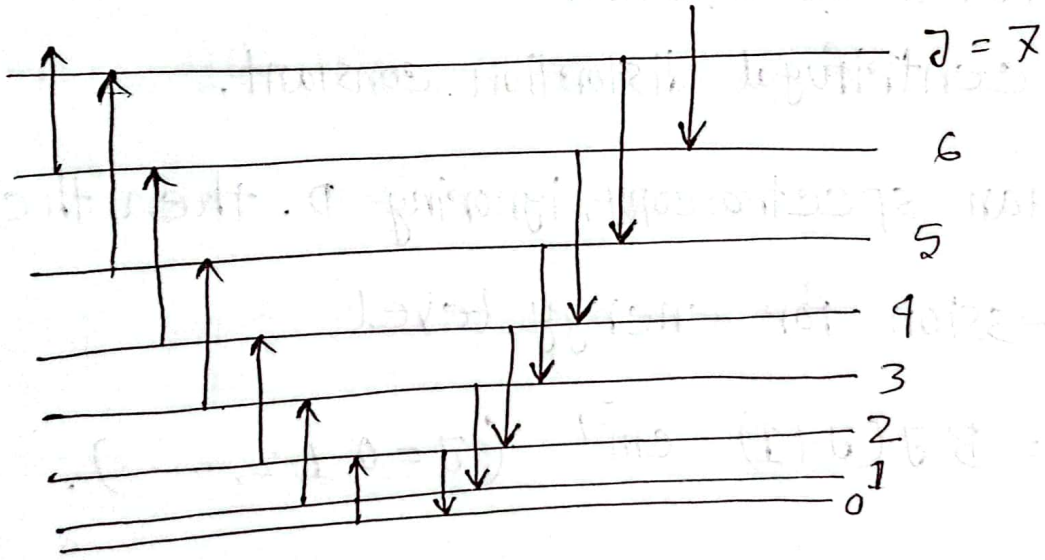
The selection rule $\Delta J = 0$ or ± 2 only.

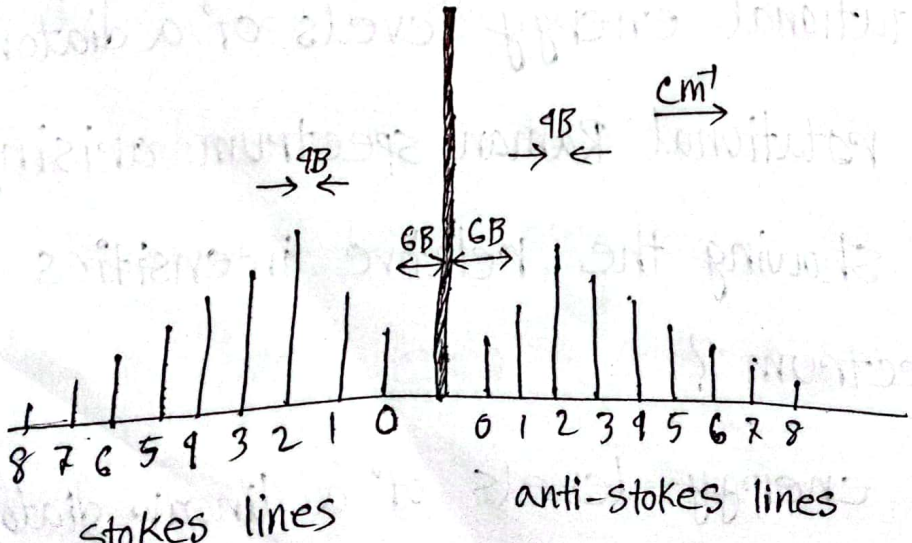
$\Delta J = 0$, corresponds to Rayleigh scattering, $\Delta J = \pm 2$ give

Rise to Raman line. when $\Delta J = \pm 2$

$$\Delta \mathcal{E} = \mathcal{E}_{J'=J+2} - \mathcal{E}_{J''=J}$$

$$= B(4J+6) \ cm^{-1}$$





2017

Explain the stokes & anti-stokes lines in the Rotational Raman spectrum of the above fig-II ?

⇒ If the molecule gain rotational energy from photon during collision, there occurs a series of S-branch lines to the two wave number side then it is stokes line and if the molecule loses rotational energy to the photon during collision, then S-branch lines appear on the high wave number side (anti stokes line).

$$\bar{\nu}_s = \bar{\nu}_{ex.} \pm \Delta E_s$$

$$= \bar{\nu}_{ex.} \pm B(4J+6) \text{ cm}^{-1}$$

where plus sign refers to anti-stokes lines, the minus to stokes lines and $\bar{\nu}_{ex.}$ is the wavenumber of the exciting radiation.

CamScanner

(★) 2017, 2015 what structural information can you get from Raman spectral analysis?

⇒ (i) Raman spectra can identify the presence of specific functional groups within a molecule, such as C=C, C≡C, C-H, C-O, & others. (ii) It provide information about the symmetry of a molecule that can helpful for distinguishing btwn different isomers. (iii) It can provide information about the conformation of polymer chains, such as whether they are crystalline or amorphous. (iv) It is used for measure the strain & stress within a material. (v) Raman can provide insights into molecular orientation in anisotropic material.

(★) 2015 write down the expression for pure raman rotational energy transition for a linear molecule?

⇒ The rotational energy levels of linear molecules,

$$\varepsilon_J = BJ(J+1) - DJ^2(J+1)^2 \ cm^{-1} \quad (J = 0, 1, 2, \ldots)$$

In raman spectroscopy, the precision of the measurements does not normally warrant the retension of the term involving D, the centrifugal distortion constant. Thus we take simpler expression,

$$\varepsilon_J = BJ(J+1) \ cm^{-1} \quad (J = 0, 1, 2, \ldots)$$

CamScanner

Selection rule : $\Delta J = 0$ or ± 2 only

for $\Delta J = +2$ the change in energy levels,

$$\Delta E = E_{J'=J+2} - E_{J''=J} = B(4J+6) \text{ cm}^{-1}$$

As $\Delta J = +2$ so we may label these lines as branch lines & write,

$$\Delta E_s = B(4J+6) \text{ cm}^{-1} \quad (J = 0, 1, 2, \ldots)$$

chapter-5

2019(H), 2021, 2020, 2018, 2017, 2016, 2015, 2021(H), 2020(H) Rajib Che-50

(A) Find the values of L, S and J for symbol:

(I) $^4S_1$ (II) $^2D_{3/2}$ (III) $^4P_{3/2}$ (IV) $^2D_{5/2}$ (V) $^3S_1$ (VI) $^2P_{3/2}$ (VII) $^3S_1$

(VIII) ~~^2P~~

⇒ (I) The term symbol is = $^{2S+1}L_J$

where, J = total angular momentum quantum number

L = orbital quantum number

S = total spin momentum

So, for $^4S_1$, we can write, $2S+1 = 4$

$\therefore S = 3/2$

And, $J = 1$

For above symbol L = 0, 1, 2, 3, 4 ----

S P D F G

So the value of L = 0

(II) For $^2D_{3/2}$, the valve of $L = 2$, $S = \frac{1}{2}$, $J = \frac{3}{2}$

(III) For $^4P_{3/2}$, the value of $L = 1$, $S = \frac{3}{2}$, $J = \frac{3}{2}$

(IV) For $^2D_{5/2}$, the valve of $L = 2$, $S = \frac{1}{2}$, $J = \frac{5}{2}$

CamScanner

(V) For $^3S_1$ the value of $L=0$, $S=1$, $J=1$

(VI) for $^2P_{3/2}$ the value of $L=1$, $S=1/2$, $J=\frac{3}{2}$

(VII) For $^3S_1$ the value of $L=0$, $S=1$, $J=1$

2019(t), 2020(t), 2021(t)

(X) Write the term symbol for: (I) $S=\frac{1}{2}$, $L=2$

(II) $S=\frac{3}{2}$, $L=1$ (III) $L=1, S=1, J=1$ (IV) $L=2, S=\frac{1}{2}$, $J=\frac{3}{2}$

(VIII) (I) For $S=\frac{1}{2}$, $L=2$

multiplicity $= 2S+1 = 2\times\frac{1}{2}+1 = 2$

$J = (L+S), (L+S-1) \ldots\ldots |L-S|$

$= \frac{5}{2}, \frac{3}{2}, \frac{1}{2}$

so the term symbol = $^2D_{5/2}$, $^2D_{3/2}$, $^2D_{1/2}$

(II) For $S=\frac{3}{2}$, $L=1$

multiplicity $= 2S+1 = 2\times\frac{3}{2}+1 = 4$

$J = (L+S), (L+S-1) \ldots\ldots (L-S)$

$= \frac{5}{2}, \frac{3}{2}, \frac{1}{2}$

CamScanner

So the term symbol = $^4P_{5/2}$, $^4P_{3/2}$, $^4P_{1/2}$

(III) for, $L=1, S=1, J=1$

multiplicity $= 3$

$\therefore$ term symbol $= {}^3P_1$

(IV) For $L=2, J=\frac{3}{2}, S=\frac{1}{2}$

multiplicity $= 2$

$\therefore$ term symbol $= {}^2D_{3/2}$

2023(t)

Find value of $\xrightarrow{L,S,J} {}^4D_{5/2}$? Write term symbol when $L=1, S=\frac{1}{2}, J=\frac{3}{2}$?

$\Rightarrow$ For $^4D_{5/2}$,

$2S+1 = 4 \quad \therefore S = \frac{3}{2}$

$L=2, \quad J = \frac{5}{2}$

$\Rightarrow$ multiplicity $= 2S+1 = 2\times\frac{1}{2}+1 = 2$

$\therefore$ term symbol $= {}^2P_{3/2}$

2017

(A) What are meant by H-like species & Helium like species? with example.

⇒ H-like species: Any atom or ion which contain only one electron in their outer shell & exhibit a spectrum like H are called H-like species.

Ex. He^+, Be^+, B^{2+} ions have one e^- in their outer shell like H & show similar spectrum like H.

$He^+(1) = 1s^1$

$Be^+(3) = 1s^2 2s^1$

$B^{2+}(3) = 1s^2 2s^1$

⇒ Any atom or ion which contain two e^- in their outer shell & exhibit a spectrum like He atom are called He-like species.

Ex. B^+, C^{2+}, Li ion have two e^- in their outer shell & they show spectrum like He atom's spectrum.

$B^+(4) = 1s^2 2s^2$

$C^{2+}(4) = 1s^2 2s^2$

$Li^+(2) = 1s^2$

2017

What is the selection rule for many e^- system in case of electronic spectroscopy?

$\Rightarrow \Delta S = 0, \Delta L = \pm 1, \Delta J = 0, \pm 1, \Delta n =$ anything

2016

Explain why the lines in atomic spectra of Na are double

$\Rightarrow$ Na(11) $\longrightarrow 1s^2 2s^2 2p^6 3s^1$

In excited state an e^- goes from 3s to 3p orbital.

Now, $L = 1$, for p orbital & $S = \frac{1}{2}$

$\therefore$ multiplicity $= 2S+1 = 2 \times \frac{1}{2} + 1 = 2$

$\therefore J = (L+S), (L+S-1), \ldots\ldots |L-S|$

$= \left(1 + \frac{1}{2}\right), \left(1 + \frac{1}{2} - 1\right) \ldots\ldots |L-S|$

$= \frac{3}{2}, \frac{1}{2}$

So the term symbol $= {}^{2S+1}L_J = {}^2P_{3/2}, {}^2P_{1/2}$

As we get two term symbol for Na so the spectral line of Na will be double/doublet.

(★) 2016 Why He^+, Be^+, B^{2+} etc show "H-like spectra" ?

⇒ He^+, Be^+, B^{2+} show H-like spectra because of the following reason:

(i) He^+, Be^+, B^{2+} all ions contain one electron in their outer shell orbital; which is similar to H.

(ii) The selection rule for these ion are the same with H, that is, Δn = anything, $\Delta S = 0$, $\Delta l = \pm 1$, $\Delta J = 0, \pm 1$.

for the same selection rule their spectra will be similar with H atom.

also The transition btwn ground state ($1s^1 2s^1$) to P-level are happen. $^2S_{1/2} \longrightarrow n\,^2P_{1/2,3/2}$ and a series of doublets similar to lymen series will be formed. From 2P state, two spectra series of lines will be seen.

$$2\,^2P_{1/2},\ ^2P_{3/2} \longrightarrow n\,^2S_{1/2} \text{ (doublet)}$$

and $$2\,^2P_{1/2,3/2} \longrightarrow n\,^2D_{3/2,5/2} \text{ (compound doublet)}$$

Thus this ions are stated as H-like species.

2021(t), 2017

(*) State & explain the building up principle for many e⁻ atoms with example?

⇒ The schrodinger eqn shows that e in atoms occupy orbitals of the same type & shape as the s, p, d.... orbitals discussed for the H-atom, but that the energies of these e⁻ differ markedly from atom to atom. There are 3 basic rules known as building-up rules, which determine how e⁻ in large atoms occupy orbitals. These may be summerized as:

1. Pauli's principle: No two e⁻ in an atom may have the same set of values for $n, l, l_z (\equiv m)$ and s_z. For example, in 1s orbital there exist two e⁻.

For 1st e⁻ → $n=1, l=0, l_z=0, s=+\frac{1}{2}$

for 2nd e⁻ → $n=1, l=0, l_z=0, s=-\frac{1}{2}$

Here for 1s orbital, value of n, l, l_z are same but their spin q.n. is different because two e⁻ occupy in opposite direction.

2. Aufbau principle: e⁻ tend to occupy the orbital with

lowest energy available. The energies of orbitals depends on the value of n & l. order of energy levels are -

$1s < 2s < 2p < 3s < 3p < 4s < 3d < 4p < 5s$

3. Hund's principle: e^- tend to occupy degenerate orbitals singly with their spins parallel. There are 3 (three) 2p orbitals each along with one co-ordinate axis, and each of these can contain two e^-. we may write the n, l, l_z & s as follows:

$n = 2, \quad l = 1, \quad l_z = 1, \quad s = \pm 1/2$

$n = 2, \quad l = 1, \quad l_z = 0, \quad s = \pm 1/2$

$n = 2, \quad l = 1, \quad l_z = -1, \quad s = \pm 1/2$

} total six e^-

electronic structure of some atoms.

	1s	2s	2p	
H	$\boxed{\uparrow}$			$1s^1$
He	$\boxed{\uparrow\downarrow}$			$1s^2$
Li	$\boxed{\uparrow\downarrow}$	$\boxed{\uparrow}$		$1s^2 2s^1$
C	$\boxed{\uparrow\downarrow}$	$\boxed{\uparrow\downarrow}$	$\boxed{\uparrow}\ \boxed{\uparrow}$	$1s^2 2s^2 2p_x^1 2p_y^1$

2017

⊛ Sketch the electronic energy levels & show transitions btwn them for single e^- of H-atom?

⇒ From the selection rules ~~we see~~ (Δn = anything, $\Delta l = \pm 1$) we see immediately that an e^- in the ground state (the 1s) can undergo a transition into any p state:

$$1s \longrightarrow np \quad (n \geq 2)$$

while a 2p electron can have transitions either into an s state or a d state:

$$2p \longrightarrow ns \text{ or } nd$$

Since s & d orbitals are here degenerate the energy of both these transions will be identical.

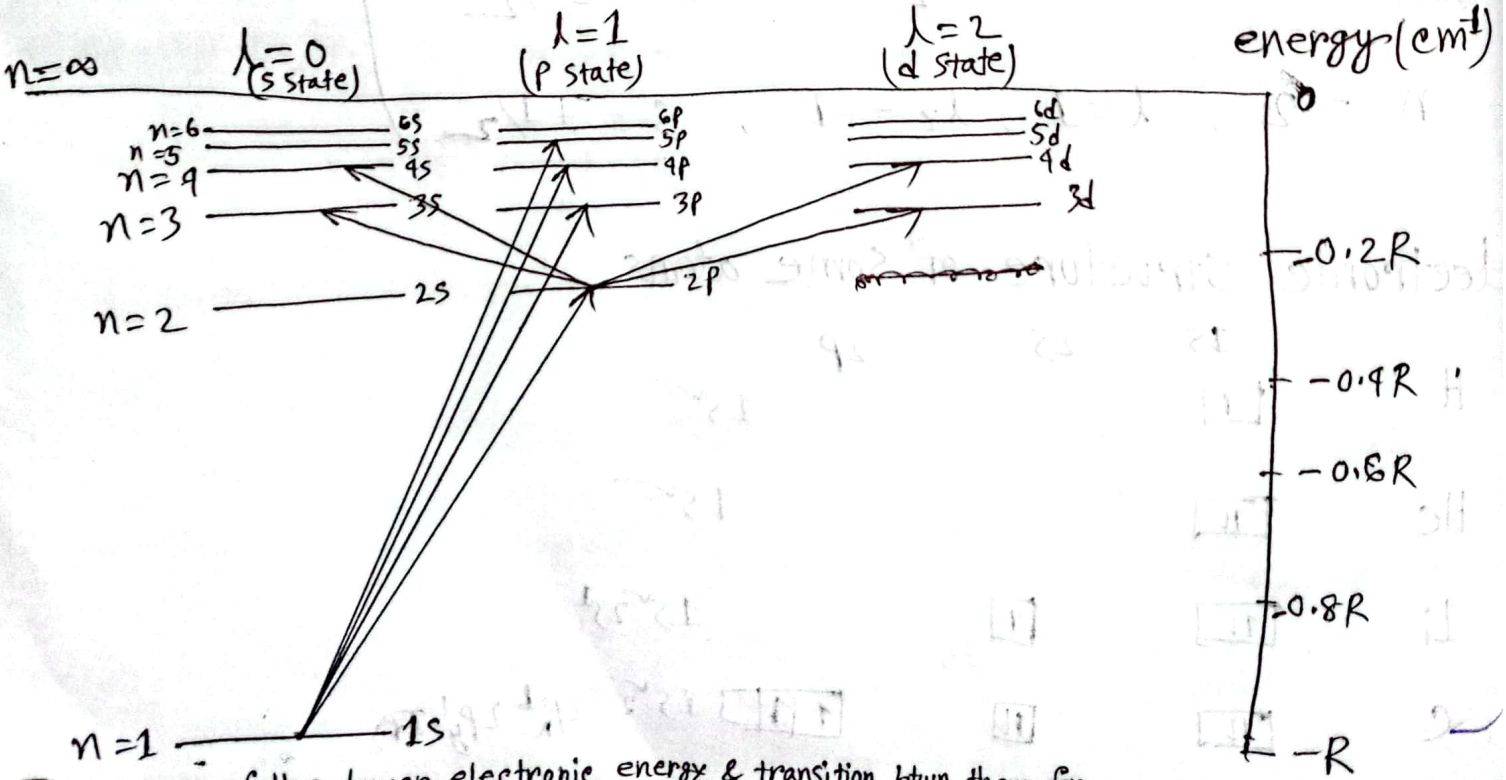


Fig: Some of the lower electronic energy & transition btwn them for the single e^- of the H-atom.

2017

⑧ State the selection rule for H-atom spectrum? which of the following transitions are allowed for H-atom spectrum?

⇒ selection rule for H-atom : Δn = anything , $\Delta l = \pm 1$ only , $\Delta J = 0, \pm 1$

⇒ for $2p \longrightarrow 4d$

for, $2p \longrightarrow n=2, l=1$

for, $4d \longrightarrow n=4, l=2$

Here, $\Delta l = 2-1 = 1$

So, the transition $2p \longrightarrow 4d$ is allowed for H-atom.

(ii) In $3p \longrightarrow 4s$

for $3p \longrightarrow n=3, l=1$

for $4s \longrightarrow n=4, l=0$

$\therefore \Delta l = 0-1 = -1$

So, the transition $3p \longrightarrow 4s$ is allowed for H-atom.

(iii) In $2p \longrightarrow 4f$,

$\Delta l = 3-1 = 2$

so the transition $2p \longrightarrow 4f$ is not allowed for H-atom.

(iv) In $2p \longrightarrow 1s$,

$\Delta l = 0 - 1 = -1$

So the transition $2p \to 1s$ is allowed for H-atom.

2017, 2020, 2021

(*) Show with spectrum that the lines of lyman series are observed at $3R/4, 8R/9, 15R/16, \ldots$ cm^{-1} in H-atom spectrum?

$\Rightarrow$ An e^- in a lower state n'' can undergo a transition into a higher state n', with absorption of energy

$$\Delta E = \varepsilon_{n'} - \varepsilon_{n''} \quad cm^{-1}$$

Therefore: $\bar{\nu}_{spect.} = -\frac{R}{n'^2} - \left(-\frac{R}{n''^2}\right) = R\left\{\frac{1}{n''^2} - \frac{1}{n'^2}\right\} cm^{-1}$

An identical spectral line will be produced in emission if the e^- falls from state n' to state n''. In both cases l must change by unity. Let us consider a few of these transitions, restricting ourselves to absorption for simplicity:

Transitions $1s \longrightarrow n'p$, $n' = 2, 3, 4, \ldots$ for these:

$$\bar{\nu}_{lyman} = R\left\{\frac{1}{1} - \frac{1}{n'^2}\right\} = R - \frac{R}{n'^2} \quad cm^{-1}$$

$$= \frac{3R}{4}, \frac{8R}{9}, \frac{15R}{16}, \frac{24R}{25}, \ldots cm^{-1} \text{ (for } n' = 2, 3, 4, 5, \ldots)$$

CamScanner

Hence we expect a series of lines at the wave numbers $3R/4, 8R/9, 15R/16, 24R/25, \ldots\ldots cm^{-1}$. Such a series is indeed observed in the atomic H-spectrum & it is called the lyman series. The spectrum of lymen series is given below:

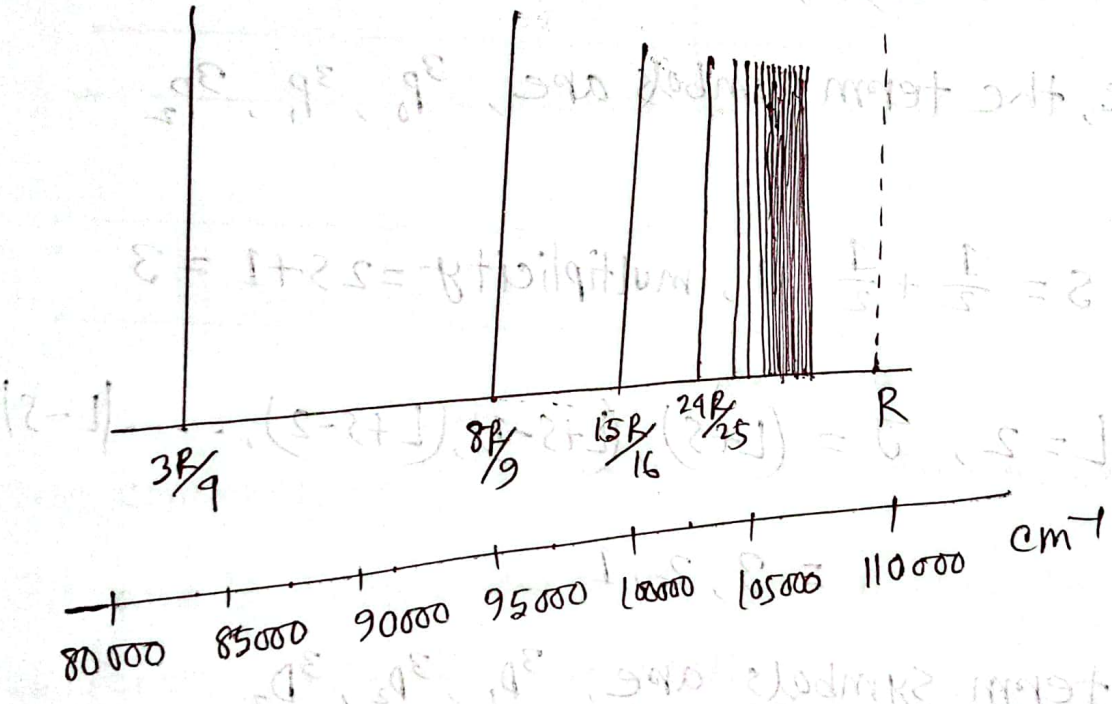


Fig: Representation of part of the lyman series of H atom, showing the convergence (ionization) point.

2017, 2020
Discuss the transitions from ^3P to ^3D states for He atom spectrum with necessary energy diagram & spectrum. why the spectrum obtained from $^3P \rightarrow {}^3D$ transitions is reffered to as compound triplet?

⇒ Transition btwn ^3P and ^3D states are very complex.

For ^3P, $S = \frac{1}{2} + \frac{1}{2} = 1$ $\therefore$ multiplicity $= 2S+1 = 2\times1+1 = 3$

$L = 1$ $\therefore J = (L+S), (L+S-1), \ldots |L-S|$

$= (1+1), (1+1-1), (1+1-2), \ldots |L-S|$

$= 2, 1, 0$

Therefore, the term symbols are, $^3P_0$, $^3P_1$, $^3P_2$

For, ^3D, $S = \frac{1}{2} + \frac{1}{2} = 1$, multiplicity $= 2S+1 = 3$

$L = 2$, $J = (L+S), (L+S-1), (L+S-2), \ldots |L-S|$

$= 3, 2, 1$

So, the term symbols are, $^3D_1$, $^3D_2$, $^3D_3$

The selection rule in this case, $\Delta J = 0, \pm 1$

We note that $^3P_2$ can go to each of $^3D_3$, $^3D_2$, $^3D_1$ but $^3P_1$ can go only to $^3D_2$, $^3D_1$ and $^3P_0$ can go only to $^3D_1$.

Thus the complete spectrum should consist of six lines. Normally the very close spacing is not resolved, & only three lines are seen, for this reason the spectrum is reffered to as a compound triplet.

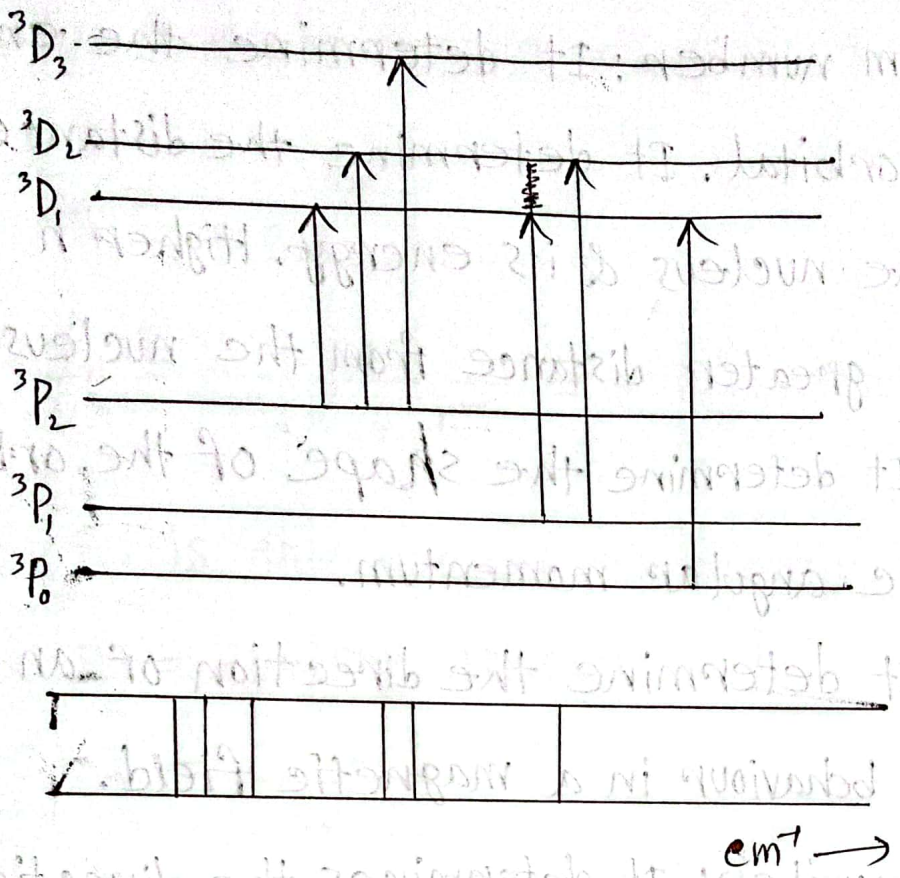


Fig: The 'compound triplet' spectrum arising from transitions btwn ^3P and ^3D levels in the He-atom. The seperations btwn levels of different J is much exaggerated.

(*) 2018 What are atomic quantum number? Discuss their significans?

⇒ Atomic Q. numbers are the set of set of numbers used to describe the position, energy, spin orientation and properties of each e^- in an atom. There are four quantum numbers: 1. principal, (n), 2. Azimuthal (l), 3. magnetic (m) 4. Spin (s) quantum number.

⇒ The significance of these are:

(*) principal quantum number: It determine the energy & size of the orbital. It determine the distance of the e^- from the nucleus & its energy. Higher n means higher energy & greater distance from the nucleus.

(2) Azimuthal (l): It determine the shape of the orbital & the electronic angular momentum.

(3) Magnetic (m): It determine the direction of an orbital & the electrons behaviour in a magnetic field.

(4) Spin (s) quantum number: It determines the direction of the e^- spin & the axial angular momentum of the e^-.

(*) 2018 write down the wave function expression & energy expression for H atom?

⇒ The wave function expression ψ_n for H-atom:

$$\psi_n = f\left(\frac{r}{a_0}\right) \exp\left(-\frac{r}{na_0}\right)$$

where $a_0 = \hbar^2/4\pi^2 me^2$, r is radial distance from the nucleus, $f(r/a_0)$ is a power series of degree $(n-1)$ in r/a_0, and "n" is the principal quantum number.

Energy expression for H-atom:

$$E_n = -\frac{me^4}{8h^2\varepsilon_0^2 n^2}\ J$$

$$\varepsilon_n = -\frac{me^4}{8h^3c\varepsilon_0^2 n^2} = -\frac{R}{n^2}\ cm^{-1} \quad (n=1,2,3,\ldots)$$

where, ε_0 is the vacuum permittivity,

R = Rydberg constant

$$= \frac{me^4}{8h^3c\varepsilon_0^2}$$

2016

(X) Sketch the energy levels of lithium atom in a figure showing the difference in energy of s, p & d states with the same value of n & also show some allowed transitions in the figure?

⇒ Electronic configuration of Li is $1s^2 2s^1$ which have one e^- in the outermost shell.

The selection rule for alkali metals are the same as for H, that is Δn = anything, $\Delta l = \pm 1$, $\Delta J = 0, \pm 1$ & so the spectra will be similar also. Thus transitions from the ground state $(1s^2 2s)$ can occur to p levels: $2S_{1/2} \rightarrow nP_{1/2,3/2}$

and a series of doublets similar to the lyman series will be formed, converging to some point from which the ionization potential can be found. From the 2p state, two separate series of lines will be seen:

$$2\,^2P_{1/2,3/2} \longrightarrow n\,^2S_{1/2}$$

and, $$2\,^2P_{1/2,3/2} \longrightarrow n\,^2D_{3/2,5/2}$$

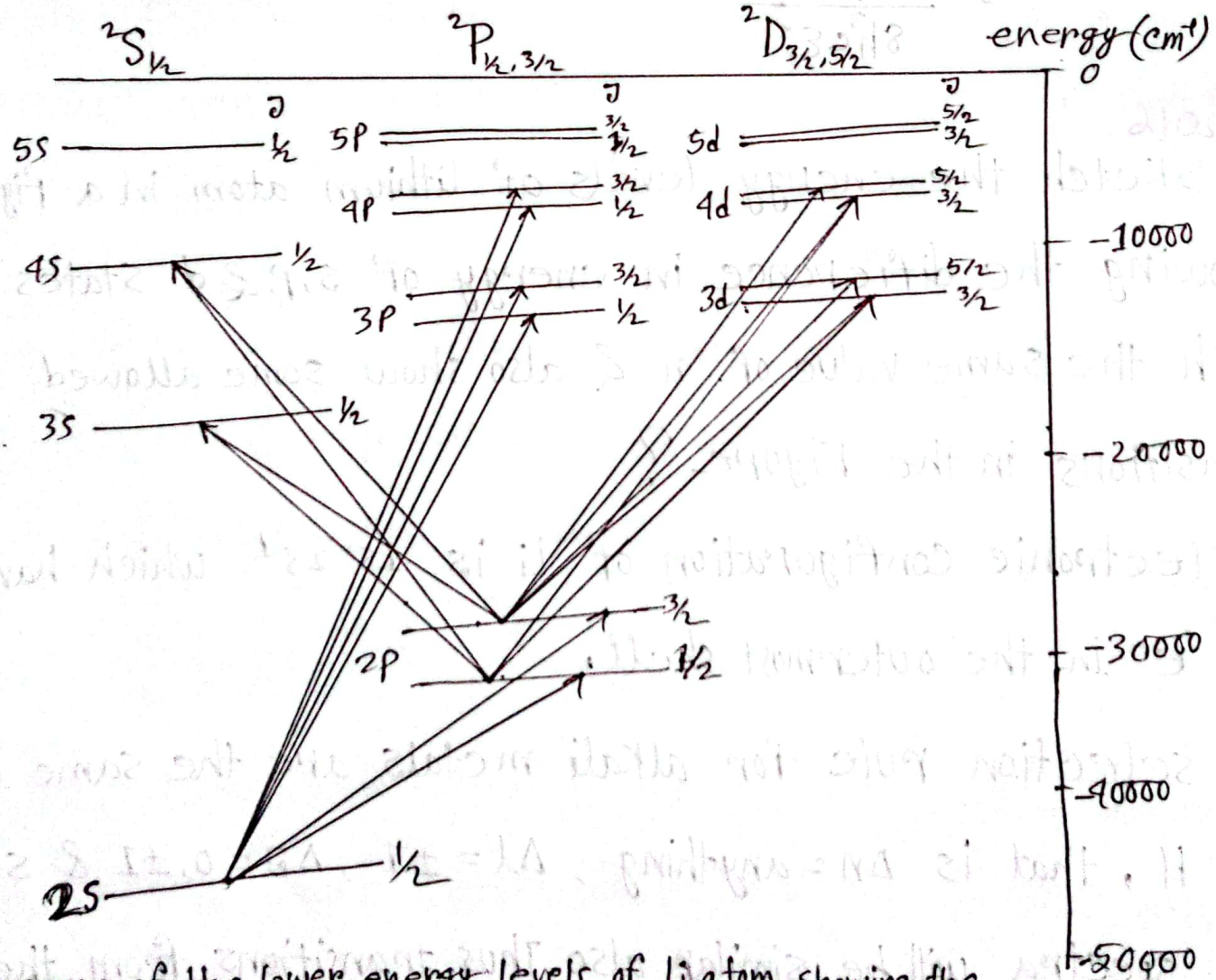


fig: some of the lower energy levels of Li-atom, showing the difference in energy of S, P & d states with the same value of n. Some allowed transitions are shown here.

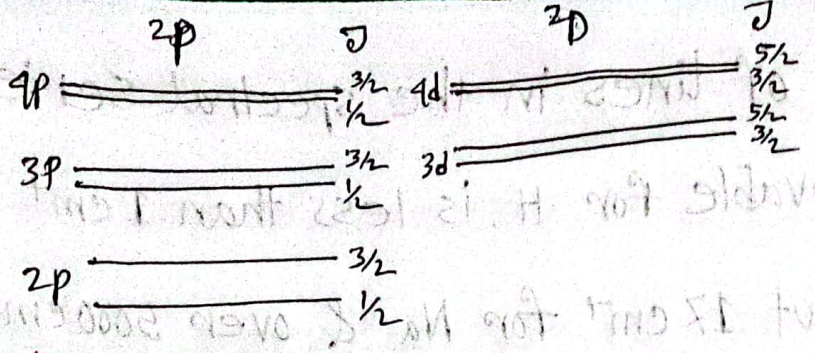
2016
⊛ The doublet seperation of lines in the spectral series, which is scarcely observable for H, is less than 1 cm^{-1} for the 2p level of Li, about 17 cm^{-1} for Na & over 5000 cm^{-1} for cesium - explain?

⇒ The doublet seperation of lines depends on the J-splitting. The J-splitting increase markedly with the atomic number due to coupling btwn l and s. For this reason, the line of seperation increases from H to Li to Na to Cs.

2016, 2023(t)
⊛ what is meant by 'J-splitting'?

⇒ There is no attempt to show the energy level splitting of the P & D states, the seperation btwn the levels differing only in J. The difference btwn J values in term as J-splitting.

The splitting of energy level due to the interaction the orbital angular momentum (l) & spin angular momentum (s) of an e^- is called J-spitting. The J-splitting decreases with increasing n & l. For example, ^2P splits into $^2P_{1/2}$ & $^2P_{3/2}$ and ^2D splits into $^2D_{3/2}$ & $^2D_{5/2}$.



2016, 2021, 2018, 2015

Explain compound doublet spectra with energy diagram & spectrum for the transition btwn ^2P & ^2D levels in the fine H-atom spectra?

⇒ Transition btwn ^2P & ^2D states are very complex.

For ^2P, $S=\frac{1}{2}$, $L=1$, so $J=\frac{1}{2}, 3/2$. the term symbols are $^2P_{1/2}$, $^2P_{3/2}$.

For ^2D, $S=\frac{1}{2}$, $L=2$, so, $J=\frac{5}{2}, \frac{3}{2}$. So the term symbols are $^2D_{3/2}$, $^2D_{5/2}$.

The selection rule in this case, $\Delta J=0, \pm 1$. Hence $^2P_{1/2}$ can go only to $^2D_{3/2}$ $[\Delta J=1]$ and $^2P_{3/2}$ can go ~~only~~ to $^2D_{3/2}$ $[\Delta J=0]$ and $^2D_{5/2}$ $[\Delta J=1]$ which are allowed but $^2P_{1/2}$ can not go to $^2D_{5/2}$ $[\Delta J=+2]$ because of ΔJ value.

$^2P_{3/2} \rightarrow {}^2D_{5/2}$ will occur close to $^2P_{3/2} \rightarrow {}^2D_{3/2}$ because the separation btwn the doublet D states is very small.

$^2P_{1/2} \rightarrow {}^2D_{3/2}$ will be more widely spaced. But $^2P_{1/2} \rightarrow {}^2D_{5/2}$ is not

allowed.

~~Thus the~~

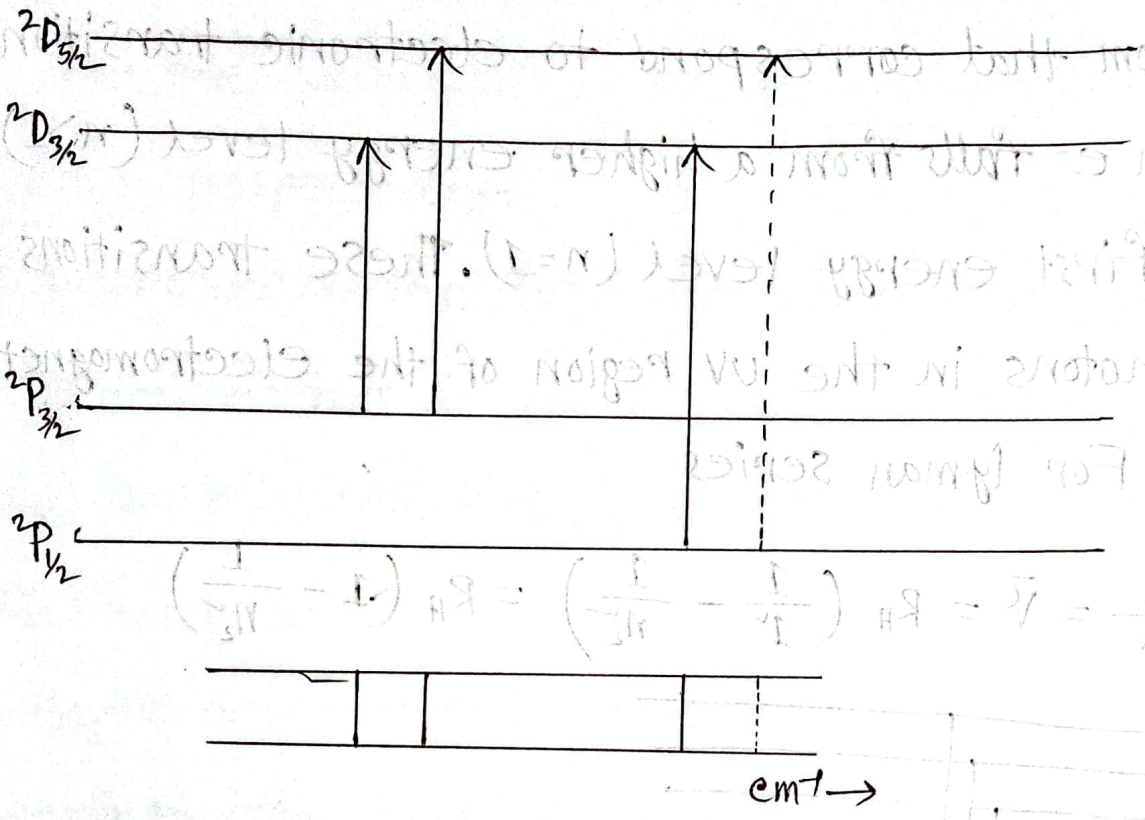


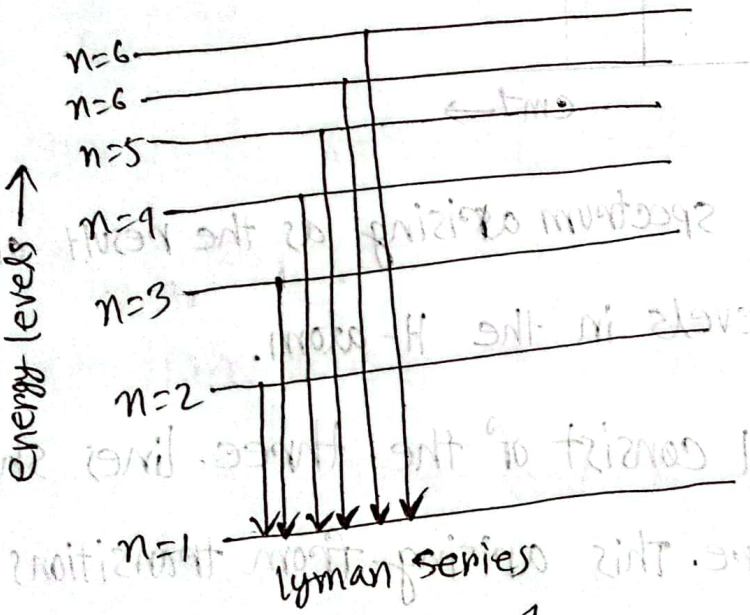
Fig: The compound doublet spectrum arising as the result of transitions btwn ^2P & ^2D levels in the H-atom.

Thus the spectrum will consist of the three lines shown at the foot of the figure. This arising from transitions btwn btwn doublet levels, is usually reffered to as a "compound doublet" spectrum.

(*) 2015 What is lyman-series of H-atom?

⇒ The lyman series is a set of spectral lines in the H-atom that correspond to electronic transitions where an e- falls from a higher energy level ($n>1$) to the first energy level ($n=1$). These transitions release photons in the UV region of the electromagnetic spectrum. For lyman series

$$\frac{1}{\lambda} = \bar{\nu} = R_H\left(\frac{1}{1^2} - \frac{1}{n_2^2}\right) = R_H\left(1 - \frac{1}{n_2^2}\right)$$



(*) 2022 Find value of L, S, & J for $^4P_{3/2}$ & $^2D_{5/2}$? What do these values signify?

⇒ For $^4P_{3/2}$, $L=1$, $J=\frac{3}{2}$, $S=\frac{3}{2}$.

for $^2D_{5/2}$, $L=2$, $J=\frac{5}{2}$, $S=\frac{1}{2}$

⇒ L represent the total orbital angular momentum of all e^-. For the values of $L=0,1,2,3....$ we use the symbols S, P, D, F respectively.

S represents the total spin angular momentum of all e^-. Each e^- has a spin of $\pm 1/2$ & 'S' is the sum of the spin contributions. Multiplicity depends on value of s which is $2S+1$.

J represents the total angular momentum of the system, including both spin (s) & orbital (L) contributions.

$$J = (L+S), (L+S-1), ..., |L-S|$$

⊗ 2023 (T), 2022
Explain the energy of an atomic orbital?

⇒ The energy of an atomic orbital depends on the principal (n) ~~and~~ quantum number & atomic number (Z). Energy eqn:

$$E_n = -13.6 \frac{Z^2}{n^2} \text{ eV} \quad \text{or} \quad \varepsilon_n = -\frac{Z^2 R_H}{n^2} \text{ cm}^{-1}.$$

The energy is negative because the e^- is bound to the nucleus, meaning that it has less energy than a free e^-. As 'n' increase, the energy becomes less negative, indicating

that the e⁻ is less tightly bounded. For H-atom the energy eqn becomes:

$$E_n = -\frac{13.6}{n^2}\ eV \quad \text{or} \quad \bar{\nu}_n = -\frac{R_H}{n^2}\ cm^{-1} \quad [As,\ Z=1]$$

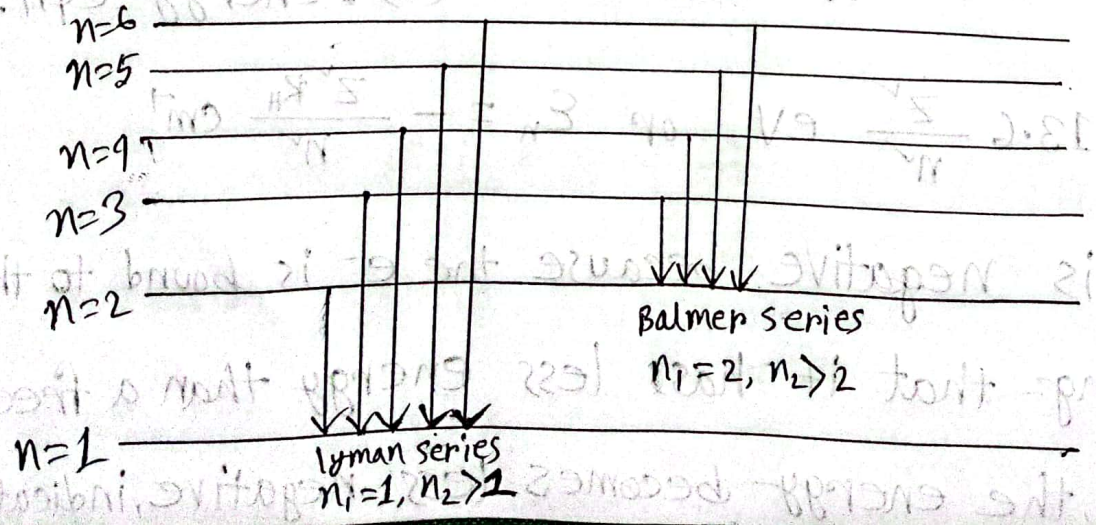
2023(t), 2022

⁂ which electronic transition can give rise to the lyman & Balmer specral lines? & Mention their absorption energy?

⇒ An electronic transitions from higher energy orbit ($n_2>1$) to the lowest energy orbit ($n_1=1$) are called lyman serie. Its spectral region is UV. $\frac{1}{\lambda} = \bar{\nu} = R_H\left(\frac{1}{1^2} - \frac{1}{n_2^2}\right)$, $\Delta E = \frac{h}{}$

The spectral lines fall in the visible region when transi occurs from higher energy level ($n_2>2$) to the first excited state ($n_1=2$) are called Balmer spectral lines.

$$\frac{1}{\lambda} = \bar{\nu} = R_H\left(\frac{1}{2^2} - \frac{1}{n_2^2}\right), \quad \Delta E = \frac{hc}{\lambda}$$



2023(t)

Q How does the compound doublet spectrum arise for H-atom?

⇒ The compound doublet spectrum arise for H-atom due to the fine structure of the H-atom's energy levels. This fine structure splitting is caused by the interaction btwn the electron's spin & its orbital angular momentum. It results in two closely spaced energy levels, leading to the observation of two distinct spectral lines in the H-spectrum. This interaction known as spin-orbit coupling, causes the energy levels to split into closely spaced doublets. when an e^- transitions btwn these closely spaced levels, it emits photons with slightly different energies, resulting in the observed doublet splitting in the spectral lines.

CamScanner

chapter-6

RAJIB
Che-50

2018, 2020, 2016, 2022

(★) State Franc-condon principle? Discuss different possibilities of transition for a molecule which undergoes electronic transition from ground to upper state both electronically & vibrationally.

⇒ State that: an electronic transition takes place so rapidly that a vibrating molecule doesnot change its internuclear distance appreciably during the transition.

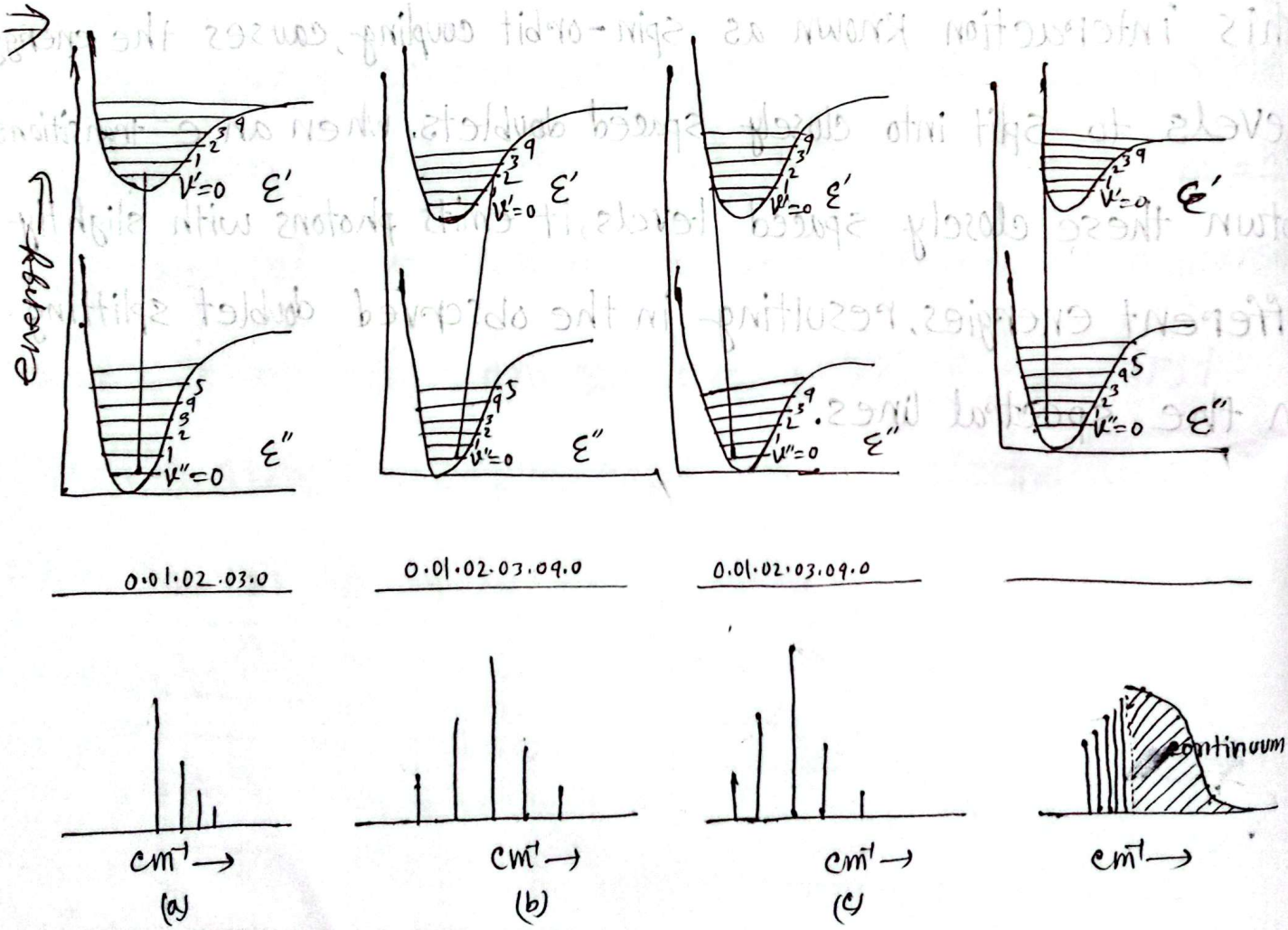


Fig: The operation of the Franck-condon principle for (a) internuclear distances equal in the upper & lower states (b) upper state internuclear distance a little less than that in the lower state (c) upper state distance a little greater than in the lower (d) upper state distance considerably greater.

In Fig-(a): the upper electronic state having the same equilibrium internuclear distance as the lower. And Franck-condon principle suggest that a transition occurs vertically on this diagram.

In Fig(b): the excited electronic state has a slightly smaller internuclear seperation than the ground state. A transition occur from $V''=0$ to $V'=2$.

In Fig(c): the excited electronic state has a slightly larger internuclear seperation than the ground state & spectrum will be similar with Fig(b).

In fig(d): The excited electronic state have greater internuclear seperation than lower state & transition take place a higher V' value.

2018, 2020, 2022

Explain why Homonuclear molecule (H_2, N_2) do not show vibrational & rotational spectra but they exhibit electronic spectra?

⇒ Because the criteria for microwave active is the molecules must be present parmanent dipole moment but Homonuclear molecule do not have any permanent dipole moment so they does not show rotation spectra. similarly for IR active during vibration dipole moment must be changed & the molecule is IR active & show vibration spectrum but homonuclear (N_2, O_2, H_2) do not

CamScanner

change its dipole moment during vibration. so Homonuclear does not show vibrational & rotational spectrum.

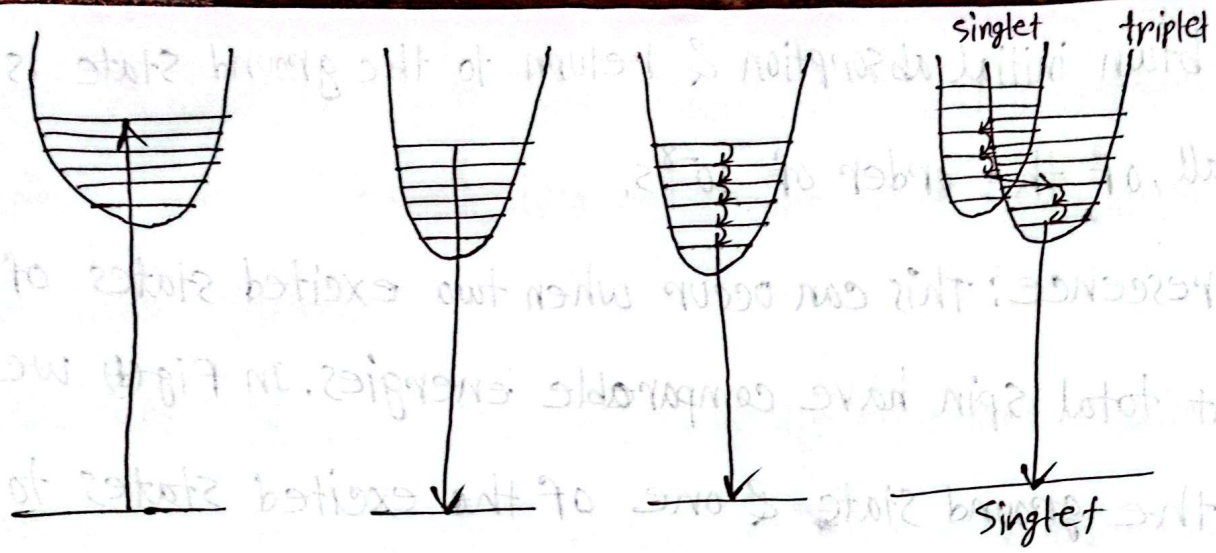
Electronic transition involve the promotion of e⁻ from one molecular orbital to another, These transitions do not depend on the presence of a dipole moment. In Homonuclear electronic transition occurs due to the rearrangement of e⁻ with in molecular orbitals which can result in absorption or emmition of UV or visible light. Hence they exhibit electronic spectra.

2018, 2020, 2021, 2017, 2016, 2022

(*) What are the different process by which excited molecules can loss its excess energy? Discuss Fluorescence process with diagram? Discuss phosphorescence with diagram?

⇒ There are several processes by which excited molecule can loss its excess energy. such as: (I) Dissociation (II) Re-emission (III) Fluorescence (IV) Phosphorescence (V) stimulated emission.

1. Dissociation: The excited molecule breaks into two fragments No spectroscopic phenomena, beyond the initial absorption spectrum, are observed unless the fragments radiate energy by one of the process mentioned below.



(a) Absorption (b) Re-emission (c) Fluorescence (d) phosphorescence

2. Re-emission: If the absorption process takes place like fig (a) then re-emission is just the reverse of this. The radiation emitted, which may be collected & displayed as an emission spectrum, is identical in frequency with that absorbed.

3. Fluorescence: If the molecule is in a high vibrational state after electronic excitation, then excess vibrational energy may be lost by intermolecular collisions. The vibrational energy is converted to kinetic energy & appears as heat in the sample. When the excited molecule has reached a lower vibrational state (for example $v'=0$), it may then emit radiation & revert to the ground state; the radiation emitted, called the fluorescence spectrum, is normally of lower frequency than that of the initial absorption, but under certain conditions it may be of higher frequency.

CamScanner

The time btwn initial absorption & return to the ground state is very small, of the order of 10^{-8} s.

4. Phosphorescence: This can occur when two excited states of different total spin have comparable energies. In Fig (d) we imagine the ground state & one of the excited states to be Singlets (that is $S=0$), while the neighbouring excited state is triplet ($S=1$). Although the rule $\Delta S = 0$ forbids spectroscopic transitions btwn singlet & triplet states, there is no prohibition if the transfer btwn the excited states occurs kinetically i.e. through radiationless transitions induced by collisions. such transfe Known as inter-system crossing, can only occur close to the cross- over point of the two potential curves & once the molecule ha arrived in the triplet state & undergone some loss of vibration energy in the state, it cannot return to the excited singlet state. It will therefore eventually reach the $V'=0$ level of th triplet state. Now although a transition from here to the the ground state is spectroscopically forbidden, it may take place but much more slowly than an allowed electronic transition. Thus it i that a phosphorescent material will continue to emit radiation

seconds, minutes, or even hours after the initial absorption. The phosphorescence spectrum, as a rule, consists of frequencies lower than that absorbed.

5. Stimulated emission: this increasingly important mechanism for removal of excess energy can lead to the production of laser radiation. The helium-neon laser emits light in the visible region of the spectrum.

2018, 2020, 2021, 2024(t), 2020(t), 2019(t)

Ⓧ Write spectral region in which following transitions may take place: $\sigma \rightarrow \sigma^*$, $n \rightarrow \sigma^*$, $\pi \rightarrow \pi^*$, $n \rightarrow \pi^*$?

⇒

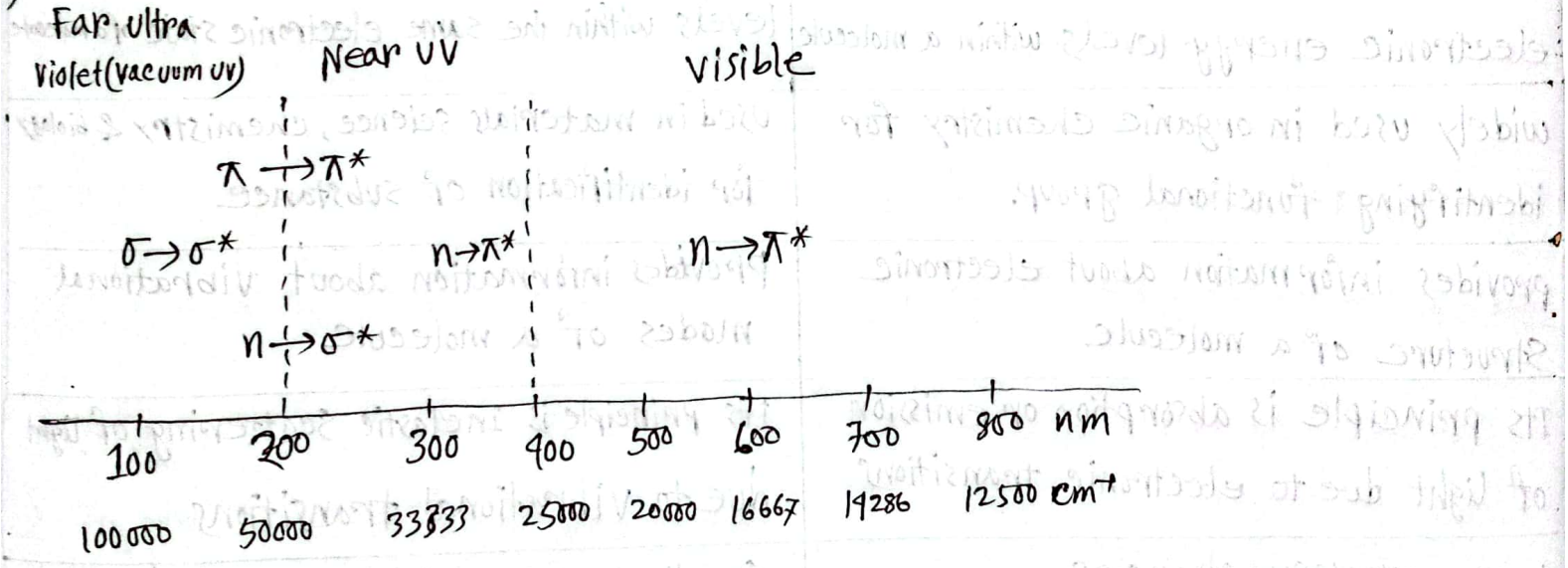


Fig: The regions of the electronic spectrum & the type of transition which occurs in each.

In general, the σ electrons are most strongly bound to the nuclei & hence require a great deal of energy to undergo transitions while π & n electrons requires less energy. The n electrons are

usually require less than the π. Thus in an obvious rotation σ → σ* transition fall into the vacuum UV, π → π* & n → σ* appears near the broadline.

(★) 2018, 2020, 2021, 2016, 2021(t) What are the basic differences btwn electronic & Raman Spectroscopy?

electronic	Raman
Its type of transition is electronic transition.	It is vibrational transitions within the same electronic state.
Its energy source typically UV or visible light.	It uses monochomatic light source
Involves transitions of e⁻ btwn different electronic energy levels within a molecule	Involves transitions btwn vibrational energy levels within the same electronic state of a molecu
widely used in organic chemistry for identifying functional group.	used in materials science, chemistry & biolog for identification of substance
provides information about electronic structure of a molecule	Provides information about vibrational modes of a molecule
Its principle is absorption or emission of light due to electronic transitions	Its principle is Inelastic scattering of light due to vibrational transitions
large energy changes	small energy changes
light is absorbed or emitted by sample	light is scattered by the sample
sensitive to electronic environment & solvent effects	less affected by solvents or external fields
The exciting radiation for e. spectroscopy must be just equivalent to the energy difference btwn the electronic state	The exciting r. for Raman S. can be any frequency except that which would induce an e transition

2021, 2018, 2017, 2016, 2015, 2021(+)

(*) Describe Born-oppenheimer approximation for electronic spectra of diatomic molecules? Explain why all molecules exhibit electronic spectra?

or, short note on "Born-oppenheimer approximation"?

⇒ The Born-oppenheimer approximation is the assumption that the electronic motion & the nuclear motion in molecules can be seperated. It assumes that the combined rotational & vibrational energy is simply the sum of the seperate energies. A. to Born oppenheimer approximation for electronic spectra of diatomic molecule.

$$E_{total} = E_{electronic} + E_{vibration} + E_{rotation}$$

which implies that the electronic, vibrational & rotational energies of a molecule are completely independent of each other. Therefore a change in the total energy of a molecule may be written as:-

$$\Delta E_{total} = \Delta E_{elec.} + \Delta E_{vib.} + \Delta E_{rot.} \quad J$$

or, $\Delta \varepsilon_{total} = \Delta \varepsilon_{elec.} + \Delta \varepsilon_{vib.} + \Delta E_{rot} \quad cm^{-1}$

The approximate orders of magnitute of these changes are:-

$$\Delta \varepsilon_{elec} \approx \Delta \varepsilon_{vib.} \times 10^{3} \approx \Delta \varepsilon_{rot.} \times 10^{6}$$

and so we see that vibrational changes will produce a 'coarse structure'

CamScanner

and rotational changes a 'fine structure' on the spectra of electronic transitions.

⇒ electronic spectra are given by all molecules since changes in the e⁻ distribution in a molecule are always accompanied by a dipole change. All the molecule give electronic spectra since all molecule contains e⁻ capable of absorbing energy & undrgoing transitions, making electronic spectra a univershal feature of molecules. All molecules contain e⁻ that can be excited from one energy level to another when they absorb energy.

2017, 2022

⊛ Sketch & discuss the UV-visible spectrum of propanone & benzene?

⇒

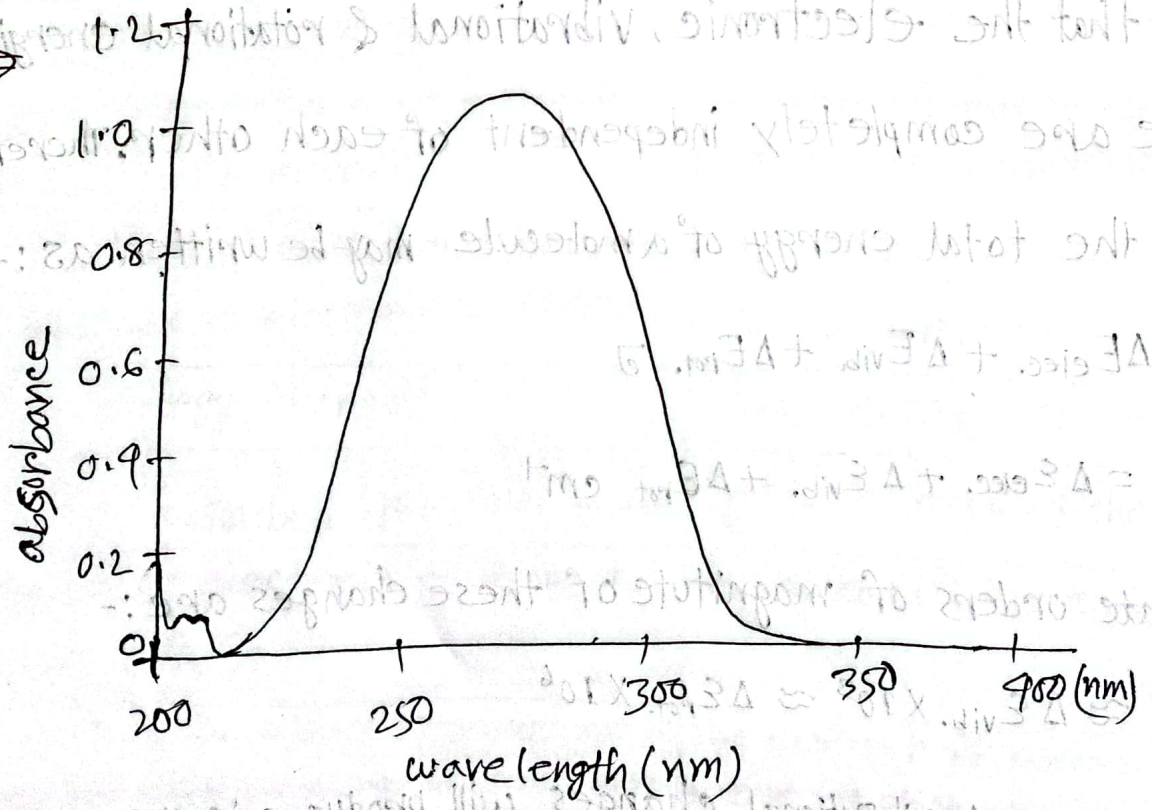


Fig: The UV spectrum of propanone.

The above Figure shows the UV visible spectrum of propanone. Then $n \rightarrow \pi^*$ transition at 272 nm is clearly seen as a broad feature. ($\lambda_{max} = 272$ nm). The UV- visible spectrum of propanone provides insight into the electronic transitions occuring in the molecule when it absorbs UV or visible light. In this transition, a non-bonding e^- from the 'O' atom is excited to the antibonding π^* orbital of the carbonyl group.

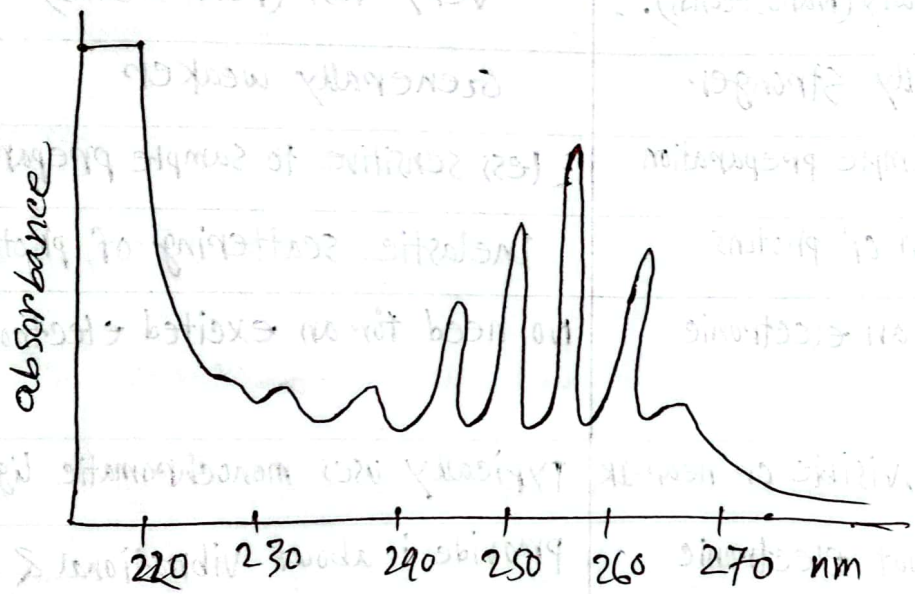


Fig: part of the UV spectrum of benzene showing vibrational Fine structure

Benzene's UV spectrum is dominated by $\pi \rightarrow \pi^*$ transitions, with fine structures that reflect its highly symmetric & conjugated nature. It shows the series of bands around 250nm being due to weak $\pi \rightarrow \pi^*$ transitions. This transitions slightly weakens the bond & gives rise to a vibrational structure.

2017
Difference btwn fluorescence process & Raman spectroscopy-write?

⇒

Fluorescence process	Raman spectroscopy
In fluorescence, radiation must be absorbed by the molecule & an exited electronic state formed.	In Raman s., energy is merely transferred from radiation to molecule but no excited electronic state is formed.
Involves real electronic transitions.	Involves virtual electronic-vibrational transitions.
wave length of emitted light longer than the incident light (stokes shift).	λ of emitted light slightly longer or shorter than incident light (Raman shift)
It occurs relatively slowly (Nanoseconds).	very fast (Femtoseconds)
Signal intensity generally stronger	Generally weaker
can be sensitive to sample preparation	less sensitive to sample preparation.
Absorption & reemission of photons	Inelastic scattering of photons
Requires excitation to an electronic excited state	No need for an excited electronic state
Its light sources are UV, visible or near-IR	Typically uses monochromatic light source
provides information about electronic transitions & environment	Provide i. about vibrational & rotatial information about molecular bonds.

2016, 2022
Discuss Woodward-Feiser's rule to calculate λ_{max} with example?

⇒ It state that: "Substituents on conjugated system also perturb the UV transitions in a systematic way, a greater deal of empirical data has led to the formulation of rules to predict the effects. These are known to as woodward-Feiser's rule".

For examaple consider butadiene $CH_2=CH-CH=CH_2$ which has a strong absorption at 217 nm due to the $\pi \rightarrow \pi^*$ transition. Any substituents or modification to this parent fragment have each been assigned a +ve/-ve value which must be added to the basic absorption at 217nm in order to arrive at the expected absorption frequency of the substituent molecule. Thus the increment for Cl atom is 5 mm & for $-CCH_3$ group 6nm; if both of these substituents occur together, the molecule absorb at $217+5+6 = 228$ nm.

2015

(*) The UV spectrum for acetone shows two peaks at $\lambda_{max}=280$ nm, $\varepsilon_{max}=15$ and $\lambda_{max}=190$nm, $\varepsilon_{max}=100$. Identify for electron transitions for each? which one of these is more intense?

⇒ acetone $CH_3-\overset{O}{\overset{\|}{C}}-CH_3$ contain $-\overset{O}{\overset{\|}{C}}-$ so the π-e⁻ and lone pair e⁻ both are present. So two types of transition also occurs:

(I) $\pi \rightarrow \pi^*$ (II) $n \rightarrow \pi^*$

The value for $\pi \rightarrow \pi^*$ transition is lower $\lambda_{max}=190$nm, $\varepsilon_{max}=100$.

The value for $n \rightarrow \pi^*$ transition is higher $\lambda_{max}=280$ nm, $\varepsilon_{max}=15$.

But $\pi \rightarrow \pi^*$ shows more intense peak. Because π & π^* both orbitals situated in the same plane. So intensity is more &

CamScanner

on the other hand $n \rightarrow \pi^*$ transition shows less intense peak.

Because n & π^* orbitals are situated perpendicular to each other.

2015
⊗ Calculate absorption maximum in the uv spectrum of 2,4-hexadiene?

⇒ $CH_3-CH=CH-CH=CH-CH_3$

2,4-hexadiene

Base value for conjugated diene, = 217 nm ($\pi \rightarrow \pi^*$ transition)

Two $-CH_3$ substituent present = 2×5 = 10 nm

Total value λ_{max} = 217 + 10 = 227 nm

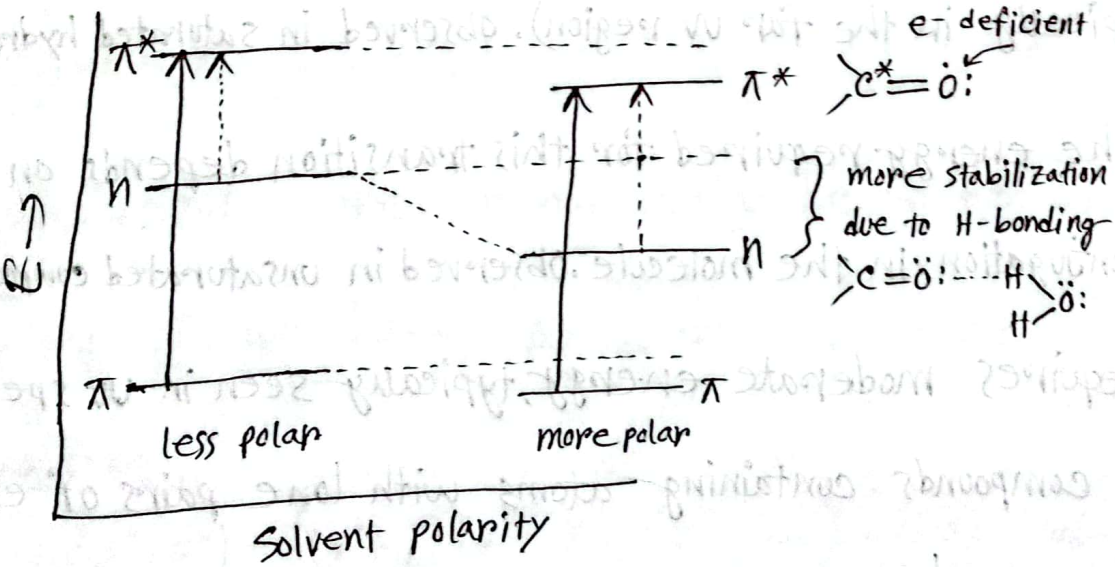
2015
⊗ Explain the effect of polar solvents on $n \rightarrow \pi^*$ & $\pi \rightarrow \pi^*$ transition?

⇒ polar solvents have distinct effects on $n \rightarrow \pi^*$ & $\pi \rightarrow \pi^*$ transitions due to their ability to interact with the electronic states of a molecule via dipole-dipole & H-bonding interactions.

In $n \rightarrow \pi^*$ transitions, an e^- is excited from n to π^*. polar solvents preferentially stabilize the ground state, which has a higher e⁻ density on the lone pair of e^- in the n orbital. This stabilization increase energy gap btwn ground & excited states, resulting in a blue shift (shift to shorter λ) of the absorption band. polar solvent

stabilize the ground state ~~(π state)~~ (n orbital) more than excited state (π* orbital) because n orbital has more localized e⁻ density that interacts strongly with the solvent. this stabilization increase ΔE btwn n & π* states, causing absorption band to shift to shorter λ (blue shift).

In π→π* transition, polar solvents typically stabilize the excited state (π*) more than the ground state due to increased dipole moment of the excited state. This stabilization lowers the energy gap btwn π & π*, causing the absorption band to shift to longer wavelength (red shift).

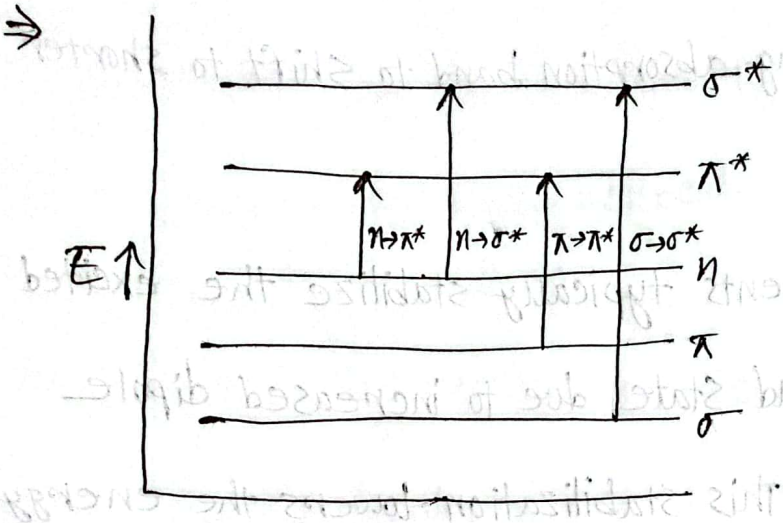


2015
⊗ what is electronic spectroscopy? Show diagrammatically various types of transition involved in this technique?

⇒ Electronic spectroscopy is a branch of spectroscopy that studies

the interaction of electromagnetic radiation with matter, specially focusing on the excitation of e^- from lower to higher energy states.

⇒



① $\sigma \rightarrow \sigma^*$: Involves the excitation of an e^- from bonding σ orbital to an antibonding σ^* orbital. Requires higher energy radiation (typically in the far UV region). observed in saturated hydrocar

② $\pi \rightarrow \pi^*$: The energy required for this transition depends on the extent of conjugation in the molecule. observed in unsaturated compounds

③ $n \rightarrow \sigma^*$: requires moderate energy, typically seen in UV spectra observed in compounds containing atoms with lone pairs of elect (e.g alc, ether, amines).

④ $n \rightarrow \pi^*$: requires lower energy radiation compared to $\pi \rightarrow \pi^*$ transitions. observed in compounds containing atoms with lone pairs of e^- conjugated with π bonds (e.g. carbonyl, nitro compounds).

CamScanner

(X) 2017 Illustrate progressions with allowed electronic transitions & spectrum?

⇒ using the Born-oppnheimer approximation the total energy of a diatomic molecule may be written as:

$$E_{total} = E_{electric} + E_{vibration} + E_{rotation} \quad \text{---(I)}$$

Ignoring rotational changes means that we rewrite eqn (I) as-

$$E_{total} = E_{electric} + E_{vibration} \quad J$$

or $$\mathcal{E}_{total} = \mathcal{E}_{electric} + \mathcal{E}_{vibration} \quad cm^{-1} \quad \text{---(II)}$$

moreover we can write

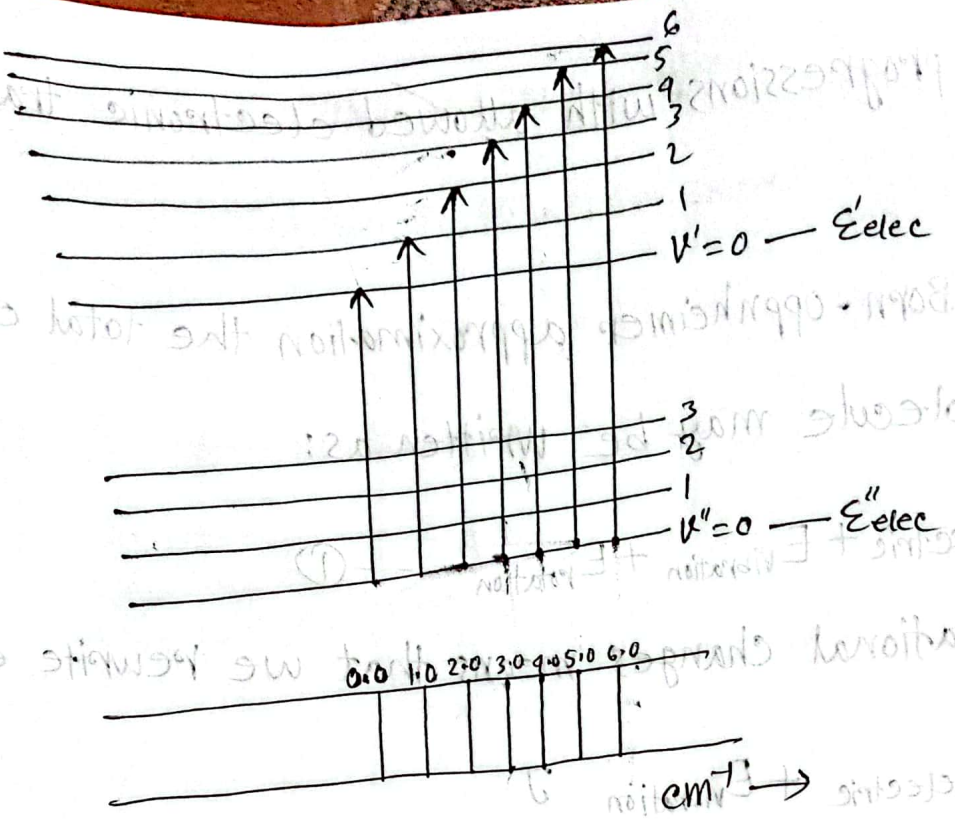
$$\mathcal{E}_{total} = \mathcal{E}_{electric} + \left(v+\tfrac{1}{2}\right)\bar{\omega}_e - x_e\left(v+\tfrac{1}{2}\right)^2\bar{\omega}_e \quad cm^{-1} \quad (v = 0, 1, 2, \ldots) \quad \text{---(III)}$$

An analytical expression can easily be written for the spectrum.

from eqn (III) we have, $\Delta\mathcal{E}_{total} = \Delta\mathcal{E}_{elec.} + \Delta\mathcal{E}_{vib.}$

Therefore,

$$\bar{\nu}_{spec.} = (\mathcal{E}' - \mathcal{E}'') + \left\{\left(v'+\tfrac{1}{2}\right)\bar{\omega}'_e - x'_e\left(v'+\tfrac{1}{2}\right)^2\bar{\omega}'_e\right\} - \left\{\left(v''+\tfrac{1}{2}\right)\bar{\omega}''_e - x''_e\left(v''+\tfrac{1}{2}\right)^2\bar{\omega}''_e\right\} cm^{-1}$$



The energy levels of the eqn (iii) are shown in this figure for two arbitrary value of $\mathcal{E}_{elec}$.

this Figure suggest that the ground state can usually undergo a transition to several excited states & each such transition will be accompanied by a band spectrum similar to this figure. such a set of transitions is called band since, under low resolution each line of the set appears somewhat broad & diffuse, & is more particularly called a v' progression, since the value of v' increas by unity for each line in the set.

(A) 2021 Explain how the dissociation energy determined by analysis of the electronic spectra of a diatomic molecule?

⇒ The excited molecule breaks into two fragments where

the equilibrium nuclear seperation in the upper state is considerably greater than that in lower. The spectrum consisted of some vibrational transitions followed by a continum representing dissociation. The lower wavenumber limit of this continum must represent just sufficient energy to cause dissociation & thus we have:

$$\bar{\nu}_{\text{continuum limit}} = D_0'' + E_{ex} \quad cm^{-1}$$

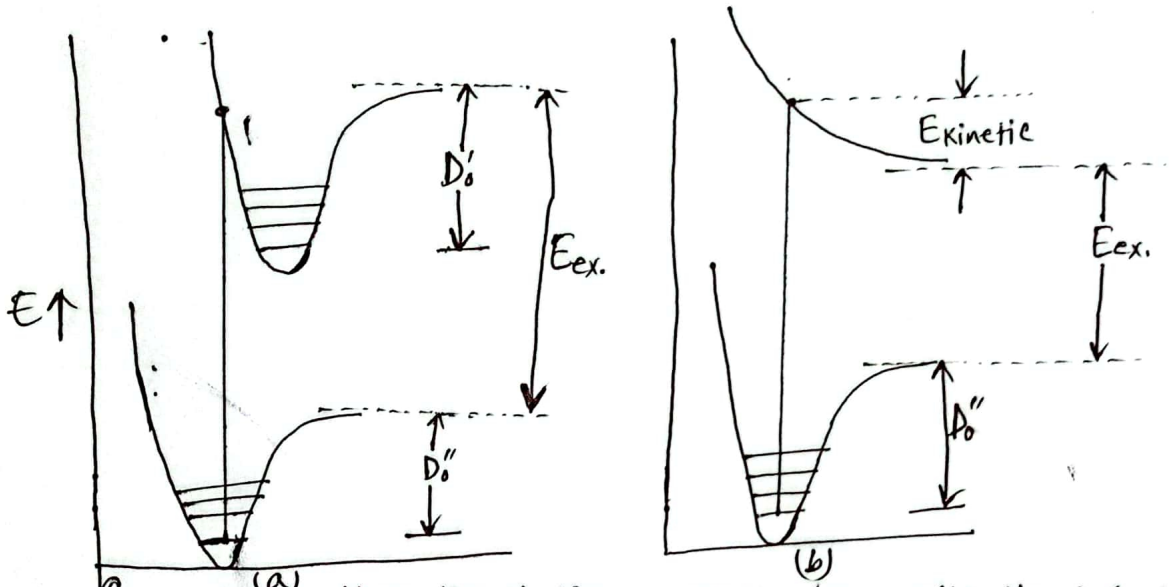


Fig: Elastrating dissociation ~~energy~~ by excitiation into (a) a stable upper state and (b) a continuous upper state

we can measure dissociation energy D_0'', if we know E_{ex}. (the exitation of the products). it is possible to derive a valve for the dissociation energy by noting how the vibrational lines converage. the vibrational energy levels may be written:

$$\varepsilon_v = \left(v + \frac{1}{2}\right)\bar{\omega}_e - x_e\left(v + \frac{1}{2}\right)\bar{\omega}_e \quad cm^{-1}$$

The seperation btwn neighbouring levels, $\Delta\varepsilon$ is

$$\Delta\mathcal{E} = \mathcal{E}_{v+1} - \mathcal{E}_v$$
$$= \bar{\omega}_e \{1 - 2x_e(v+1)\} \text{ cm}^{-1}$$